THREE PERSPECTIVES / ONE MEMORY CYCLE

存储周期的三重视角

从 AI 持续学习、价格证据到中国供给

依据 Citi、Bernstein 与 Morgan Stanley 三份报告，建立从概念到模型、从数据到判断的学习路径。

三个视角，理解同一个存储周期

需求 | Citi	价格 | Bernstein	供给 | Morgan Stanley
持续学习 训练 + 推理 时间：2027—2031E	现货 / 合约 / 长约 成本与降配 时间：2026年8月 / Q3	中国扩产与技术进展 边际价格与跨周期回报 时间：2028年及以后

24	89	75
系统课程	原图表导读	完整原文页

原报告日期：2026年8月26日、9月3日、9月14日。所有预测保留原机构与时间；中文课程为忠实转述和教学整理，完整原件附于书后。

LEARNING MAP

从入门，走向独立判断

01 **读懂语言 / 第1—4课**

先分清产品、单位、价格口径，再进入预测。

02 **理解 AI 需求 / 第5—8课**

从持续学习理解 HBM、服务器 DRAM 与 eSSD 的关系。

03 **拆开供需模型 / 第9—13课**

把晶圆、位元、资本开支与缺口连接起来。

04 **读懂市场证据 / 第14—16课**

价格分化、降配和长约共同约束这一轮周期。

05 **理解中国供给 / 第17—20课**

从扩产、技术壁垒到 2028 年情景压力测试。

06 **形成研究判断 / 第21—24课**

产业史、估值、数据核验与采购迁移。

三种内容标记

报告观察：保留原报告记录的市场数据。机构预测：保留E、TBD、可能等不确定性。教学整理：解释概念、公式与采购迁移，情景参数不冒充机构新预测。

推荐方式：第一次按课程顺序建立主线；第二次使用供需表和第23课检查口径；第三次回到图表索引与原文页面，逐项验证假设。交互版可查看原图、复算供需、操作大摩矩阵和记录学习进度。

READ THE DENOMINATOR

先记住口径，再记住数字

报告与指标	原文数值	适用范围
Bernstein：Q3常规DRAM	约+19% QoQ	8月样本所指向的季度涨幅，非完整Q3实现值
Bernstein：Q3 NAND	样本+9.4%；行业接近+20%	样本不含SSD；行业值纳入机构对SSD的判断
Citi：2027 HBM	需求75.157bn Gb；缺口约21%	持续学习和ASIC需求驱动的模型预测
Citi：2027 DRAM / NAND	缺口约8.7% / 6.1%	按其图20/36供需表口径
MS：2027 HBM	供给56.300；需求56.085bn Gb	接近平衡，与Citi需求假设有显著分歧
MS：长鑫2028产能	500KWPM	晶圆能力口径，不等于收入或技术全面领先
MS：2028 NAND情景	从缺口6%到过剩9%	取决于YMTC产能和AI SSD增速组合

Bernstein · p.6 Citi · p.11 Citi · p.13 Citi · p.17 Morgan Stanley · p.4 Morgan Stanley · p.5 Morgan Stanley · p.12

为什么课程保留分歧

同一个市场可以因需求估计、供给爬坡、产品范围与时间口径不同，得到不同预测。图表重绘以明确原始表为依据；摘要增速与表内绝对量不能对应的地方，会单独记录而不擅自修正。

LESSON 01 / 学习方法

三份报告，三个观察位置

为什么“现在紧缺”和“未来竞争加剧”可以同时成立?

花旗从需求端出发，讨论持续学习如何让训练和推理同时增长，并把供需预测延伸到 2031 年。伯恩斯坦从交易证据出发，观察 2026 年 8 月现货、合约与各应用价格，再讨论成本压力和长约上限。大摩从供给端出发，研究长鑫与长江存储扩张，以及它们如何影响全球竞争和跨周期回报。

阅读顺序宜先建立产品与价格语言，再学习花旗的需求逻辑，随后用伯恩斯坦的价格证据和大摩的扩产情景检验它。三份报告共同认可 AI 需求与供给配置的重要性，但预测的时间范围、数据样本和假设并不相同。这里所有市场数字均保留在各报告的观察日期，2026E、2027E 等 E 均表示机构估计或预测。

Citi · p.1 Bernstein · p.1 Morgan Stanley · p.1



进阶研读 / 检查结论的边界

进阶阅读要把“观察”“解释”“预测”拆开。例如 8 月报价属于报告记录的数据，降配是资源优化属于机构解释，紧缺持续至 2031 年属于条件性预测。课程中的示意图、公式解释和采购迁移是依据材料编制的教学内容，均不冒充原报告新增结论。

LESSON 02 / 入门概念

存储系统的三层分工

HBM、DDR5 与 eSSD，分别解决什么问题?

花旗把 HBM 对应到训练和持续模型更新，把服务器 DRAM、SoCAMM2 对应到推理与 CPU 工作负载，把 eSSD 对应到需要长期保存的数据。理解这一区分，可以把系统想成靠近加速器的高速工作区、服务器侧的运行内存，以及容量更大的持久存储。这个分层是学习用的简化模型，真实系统会把数据按访问需求在不同层之间移动。

HBM 属于 DRAM 家族，DDR5、LPDDR5/5X 也是不同使用场景下的 DRAM 产品，而 NAND 是 SSD、eMMC 和 UFS 所依赖的存储介质。eMMC/UFS 将 NAND 与控制器封装在一起，eMCP/uMCP 又加入 DRAM，因此整包价格不能当作单一 DRAM 或 NAND 的单价。报告同时谈论产品需求和介质位元需求，读图时需要分清这两个层次。

Citi · p.3 Citi · p.6 Citi · p.16 Bernstein · p.6



产品	材料中的主要角色	阅读重点
HBM	训练、模型更新、GPU/ASIC	位元需求、堆叠代际、TSV 与认证
服务器 DDR5 / SoCAMM2	推理、CPU、外部内存工作负载	每台容量与服务器数量
eSSD / QLC SSD	持久数据、近 GPU 存储、部分卸载	企业级产品与 NAND 总位元
LPDDR / eMMC / UFS	移动与终端设备	终端减产、规格与产品组合

进阶研读 / 检查结论的边界

NAND 进入更多 AI 工作负载，不表示它在所有性能维度上可以替代 HBM。花旗讨论的是分层存储、数据卸载和更靠近 GPU 的容量池，其商业价值仍取决于具体工作负载。带宽、延迟、容量、持久性和成本之间的取舍，比简单按“先进程度”排列产品更有解释力。

LESSON 03 / 入门概念

单位，是研究的第一道门槛

晶圆产能、位元出货和营收份额为什么不能混用?

DRAM 表常用百万个 1Gb 等效单位，NAND 表常用百万个 8Gb 等效单位；这里的小写 b 是 bit，大写 B 是 byte。1GB 等于 8Gb，因此 16Gb DRAM 芯片对应 2GB，不能把 16Gb 当成 16GB。伯恩斯坦总表多用美元/GB，但部分图用美元/颗或美元/条，eMCP/uMCP 还存在整包计价例外。

KWPM 表示每月千片晶圆，300KWPM 就是 30 万片/月。位元产出还受每片晶圆可用 die 数、单 die 容量、良率和利用率影响，营收又取决于价格和产品组合。因此大摩给出的长鑫晶圆份额、位元份额和收入份额不同，是三种度量，不能相互替代。

Bernstein · p.6 Morgan Stanley · p.3 Morgan Stanley · p.5 Morgan Stanley · p.9 Citi · p.13

同一个数字，先看单位与分母

16Gb = 2GB

芯片标称容量
bit 与 byte 相差 8 倍

300KWPM

每月 300,000 片晶圆
不是全年产出

供需比

供给 ÷ 需求 − 1
负值：不足；正值：过剩

原始表示	正确读法	常见误读
75,157 Mpcs, 1Gb eq	751.57 亿 Gb，即 75.157bn Gb	当成 751.57 亿 GB
1,581,792 mn, 8Gb eq	1,581.792bn 个 8Gb 等效单位	直接与 DRAM 1Gb 表比较
300KWPM	300,000 片/月	当成全年 30 万片
US$/GB	每 GB 价格	当成整条内存或整颗芯片价格

进阶研读 / 检查结论的边界

花旗预测长鑫 2027 年末 TSV 产能为 50KWPM，而大摩使用 40KWPM；两者对应的 HBM 年度产出预测也差异很大。年末能力不等于全年平均产能，TSV 产能也不等于通过客户认证后可交付的位元。模型复算应明确期间、单位和是否包含 HBM，避免在不同口径之间拼接。

LESSON 04 / 入门概念

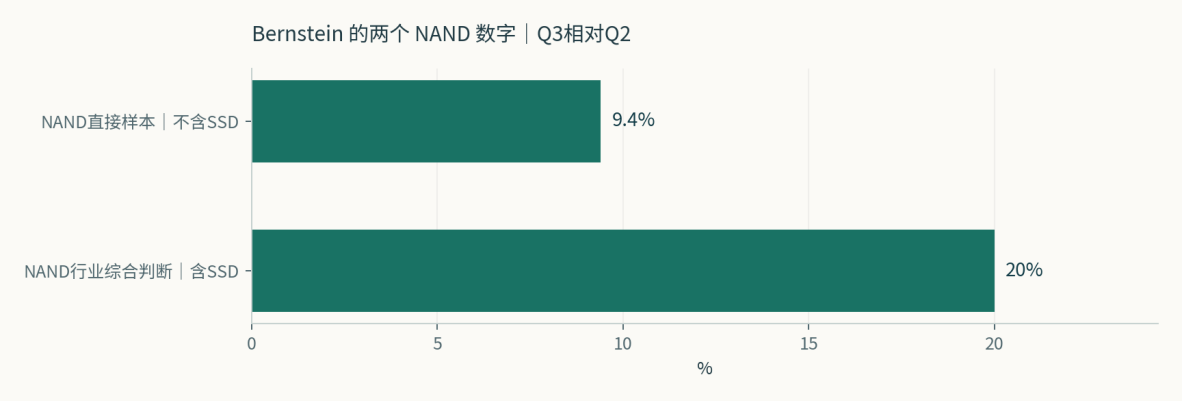
涨价，要先问是哪一种价格

为什么同一季度可以同时出现 NAND +9.4% 和接近 +20%?

伯恩斯坦直接跟踪的 NAND 样本由晶圆与 eMMC/UFS 构成，8 月数据指向相对第二季度上涨 9.4%。该样本未覆盖 SSD，而机构判断客户端和企业级 SSD 价格均可能上涨约 20%，纳入后行业涨幅才被估为接近 20%。这两个数表达的市场范围不同，不应被当成同一指标的两个版本。

同样，常规 DRAM 合约样本上涨 19.1%，不等于厂商财报中的混合 ASP 也上涨 19.1%。HBM 年约、产品组合、合同签订与出货的时间差，会改变公司最终确认的平均售价。报告称现货市场约占总量一成且噪声较大，因此现货更适合提供方向线索，需要与合约和终端需求共同阅读。

Bernstein · p.3 Bernstein · p.5 Bernstein · p.6 Bernstein · p.17



进阶研读 / 检查结论的边界

月环比、季度环比、季度迄今均价以及年度平均价格同比不能直接并列。伯恩斯坦 9 月 3 日的总表中，9 月合约一栏仍为缺失，因此其 Q3 判断不是完整季度实绩。报告表内的厂商模型注明上次更新在 2026 年 6 月，也不能误认作 8 月重新校准后的结果。

LESSON 05 / 核心机制

持续学习如何改变需求

AI 获得“新知识”，为什么可能同时需要更多内存与存储?

花旗将持续学习定义为在吸收新任务、新知识的同时保留已有知识的增量训练方法。难点在于学习新信息时避免“灾难性遗忘”，因此既可能需要不断更新模型，也需要再次访问过去的数据。由此，花旗推导出持续训练、推理和历史数据保留三方面的长期资源需求。

这条逻辑把存储需求从一次训练后的阶段性采购，扩展到模型持续更新和长期运行。HBM 与模型更新相关，服务器 DRAM 支持推理和 CPU 工作负载，eSSD 则承接持久数据需求。这里的长期增长是花旗的产业判断，具体增量仍取决于持续学习技术落地、工作负载规模和资源效率。

Citi · p.1 Citi · p.2 Citi · p.3

持续学习的需求链｜按花旗逻辑整理

学习新任务	执行新任务	保留已有经验
持续训练和模型更新 对应 HBM 需求	更复杂推理与CPU负载 对应服务器 DRAM	需要持久保存与访问数据 对应 eSSD / NAND

进阶研读 / 检查结论的边界

需要区分“模型参数获得更新”和“应用可以查阅历史数据”。报告前者对应持续学习，后者是其涉及的数据保留与访问需求之一，两者不能简单画等号。学习时可追问：新业务量、单次任务资源强度与系统效率分别如何变化，最终净需求才是三者共同作用的结果。

LESSON 06 / 核心机制

训练、推理与个人 AI 的时间轴

为什么 HBM 的阶段性降温不等于整个 AI 存储周期结束?

花旗把 2022 年至 2025 年上半年概括为训练主导阶段，HBM 需求强，而常规 DRAM/NAND 相对平淡。2025 年下半年至 2026 年，推理与 KV cache 的复杂度上升，服务器 DRAM 和 eSSD 需求更受关注。它进一步预测，2027 年至 2028 年上半年进入持续学习第一阶段，训练与推理相关存储共同增长。

花旗将 2028 年下半年以后定义为个人 AI、实体 AI 的潜在扩展阶段，端侧执行和本地数据管理的重要性提高。图中的时间边界是一条机构预测路径，并非已经发生的行业分界。其核心启发是观察需求在不同存储层之间迁移，而非只用某一种存储产品的热度判断整个周期。

Citi · p.2 Citi · p.3 Citi · p.7

花旗预测的 AI 工作负载演进



进阶研读 / 检查结论的边界

图 3 的 token 使用趋势来源于 OpenRouter 与花旗，报告标示 2025 年 1 月至 2026 年 8 月月复合增速 31%，2026 年 8 月同比 2,434%。这说明其所跟踪序列增长很快，但原图不足以证明它代表全球全部 AI token。也不能把 token 增速按同一比例外推为 HBM、DRAM 或 NAND 位元增速。

LESSON 07 / 模型研读

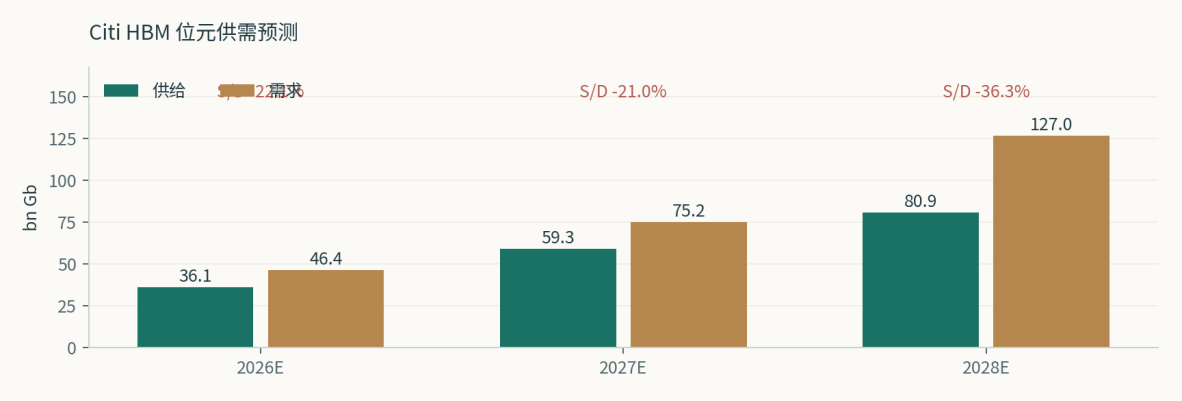
HBM：把客户需求拆开看

谁在推动花旗 2027—2028 年 HBM 需求上修?

花旗预测 HBM 需求由 2026 年的 46.409bn Gb 增至 2027 年的 75.157bn Gb，再增至 2028 年的 126.990bn Gb，对应约 62% 和 69% 增长。除 Nvidia 外，Broadcom、Google 及其他 ASIC 相关需求也构成重要贡献。仅看单一 GPU 产品配置变化，容易漏掉客户数量、加速器出货量以及 ASIC 的扩展。

在花旗表中，Broadcom 需求由 2027 年的 13.001bn Gb 增至 2028 年的 41.003bn Gb，成为该年非常重要的增量来源。报告同时列示 HBM3E、HBM4、HBM4e 等代际与堆叠组合，显示增长既包括总量也包括结构变化。客户分类沿用花旗模型口径，不能另把 Google 等自研 ASIC 全部重复加进 Broadcom 一栏。

Citi · p.4 Citi · p.13 Citi · p.14



进阶研读 / 检查结论的边界

花旗对 HBM 总量的计算比“单颗配置降低”更关注全系统出货。教学关系式是总位元需求等于设备数量乘以平均配置，再按客户与产品加总。课程实验室允许改变数量和配置，展示降配后总量上升、持平或下降所需的不同条件，参数变化仅为教学示例。

LESSON 07 / 结构化研读

HBM：把客户需求拆开看 · 数据与对照

客户口径	2026E	2027E	2028E
Nvidia	22.003	32.503	40.939
Broadcom	9.002	13.001	41.003
Google	4.702	10.300	14.003
AMD	4.200	7.000	8.200
AWS	2.501	2.802	6.391
Others	4.001	9.550	16.455
总量，bn Gb	46.409	75.157	126.990

Citi · p.4 Citi · p.13 Citi · p.14

回到原图，看完整上下文

Citi #4 · HBM 客户与代际需求表

拆分客户、HBM代际和需求占比；与图28为重复呈现，保留原表。

Citi #26 · HBM 供需总表

2027E 59.347/75.157bn Gb，对应约-21%。

Citi #27 · HBM TSV 能力与厂商供给

年末TSV能力、年度位元供给和市场份额分开阅读。

Citi #28 · HBM 客户与代际需求表（再列）

对照图4可核验重复数据；个别历史分项存在差别。

LESSON 08 / 产品研读

HBM 路线图：配置不是总需求

如何把产品路线图读成可验证的需求假设？

花旗图 5 列出 Nvidia、AMD、AWS、Google、Microsoft、Meta、OpenAI、Apple 与 Anthropic 等客户的 GPU/ASIC 路线图。表中既有已列示产品，也有 2027—2028 年预测、可能的设计伙伴与 TBD，因此不能把整张表当作厂商正式发布承诺。读图时应同时检查时间、HBM 代际、堆叠层数和单颗加速器容量。

例如，图中列示 B300 的 HBM3E 12hi 配置为 288GB，AMD MI400 的 HBM4 12hi 配置为 432GB；这些数字必须保留在该报告的模型语境内。个别未来产品配置下降，可能与花旗讨论的从 scale-up 向 scale-out 转变有关。其是否拉低总需求，仍取决于设备数量以及其他客户、其他产品的需求增量。

Citi · p.5 Citi · p.4



报告中的产品例子	列示 HBM 配置	阅读方式
B300	HBM3E 12hi，288GB	与 GB300 的双 GPU 口径区分
Rubin / Rubin Ultra	288GB / 192GB	报告预测，不能仅据此推断总量
MI400	HBM4 12hi，432GB	高配置需与出货量一起估算
TPU v9i / v9t	HBM4e 8hi，384GB	后续代际与设计伙伴需验证
OpenAI Jalapeño	HBM4 12hi，216GB	沿用该研报路线图表述

进阶研读 / 检查结论的边界

原图包含多处“possibly”和“TBD”，这些不确定性本身也是信息。课程只选取例子帮助理解，完整客户、产品、设计伙伴和时间信息可在原文第 5 页查看。对采购建模而言，应把发布、量产、认证、客户部署作为不同里程碑，而不要把预测年份直接变成确定交付年份。

LESSON 09 / 供给机制

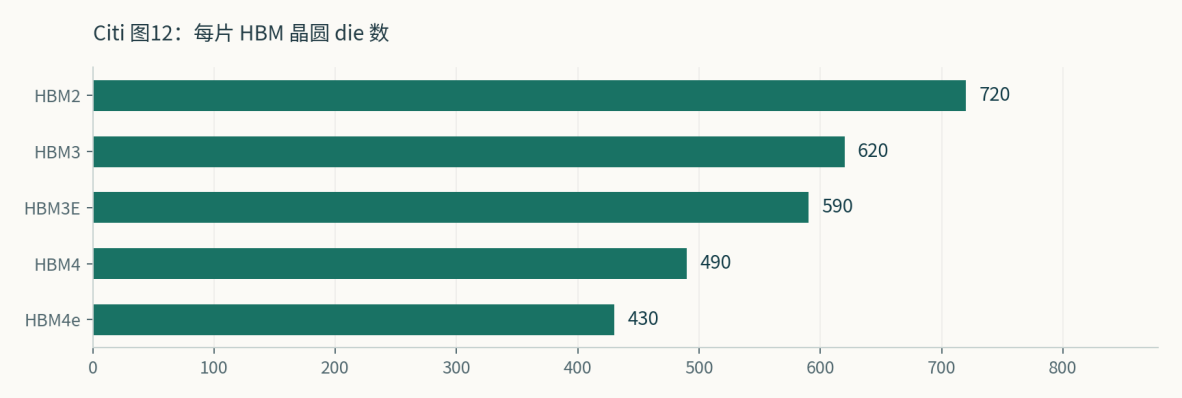
HBM 的晶圆机会成本

同样增加一片晶圆，为什么可交付位元不同?

大摩指出，HBM 需要较大的 DRAM die，并叠加 TSV、封装、测试和客户认证等环节，单位位元对应的晶圆负担较大。它估计这一负担由 2021—2023 年约 2.5 倍常规 DRAM，提高至 2028 年约 3 倍。HBM 对先进 DRAM 晶圆的占用比例，也被预测从 2023 年约 6% 提高到 2028 年约 34%。

花旗图 12 从另一角度展示压力：每片 HBM 晶圆的 die 数由 HBM2 的 720，降至 HBM3 的 620、HBM3E 的 590、HBM4 的 490 和 HBM4e 的 430。晶圆变多未必带来同幅度位元增长，还要考虑 die 容量、良率和产品组合。因此扩产、制程迁移与最终合格产品交付应分开跟踪。

Morgan Stanley · p.4 Morgan Stanley · p.10 Citi · p.9



进阶研读 / 检查结论的边界

大摩的 34% 是“先进 DRAM 晶圆”占用比例，不是 HBM 营收份额，也不是全行业全部晶圆的同口径占比。花旗的 die 数也不能单独换算成最终 HBM stack 数量。要从前端产能走到客户可用供给，还需加入堆叠层数、封装良率和认证限制。

LESSON 10 / 模型研读

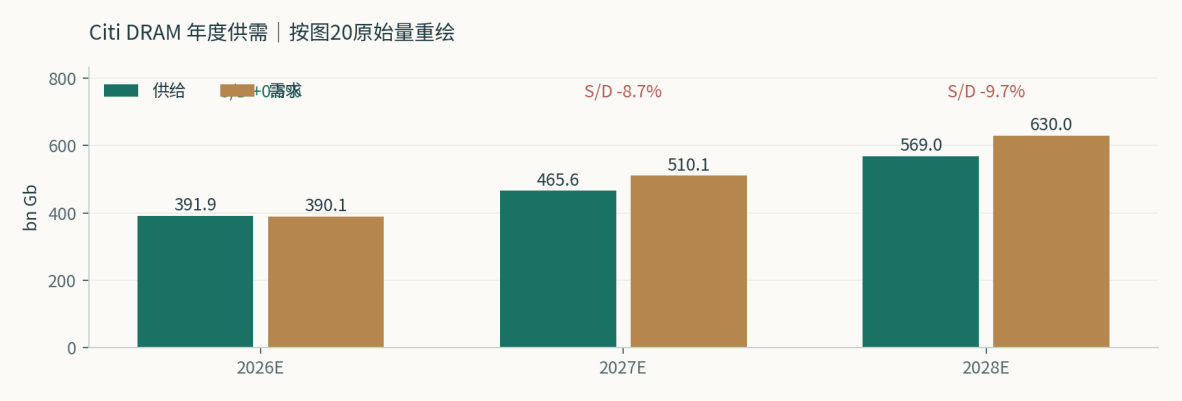
DRAM：从供需总表推导缺口

增长率差，为什么不等于供需缺口?

花旗 2027 年 DRAM 供给预测为 465,560，需求为 510,101，单位均为百万个 1Gb 等效单位。按“供给÷需求－1”计算，缺口约为 -8.7%，与表内 S/D 一致。2026 年表内供需大体平衡，而 2027 年季度缺口由第一季度的 -3.3% 扩大到第四季度的 -13.9%。

需求结构方面，花旗预计服务器 DRAM 需求由 2026 年的 226.3bn Gb 增至 2027 年的 341.7bn Gb，约增长 51%。它预测服务器平均容量由 949GB 提高至 1,345GB，而行业平均晶圆产能仅由 2,129KWPM 升至 2,295KWPM。需求密度与可用供给不同步，构成该报告看多 DRAM 的核心机制。

Citi · p.9 Citi · p.10 Citi · p.11



进阶研读 / 检查结论的边界

供给同比减去需求同比，只能提示增长速度差，不能替代 S/D 的水平计算。还应注意花旗图 20 的 2028 年供需水平对应需求增长约 23.5%，与摘要所述约 35% 不一致。课程图按表内原始水平重绘，摘要的增速判断则另行保留在核验页。

LESSON 11 / 需求机制

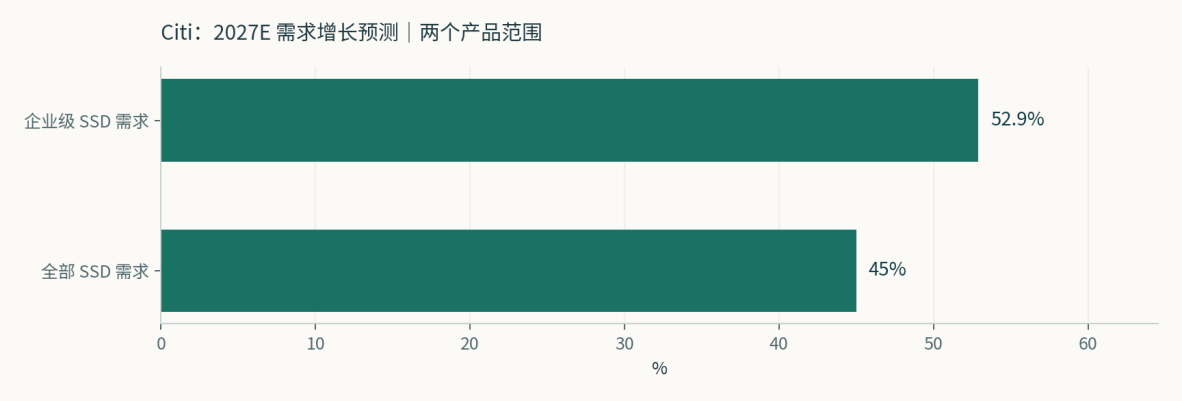
NAND：AI 推理如何传导到 eSSD

消费端偏弱，NAND 为什么仍可能紧张?

花旗预计持续学习、KV cache 卸载与更靠近加速器的大容量存储，推动企业级 SSD 需求。其文字预测 2027 年 eSSD 需求增长 52.9%，全部 SSD 需求增长 45.0%，而消费者端 NAND 面临手机和 PC 市场偏弱。由此，NAND 的总量判断越来越依赖企业级业务能否抵消消费需求压力。

报告还讨论 QLC SSD、XL Flash、HBF 与 CXL 等方向，并提出中国 AI 数据中心考虑更多使用 SSD 替代 HDD 的可能性。这里 XL Flash、HBF 属于其提及的先进存储方向，CXL 是互连和扩展相关技术，不能都归为同一种 NAND 芯片。技术方向、产品可用性和实际采购放量之间，仍需要供应商与客户验证。

Citi · p.6 Citi · p.7 Citi · p.14 Citi · p.15 Citi · p.16



进阶研读 / 检查结论的边界

花旗图 7 标题是 SSD Demand Outlook，2023—2028E CAGR 为 34%，并不等同于 eSSD 单独的增长曲线。图 29、30 对 SSD 总需求和平均容量的讨论，也应与 eSSD 的 52.9% 分开。用“企业级增长快”解释结构变化时，仍需保留总盘子、份额和客户级需求的区别。

LESSON 12 / 模型研读

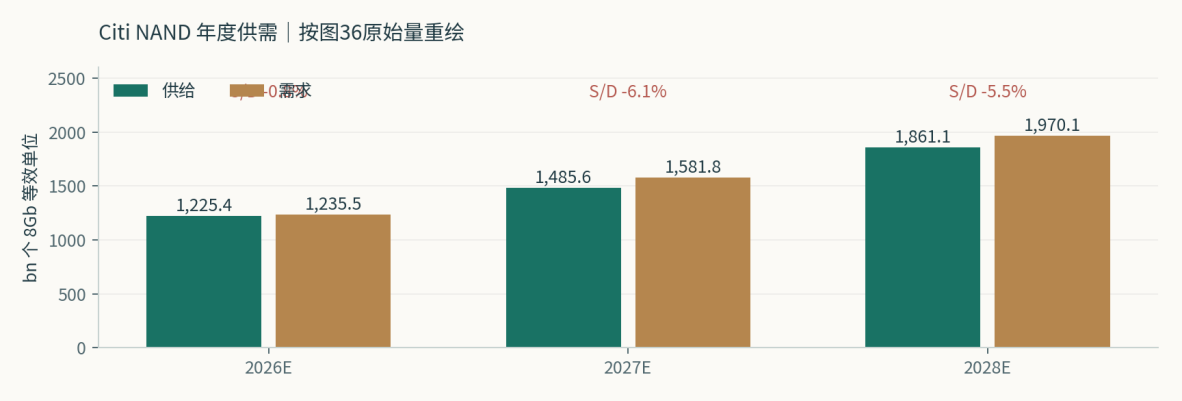
NAND：把紧缺与价格预测分开

NAND 供需表能够支持哪些结论?

花旗图 36 预测 2027 年 NAND 供给为 1,485,636、需求为 1,581,792，单位为百万个 8Gb 等效单位，对应缺口约 -6.1%。2028 年供给为 1,861,081、需求为 1,970,128，对应缺口约 -5.5%。这一组绝对量支持“连续两年存在缺口”的表内结论，但其增长率与部分摘要文字需要分开核验。

在价格方面，花旗预计 2026 年第三、第四季度混合 NAND ASP 分别环比上涨 34% 和 15%，2027 年全年平均 ASP 同比上涨 45.3%。这些预测不同于伯恩斯坦 9 月 3 日基于样本和 SSD 判断得到的第三季度接近 20%。两者报告日期与预测范围不同，不能把差异直接理解为某一家已经被市场证伪。

Citi · p.15 Citi · p.16 Citi · p.17 Bernstein · p.5



进阶研读 / 检查结论的边界

花旗图 36 的年度供给和需求基本可由季度数加总复现，个别存在百万等效单位级尾差，S/D 也可复算。问题主要出在部分文字所述的增速与需求水平，因此应保留这些差异，而不是为追求一条整齐的故事线擅自修改原表。课程实验室按表内绝对量计算，展示透明的数值路径。

LESSON 13 / 供给机制

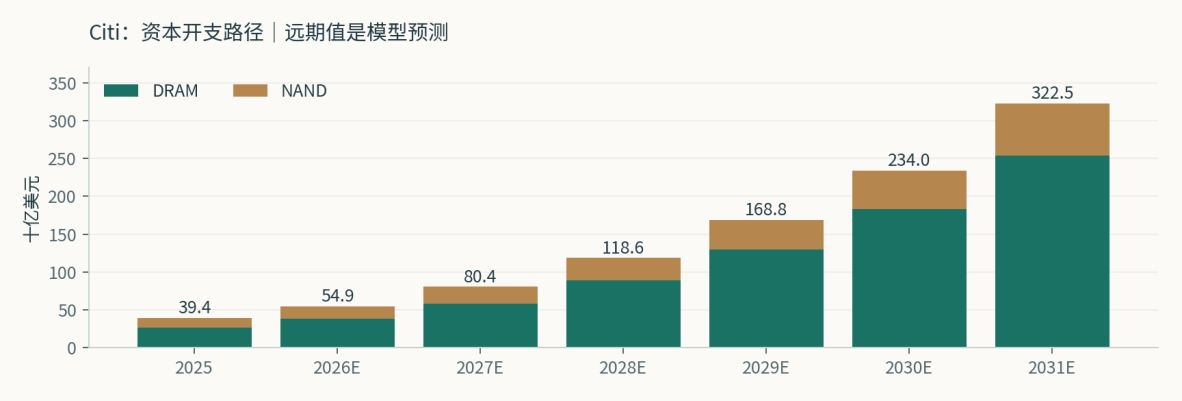
资本开支不等于即时供给

投入大增，为什么缺口仍可能扩大?

花旗预测 2027 年存储资本开支达到 804.17 亿美元，同比增长 46.5%，其中 DRAM 为 585.94 亿美元，NAND 为 218.23 亿美元。DRAM 的主要增量投向 HBM 和先进制程，NAND 厂商对新增晶圆产能仍较克制。新厂建设、装机、爬坡和认证需要时间，因此年度投入与当年可交付位元不会同步增长。

在花旗预测中，2027 年 DRAM 平均晶圆产能增加约 8%，NAND 增加约 3.2%，都显著低于资本开支涨幅。大摩也预计绝对资本开支上升，但因价格推高营收，资本开支占营收的比例可能下降。分母效应解释了“花钱更多”和“资本强度降低”为何可以同时出现。

Citi · p.10 Citi · p.15 Citi · p.18 Morgan Stanley · p.8



进阶研读 / 检查结论的边界

花旗第 18 页同时使用 Total Memory Capex 和 Memory WFE Capex 表述，未在该页完整澄清边界，跨机构比较应先确认是否包含厂房与封装等。其 2031 年合计 3,225.31 亿美元是远期模型值，不能视为已公告投资。表中列出的头部公司也未穷尽总额，未列示部分不应误判为加总错误。

LESSON 14 / 价格证据

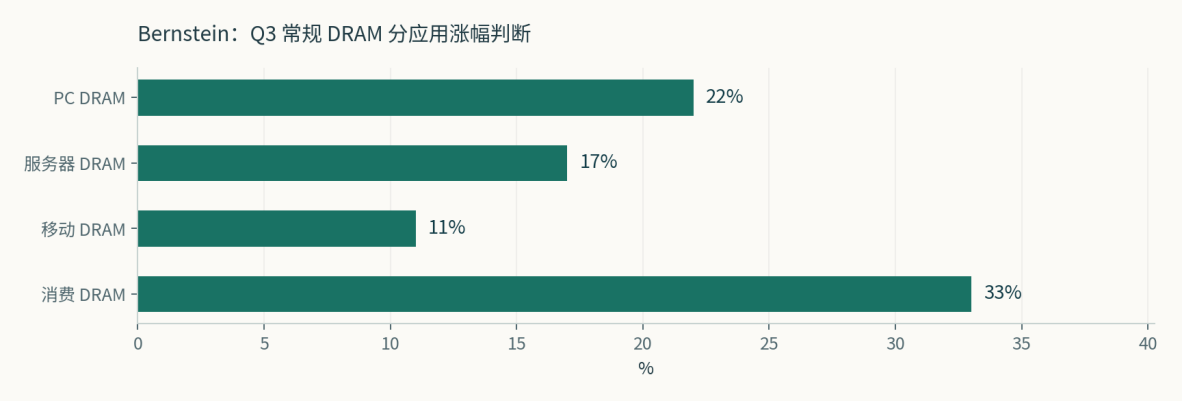
8 月 DRAM：紧张中的分化

服务器、手机与消费 DRAM 为什么涨幅不同?

伯恩斯坦 8 月合约样本指向第三季度常规 DRAM 环比约上涨 19%，分应用看 PC 约 22%、服务器约 17%、移动约 11%、消费约 33%。DDR4 的部分长尾需求仍在，但大厂退出和供应重心转移使供给更难匹配。报告认为消费 DRAM 的驱动正由传统电视、机顶盒扩展到 SSD、服务器和网络设备。

8 月 PC DDR5 与 DDR4 芯片现货分别上涨 5.8% 和 5.4%，服务器 DDR5 模组上涨 3.9%，服务器 DDR4 模组则下降约 1%。PC DDR5 相对 DDR4 便宜约 9%，服务器 DDR5 相对 DDR4 却贵约 16%，说明价差受应用和产品形态影响。采购比较必须使用相同容量、形态和交易口径，不能拿一个市场的倒挂套用全部 DDR 产品。

Bernstein · p.3 Bernstein · p.4 Bernstein · p.6 Bernstein · p.10 Bernstein · p.12 Bernstein · p.13



进阶研读 / 检查结论的边界

报告指出 128GB 服务器模组的价格溢价收窄，因为客户转向 32GB/64GB 配置以支持更多服务器并改善 TCO。这个变化可以与行业位元仍紧缺并存。材料也转述南亚科 2026 年第二季度 LTA 覆盖已达 50%，但该比例不能外推给整个消费 DRAM 市场。

LESSON 15 / 价格证据

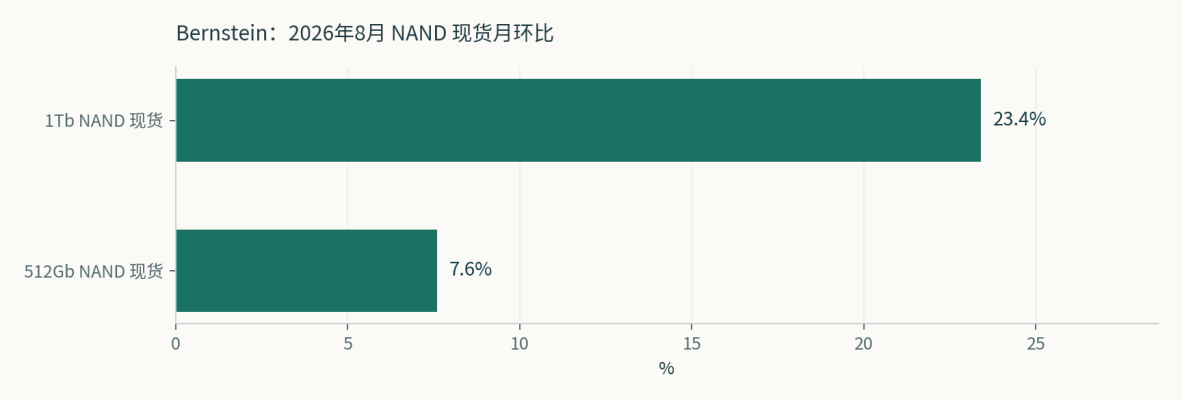
8 月 NAND：现货热，合约慢

为什么不能根据 1Tb 现货上涨就判断所有 NAND 都在同幅涨价?

伯恩斯坦记录 8 月 1Tb NAND 现货上涨 23.4%，512Gb 上涨 7.6%，但晶圆合约环比只上涨约 1%—3%。模组厂对消费者需求保持谨慎，交易量很低，并更多采购较低等级的芯片控制成本。现货与合约表现不一致，反映了客户结构和交易范围的差异。

移动 NAND 第三季度涨幅约 20%，但手机厂商削减生产并更积极抵抗涨价，报告预计后续涨幅趋缓。伯恩斯坦还观察到长江存储的涨价要求较温和，因而赢得更多销量。其对 NAND 的长期判断比 DRAM 更谨慎，认为中国竞争和高价格对需求的抑制可能使价格在 2027 年某个时点见顶。

Bernstein · p.5 Bernstein · p.6 Bernstein · p.14 Bernstein · p.15 Bernstein · p.16



进阶研读 / 检查结论的边界

伯恩斯坦只是提出 1Tb 现货走强“是否与更多 eSSD 需求流向模组厂有关”的疑问，并未给出确定因果结论。总表的 1Tb 现货类别同时包含 TLC/QLC，也不能直接与纯 TLC 合约作无差别溢价计算。课程保留机构的疑问语气和原始分类。

LESSON 16 / 经营机制

长约、成本与降配的反馈

长约可以稳定什么，又不能保证什么？

花旗认为客户把 LTA 从三年延长至五年，与多年度需求能见度提高一致。伯恩斯坦则发现，一些 LTA 设有价格上限，供应商尝试重谈上限或对约外增量收取更高价格。长约同时影响供货、定价和利润可见性，其作用取决于具体条款，不能仅用年限判断保护力度。

价格上涨也会反过来改变终端行为：PC 与手机减产，服务器客户降低部分模组容量，以既定资源支持更多设备。花旗和伯恩斯坦都认为降配不必然导致总位元需求下降，但这依赖设备数量的增长能够抵消单机容量减少。这个条件可用数量乘配置的简单模型检验。

Citi · p.6 Citi · p.9 Bernstein · p.4 Bernstein · p.5

降配是否压低总需求？ | 教学情景

设备数量 +30%	单机容量 −20%	总位元需求 +4%
数量系数 1.30 并非报告的市场预测	配置系数 0.80 总量取决于二者乘积	1.30 × 0.80 = 1.04 低于此数量增长，结果会变

进阶研读 / 检查结论的边界

以教学情景为例，设备数量增长 30%、单机容量降低 20%，总需求变为 1.30×0.80=1.04，即增长 4%。若数量仅增长 10%，相同降配就使总需求下降 12%。采购迁移时可关注基础量、增量、价格上限、重谈条件和履约节奏，具体合同安排不在三份报告的可验证范围内。

LESSON 17 / 中国供给

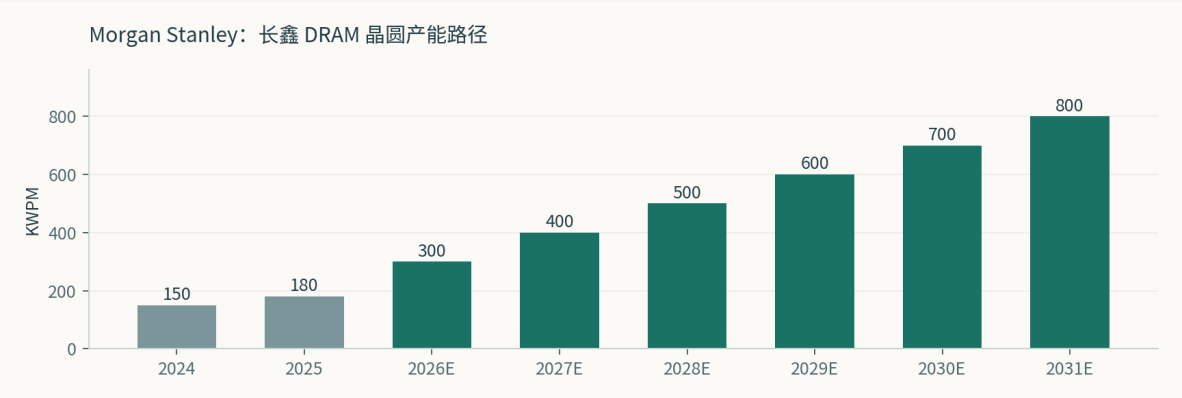
长鑫扩张：规模先于全面领先

为什么技术仍有差距，也能影响价格?

大摩预测长鑫产能由 2025 年的 180KWPM，提高到 2026 年的 300KWPM，2028 年达到 500KWPM，2031 年达到 800KWPM。其 2026 年晶圆份额约 13%、位元份额约 11%，而收入份额图给出约 6%。这些差别说明产品结构、每片位元产出和售价，会共同影响规模如何转化成收入。

大摩的核心判断是，影响商品化存储的边际价格，不必先在所有技术和客户市场实现领先。国内新增供给可以替代进口，使国际厂商把资源重新投向其他市场，从而传导竞争压力。短期中国供给能缓解缺货，到了需求放缓阶段，同样的产能可能加大行业再平衡的难度。

Morgan Stanley · p.1 | Morgan Stanley · p.2 | Morgan Stanley · p.3 | Morgan Stanley · p.4 | Morgan Stanley · p.9 | Citi · p.10



进阶研读 / 检查结论的边界

报告首页称长鑫可能在 2028 年超过美光，结合正文应按产能扩张语境理解，不能改写为收入、技术或 HBM 地位全面超越。大摩预测 2028 年全球 DRAM 晶圆产能为 3,374KWPM，长鑫约占 15%。花旗图 16 使用平均产能路径，跨报告比较前要先确认是平均还是期末口径。

LESSON 18 / 技术壁垒

从可用产品到合格供应

良率超过 90%，为什么仍不能说明所有技术问题已经解决?

大摩估计长鑫在 2026 年第二季度完成向 Gen4B、约 16nm 级工艺的生产转移，良率超过 90%，并提供主流 DDR5、LPDDR5/5X 和部分服务器产品。它同时强调节点标签不能直接比较，部分可见规格相近，不代表成本、可靠性、每片位元数与高级产品占比相同。浸没式 DUV 与多重图形化还会增加掩膜、工序和制造周期。

HBM 另有 TSV、减薄、热与翘曲控制、底层逻辑 die、封装、测试及加速器认证等壁垒。大摩预测长鑫 HBM3E 12H 于 2027 年上半年量产，2027 年可生产约 300万—400 万颗 stack，支持约 80万—100 万颗国内 AI GPU。这是报告的产能与资格爬坡判断，不能拿普通 DRAM 的良率直接替代 HBM 综合良率。

Morgan Stanley · p.4 | Morgan Stanley · p.5 | Morgan Stanley · p.10 | Morgan Stanley · p.11 | Citi · p.13



进阶研读 / 检查结论的边界

大摩 HBM 模型为长鑫设置 2026/2027 年 40%/50% 的良率假设，另列 20/40KWPM TSV 能力与年度产出。花旗则采用明显更低的长鑫 HBM 位元供给预测，显示“建成能力”和“形成可交付产出”存在较大判断空间。国产化研究因此应按具体材料、工艺和客户认证拆分，而不只看节点或层数。

LESSON 19 / 中国供给

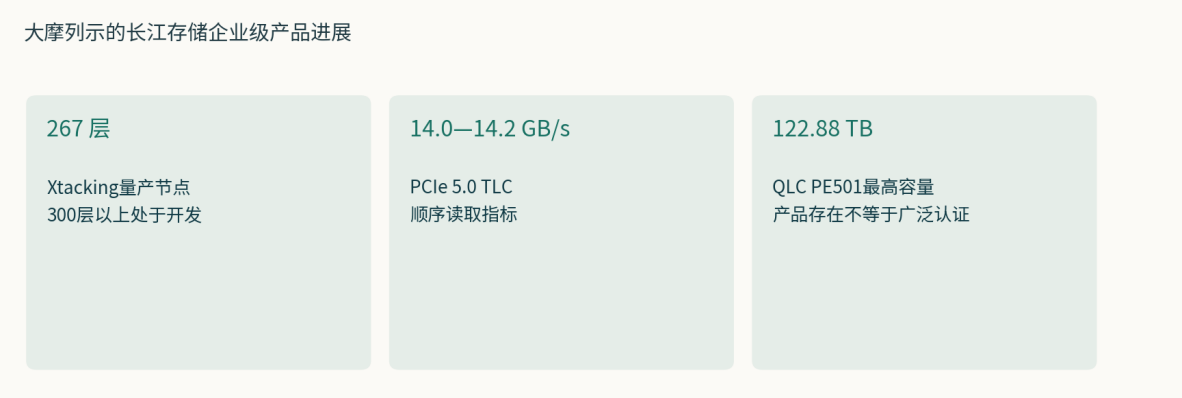
长江存储：架构、产品与设备

NAND 的技术突破为什么还要经过企业级认证?

大摩将长江存储的突破主要归于以混合键合为基础的 Xtacking 架构，报告称 267 层已量产、300 层以上产品在开发。其列示 PCIe 5.0 TLC 企业级 SSD 顺序读取可达 14.0—14.2GB/s，随机读取约 310万—330 万 IOPS。QLC PE501 则列示最高 122.88TB、5,000 次 P/E 和 0.6 DWPD，体现容量、耐久与系统整合的进展。

这些参数属于该报告引用的产品信息，并不能证明已在所有大客户完成广泛认证。大摩强调可用产品、客户信任、批量资格和高价值产品占比之间仍存在距离。对于中国设备生态，报告认为刻蚀、沉积、清洗、CMP 等环节推进较快，而 DUV、先进量测、制造软件、备件与服务仍是重要薄弱环节。

Morgan Stanley · p.1 Morgan Stanley · p.11 Morgan Stanley · p.12



进阶研读 / 检查结论的边界

报告称长江存储新厂本地化渗透接近 60%，这是特定新厂口径，不能外推为全部存量生产线或所有关键设备均已国产化。P/E 次数与 DWPD 也是不同尺度：前者描述擦写循环，后者描述整盘每日写入量相对于盘容量的指标。课程仅解释阅读方法，不补入三份材料之外的实际认证名单或最新产品变更。

LESSON 20 / 情景推演

2028 年 NAND 的二维压力测试

什么条件下，紧缺会转为过剩?

大摩以长江存储 2028 年产能和 AI SSD 需求增长为两个变量，测试 NAND 供需状态。其余厂商扩产沿用团队预测，非 AI NAND 需求增长固定为 5%，因此矩阵是在这些前提下的敏感性分析。产能轴从 310KWPM 到 470KWPM，需求轴从同比增长 30% 到 60%。

若长江存储为 310KWPM、AI SSD 需求增长 50%，表中缺口为 -4%；同样需求增长下，产能达到 430KWPM 时，结果变为过剩 3%。若产能固定为 390KWPM，需求增长由 30% 提高到 60%，表中结果从过剩 4% 转为缺口 -1%。矩阵显示的是供需分界如何移动，而非某个单点必然发生。

Morgan Stanley · p.12

Morgan Stanley：2028 NAND 供需情景格点

YMTC 产能（KWPM） \ AI SSD 需求同比增速	30%	40%	50%	60%
310	-1%	-2%	-4%	-6%
330	0%	-1%	-3%	-4%
350	+2%	0%	-2%	-3%
370	+3%	+1%	-1%	-2%
390	+4%	+2%	+1%	-1%
410	+5%	+4%	+2%	0%
430	+7%	+5%	+3%	+1%
450	+8%	+6%	+4%	+3%
470	+9%	+7%	+5%	+4%

进阶研读 / 检查结论的边界

负值表示模型供给不足，正值表示过剩，零附近仅表示表内取整后的接近平衡。矩阵中的格点全部来自大摩原表，课程不做插值，也不赋予概率。报告另称五座已宣布工厂若全部投入 NAND，最终份额可能达到 24%，这一长期潜在值不能当作 2028 年确定份额。

LESSON 21 / 竞争机制

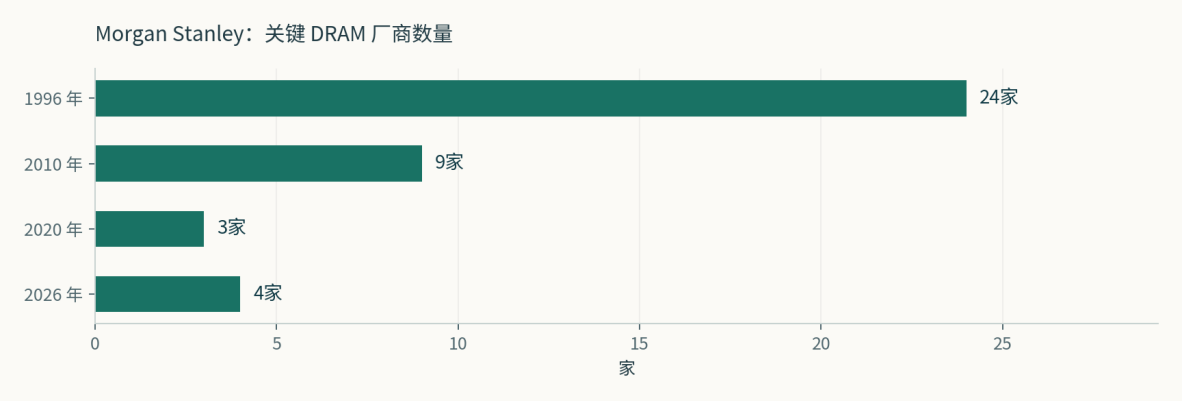
产业史：谁决定边际价格

四家竞争，与三家竞争最大的差别在哪里？

大摩回顾了美国以技术商业化、日本以制造质量、韩国以规模投资与成本领先取得地位的过程。它认为中国正在通过国内需求、替代进口和连续扩产，在尚未全面追平前沿技术时先形成规模。历史的共同点是竞争基础发生变化，领先并不只由某一代产品性能决定。

大摩给出的关键 DRAM 厂商数量由 1996 年的 24 家，降至 2020 年的 3 家，2026 年随长鑫加入变为 4 家。新变化在于企业目标可能不完全一致：传统厂商更强调利润与供给纪律，长鑫还可能重视替代进口、技术学习和供应安全。若需求放缓时新增厂商继续扩张，恢复平衡所需的减产负担就可能重新分配。

Morgan Stanley · p.2 Morgan Stanley · p.5 Morgan Stanley · p.6 Morgan Stanley · p.7 Morgan Stanley · p.8



进阶研读 / 检查结论的边界

大摩描述的可能终局先是市场分层：通用产品边际价格受中国供给影响，高端 HBM、服务器 DRAM 与企业级存储仍保留差异化利润。真正更深层的挑战，是中国厂商进一步进入这些高利润产品并完成广泛客户认证。这里属于报告提出的竞争机制，不是对所有企业未来行为的确定判断。

LESSON 22 / 进阶研读

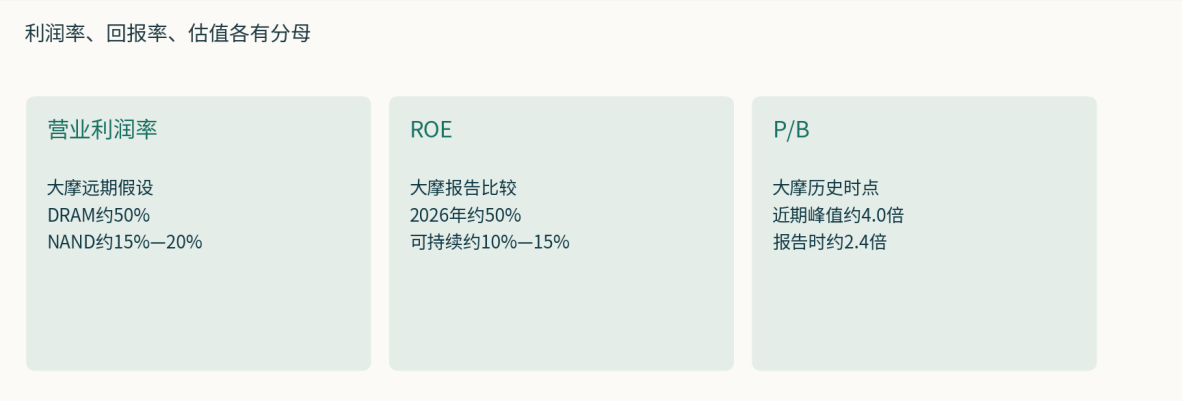
把供需判断连接到回报与估值

为什么短期高盈利与长期回报承压能同时出现?

大摩认为 AI 与涨价可推动周期内盈利和估值创新高，但中国扩产可能降低长期可持续利润率。其十年模型把 DRAM 营业利润率向远期约 50% 归一化，NAND 则更快回落至 15%—20%。因此“这一轮利润很高”和“下一轮资本回报承压”属于不同时间层次的判断。

报告称 DRAM 企业平均未来十二个月市净率从近期约 4 倍回落至约 2.4 倍，历史典型高点区间为 1.8—2.1 倍。它同时比较 2026 年约 50% 的 ROE 与 10%—15% 的可持续 ROE，并以三星净经营资产回报作为产业代理。ROE、营业利润率与 RNOA 分母不同，课程保留其定义差异。

Morgan Stanley · p.8 Morgan Stanley · p.9 Citi · p.1 Citi · p.19 Bernstein · p.2 Bernstein · p.19 Bernstein · p.20



原材料指标	报告数值	口径
DRAM 远期营业利润率	约 50%	大摩十年模型归一化假设
NAND 远期营业利润率	15%—20%	大摩更谨慎的竞争假设
ROE	2026 年约 50%；可持续 10%—15%	股东权益回报
12M forward P/B	近期峰值约 4.0x；报告时约 2.4x	历史时点估值，非实时行情

进阶研读 / 检查结论的边界

花旗对主要存储厂商及设备材料链保持积极；伯恩斯坦对三星、海力士、美光、闪迪给出跑赢评级，而对铠侠为跑输，表明看好行业并不等于看好所有公司。伯恩斯坦在第 19—20 页给出估值倍数和风险，包括供需逆转、中国竞争及 NAND 周期回落。这里保留评级仅为解释原报告，不把历史评级变成当下交易建议。

LESSON 23 / 进阶研读

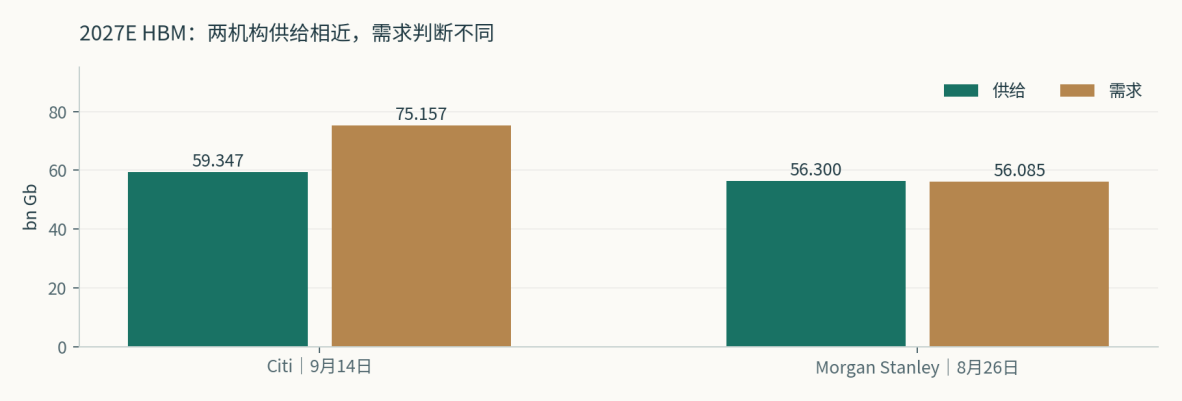
共识、分歧与原文核验

如何避免把不兼容的数据拼成一个模型?

三份报告共同关注 AI 需求、HBM 对资源的配置影响，以及高价格对终端行为的反馈。花旗更看好持续学习驱动的多年增长，伯恩斯坦强调涨幅收敛和 NAND 的周期风险，大摩更关注 2028 年以后中国新增供给的边际作用。真正需要比较的是需求强度、有效供给和扩产行为，而不是只比较结论的乐观程度。

在 HBM 上，花旗 2027 年供给与需求为 59.347/75.157bn Gb，缺口约 -21%；大摩为 56.300/56.085bn Gb，接近平衡。两者供给水平相对接近，需求预测差别显著，是判断分歧的重要来源。长鑫 2027 年 HBM 产出在两份报告中分别为 0.336 和 2.462bn Gb，差距也提示爬坡假设需要单独追踪。

Citi · p.1 | Citi · p.11 | Citi · p.13 | Citi · p.15 | Citi · p.17 | Morgan Stanley · p.5 | Bernstein · p.5



进阶研读 / 检查结论的边界

原材料本身还存在数值不一致：花旗图 20 的 2028 年需求同比为约 23.5%，摘要却写约 35%；图 36 的 2027/2028 年 NAND 需求同比约为 28.0%/24.6%，摘要写约 29%/33%。课程同时保留原文叙述与表内重算值，不擅自判定哪一版才是作者最终模型。图表旁的核验标记和专门核验表，帮助读者带着差异继续阅读。

LESSON 24 / 综合练习

把研究迁移到采购判断

怎样把宏观研报转成可追踪的材料问题?

对于电信与服务器采购，可从三个方向建立跟踪问题：AI 需求是否继续增加、旧产品供给是否持续退出，以及中国新增产能能否变成经过认证的可交付供给。DDR4 应更重视长尾需求与供给收缩，DDR5 应结合平台切换与服务器数量，NAND 则要把 eMMC/UFS、消费 SSD 和 eSSD 分开。这样才能将行业判断映射到材料，而不是用一个存储涨幅覆盖全部 BOM。

下一步可把价格证据与实际供货信息对照：同料号报价、现货成交、合约有效期、交付承诺、在手库存和客户消耗各回答不同问题。长约可重点拆分基础量与追加量、价格机制和交付节奏，而中国供应商验证应落实到具体产品与系统适配。以上是基于三份报告的教学迁移，并未使用企业内部采购数据。

Bernstein · p.4 Bernstein · p.5 Citi · p.9 Citi · p.15 Morgan Stanley · p.10 Morgan Stanley · p.12



材料方向	原文支撑的关键矛盾	学习迁移中的跟踪问题
DDR4 / 消费 DRAM	退出与长尾需求并存	认证替代供给是否补得上退出?
服务器 DDR5	高需求、降配、LTA 上限	位元需求、追加量价格如何变化?
eMMC/UFS	终端减产与供应方涨价	容量配置和订单消耗是否下降?
eSSD / NAND	AI 拉动与中国扩产	需求能否吸收新产能与认证放量?
国产 HBM / 企业存储	产品进展与商业资格差距	良率、可靠性、客户资格何时达标?

进阶研读 / 检查结论的边界

完成学习后，应能分别解释：为什么降配不一定压低总需求，为什么晶圆扩产不等于位元同幅增加，为什么现货与合约会分化，以及为什么短期缺货与长期竞争可以并存。然后为每个判断写出一个可能推翻它的证据，例如需求减速、爬坡加快或库存持续累积。能说明结论的条件和失效路径，才算从读懂研报走到形成判断。

SOURCE AUDIT

保留原文，也保留不一致

下列记录区分原文表述、原始数据复算和跨机构比较。数值差异可能来自版本或编辑问题，但课程不代替作者判断原意。

01 Citi：2027 年 DRAM 增速

正文 p9：需求 +30.2%；图20：510,101 ÷ 390,057 − 1 = +30.8%。

相差约 0.6 个百分点；表内复算采用原始水平，不覆盖正文。

Citi · p.11

02 Citi：2028 年 DRAM 增速

摘要 p1：需求约 +35%；图20：630,035 ÷ 510,101 − 1 = +23.5%。

明显不一致，不能直接把摘要增速代入表内下一年需求。

Citi · p.11

03 Citi：NAND 需求增长

摘要 p1：2027/28 年约 +29%/+33%；图36复算为 +28.0%/+24.6%。

2027 年正文的 +29.1% 也与表内复算不完全一致。

Citi · p.17

04 Citi：2027 年 NAND 需求水平

p15 文字写约 1,486bn 个8Gb等效单位；图36需求为 1,581.792bn，供给为 1,485.636bn。

文字接近供给数值，疑似供需列误指；课程不擅自确定作者原意。

Citi · p.15

05 Citi：NAND 晶圆产能

p15 正文/图31：2027 年 1,460KWPM；图32的 Total 柱标为 1,440KWPM。

图与文保留原貌，课程叙述注明采用正文和图31。

Citi · p.15

SOURCE AUDIT

核验记录 · 续

06 Citi：资本开支表尾差

p18 2025 年 DRAM+NAND 分项加总为 39,363，总表39,364；2026、2028年也有1百万美元尾差。

保留各表原数，不把小尾差当作额外增长；WFE/Total 边界另需确认。

Citi · p.18

07 Bernstein：权重与样本

p5 常规 DRAM 权重34%+34%+9%+15%=92%；p17图26另列 HBM 8%。

图26补足到100%，但文字说HBM另行考虑；不能直接用图26重算常规DRAM指数。

Bernstein · p.17

08 两机构：2027 年 HBM

Citi 供需59.347/75.157bn Gb；MS供需56.300/56.085bn Gb。

同单位下需求差约34%，供给差约5%；不是四舍五入能解释的分歧。

Morgan Stanley · p.5

09 两机构：长鑫 HBM 爬坡

2027年供给：Citi 0.336bn Gb，MS 2.462bn Gb；TSV分别50/40KWPM。

应检查年末/平均能力、良率、爬坡与认证，不能按晶圆比例平移预测。

Citi · p.13

FIGURE INDEX

每幅原图，都有阅读入口

本索引保留原报告编号。点击机构页码可跳到书后对应原页；交互版可按题名或关键词筛选。重复出现的原图也保留其独立编号。

Citi 花旗 / 42 幅

#01 持续学习的机制

新知识学习与旧知识保留同时发生，是整篇报告的需求起点。 Citi · p.2

#02 AI 工作负载与存储需求演进

按四个阶段区分HBM、服务器DRAM、eSSD和端侧需求；未来阶段为预测。 Citi · p.3

#03 月度 AI token 使用趋势

来源为OpenRouter和花旗，不宜当成全球全部token的精确统计。 Citi · p.3

#04 HBM 客户与代际需求表

拆分客户、HBM代际和需求占比；与图28为重复呈现，保留原表。 Citi · p.4

#05 GPU / ASIC 客户 HBM 路线图

同时核对代际、堆叠、容量与时间；possibly和TBD不是确定承诺。 Citi · p.5

#06 服务器 DRAM 需求展望

图示2023—2028E CAGR约41%，核心是服务器位元密度与出货。 Citi · p.6

#07 全部 SSD 需求展望

图题为SSD总量，2023—2028E CAGR34%，不能直接改称eSSD。 Citi · p.7

#08 DRAM 位元供给增长拆分

区分每晶圆位元、晶圆产能和总位元供给增长。 Citi · p.8

#09 NAND 位元供给增长拆分

层数/技术进步与晶圆扩产共同影响位元产出。 Citi · p.8

#10 DRAM 供需长期展望

区分左右轴，并对照图20的近期绝对量。 Citi · p.8

FIGURE INDEX

Citi 花旗 · 图表导读

#11 NAND 供需长期展望

2027年缺口-6.1%，长期判断依赖需求与供给路径。 Citi · p.8

#12 每片 HBM 晶圆的 die 数

HBM2至HBM4e从720下降到430，但不能只据此计算成品位元。 Citi · p.9

#13 移动端 Wide I/O / Mini HBM 示意

SK hynix来源的技术示意，报告未给出成熟商用规模。 Citi · p.9

#14 DRAM 各应用需求

服务器增长显著；读单位为位元等效量。 Citi · p.10

#15 DRAM 单机平均容量

服务器2027E约1,345GB；容量与台数共同决定需求。 Citi · p.10

#16 DRAM 行业平均晶圆产能

2027E为2,295KWPM，平均与期末产能勿混用。 Citi · p.10

#17 DRAM 厂商产能变化

三星、海力士、美光以外的供给仍在行业总计内。 Citi · p.10

#18 DRAM 分应用需求长期走势

累计柱与增速轴反映不同量，不要从图形高度推算未标数值。 Citi · p.11

#19 DRAM 各应用占比

占比变化表达结构迁移，需与总需求同时阅读。 Citi · p.11

#20 DRAM 季度与年度供需总表

课程复算依据；摘要增速与表内水平存在差异。 Citi · p.11

#21 DRAM 供需长期图

与图10重复论点，展示长期预测，不能当成实现数据。 Citi · p.11

FIGURE INDEX

Citi 花旗 · 图表导读

#22 DDR5 PC 现货与合约

区分现货溢折价、绝对价格与交易范围。 Citi · p.12

#23 服务器 DDR4 / DDR5 价格

以图示容量和模组类型进行同口径比较。 Citi · p.12

#24 DRAM 平均售价展望

混合ASP体现产品组合，不是单一SKU价格。 Citi · p.12

#25 DRAM 年度价格变化

2026/27E分别+242.4%/+23.1%，为花旗年度平均预测。 Citi · p.12

#26 HBM 供需总表

2027E 59.347/75.157bn Gb，对应约-21%。 Citi · p.13

#27 HBM TSV 能力与厂商供给

年末TSV能力、年度位元供给和市场份额分开阅读。 Citi · p.13

#28 HBM 客户与代际需求表（再列）

对照图4可核验重复数据；个别历史分项存在差别。 Citi · p.14

#29 NAND 各应用需求

SSD总量与eSSD增长不是同一指标。 Citi · p.15

#30 NAND 单机平均容量

SSD为客户端与企业端混合平均，不能当作典型服务器规格。 Citi · p.15

#31 NAND 行业平均晶圆产能

正文与该图2027E为1,460KWPM。 Citi · p.15

#32 NAND 厂商产能变化

Total柱2027E标1,440，与正文/图31不同，保留待核验。 Citi · p.15

FIGURE INDEX

Citi 花旗 · 图表导读

#33 NAND 各应用需求长期走势

观察SSD对总位元的影响，不外推精确未标注数值。 Citi · p.16

#34 NAND 分应用占比

SSD占比上升为机构预测，与消费端走弱相联系。 Citi · p.16

#35 NAND 长期供需平衡

远期缺口判断依赖AI需求与厂商扩产纪律。 Citi · p.16

#36 NAND 季度与年度供需表

复算供需比，另记正文增速和需求水平的不一致。 Citi · p.17

#37 512Gb TLC NAND 现货价格

该图使用Spectrum现货指数，来源口径与其他图可能不同。 Citi · p.17

#38 NAND 混合 ASP 展望

产品组合、价格、时间三者共同影响混合ASP。 Citi · p.17

#39 NAND 年度价格变化

2027E+45.3%为花旗判断，不等于伯恩斯坦的季度价格。 Citi · p.17

#40 DRAM + NAND 总资本开支

2027E804.17亿美元；保留原表远期值与WFE口径疑问。 Citi · p.18

#41 DRAM 厂商资本开支

总额不只包含图中单列的三家厂商。 Citi · p.18

#42 NAND 厂商资本开支

总额保留原数；分项与总表有百万美元级尾差。 Citi · p.18

Bernstein 伯恩斯坦 / 27 幅

#01 DRAM 现货、合约与模型总表

区分美元/GB、整包例外和模型最后更新日期。 Bernstein · p.6

FIGURE INDEX

Bernstein 伯恩斯坦 · 图表导读

#02 NAND 现货、合约与模型总表

9.4%为不含SSD的样本，行业接近20%是另行估计。 Bernstein · p.6

#03 PC DDR5 芯片现货与合约

图以单颗美元计价，总表则折算为美元/GB。 Bernstein · p.7

#04 PC DDR5 现货日涨跌

日波动提供交易节奏，不等于季度财报ASP。 Bernstein · p.7

#05 PC DDR4 芯片现货与合约

长尾需求与供给退出支持旧规格价格。 Bernstein · p.8

#06 PC DDR4 现货日涨跌

8月增速放缓仍可能伴随价格继续上行。 Bernstein · p.8

#07 PC DDR5 16GB 模组合约

8月约+4%月环比，季度谈判已接近结束。 Bernstein · p.9

#08 PC DDR4 8GB 模组合约

8月约+2%月环比，容量不同先换算口径。 Bernstein · p.9

#09 PC DDR4 16GB 模组合约

与8GB产品涨幅接近，勿把模组价格当每GB单价。 Bernstein · p.10

#10 PC DDR5 相对 DDR4 价差

DDR5折价约9%，与服务器市场不同。 Bernstein · p.10

#11 服务器 DDR5 64GB 现货与合约

现货显著高于合约，反映渠道紧张与定价差异。 Bernstein · p.10

#12 服务器 DDR4 64GB 现货与合约

8月现货约-1%，并不否定其他DRAM仍上涨。 Bernstein · p.11

FIGURE INDEX

Bernstein 伯恩斯坦 · 图表导读

#13 服务器 DDR5 合约与月变化

8月小幅+1%左右，图单位为美元/GB。 Bernstein · p.11

#14 服务器 DDR4 合约与月变化

8月持平，旧平台需求与退出共同影响供需。 Bernstein · p.12

#15 服务器 DDR5 相对 DDR4 溢价

溢价约16%，不能套用PC的折价。 Bernstein · p.12

#16 移动 LPDDR5 合约价格

季度估计约+11%，终端抵抗涨价。 Bernstein · p.12

#17 移动 LPDDR4X 合约价格

季度估计约+12%，保留季度更新频率。 Bernstein · p.13

#18 低端 eMCP 整包合约价格

季度约+2.3%，整包含DRAM、NAND和控制器。 Bernstein · p.13

#19 消费 DDR4 4Gb 合约价格

8月约+5%月环比，长尾供给仍偏紧。 Bernstein · p.13

#20 512Gb NAND 现货与合约

8月现货反弹7.6%，不代表模组厂积极补库。 Bernstein · p.14

#21 1Tb NAND 现货与合约

现货8月上涨23.4%，类别含TLC/QLC。 Bernstein · p.14

#22 NAND TLC 晶圆合约

合约涨幅温和，需结合消费者需求和成交量。 Bernstein · p.15

#23 NAND QLC 晶圆合约

1Tb QLC合约总体平缓。 Bernstein · p.15

FIGURE INDEX

Bernstein 伯恩斯坦 · 图表导读

#24 eMMC / UFS 合约

季度涨幅约20%，但终端削减产量并抗价。

Bernstein · p.16

#25 高容量 512GB UFS 合约

季度约20%，高容量产品需独立跟踪。

Bernstein · p.16

#26 DRAM 产品权重

图另含HBM8%，常规样本与混合ASP处理需区分。

Bernstein · p.17

#27 NAND 产品权重

eMMC/UFS40%、晶圆60%；SSD另行考虑。

Bernstein · p.17

Morgan Stanley 摩根士丹利 / 20 幅

#01 按存储类型划分的供应商地位

定性示意，不是打分或可交付能力的量化承诺。

Morgan Stanley · p.3

#02 全球 DRAM 晶圆产能

2028E3,374KWPM，先核对与其他机构的期间口径。

Morgan Stanley · p.4

#03 长鑫 DRAM 产能路径

180→300→500→800KWPM分别对应2025/26/28/31。

Morgan Stanley · p.4

#04 HBM 晶圆产出负担

单位位元约2.5倍走向2028E约3倍，分母为常规DRAM。

Morgan Stanley · p.4

#05 HBM 对先进 DRAM 晶圆的占用

2023约6%、2028E约34%，不是营收份额。

Morgan Stanley · p.4

#06 全球 HBM 产能建设

与图8绝对量一起读，勿只凭堆叠柱外推需求。

Morgan Stanley · p.5

#07 HBM 位元市场份额

长鑫2026/27E约3.8%/4.4%，与花旗假设不同。

Morgan Stanley · p.5

FIGURE INDEX

Morgan Stanley 摩根士丹利 · 图表导读

#08 HBM 和 DRAM 供需模型

包含TSV、良率、利用率与位元供需，是机构分歧的比较基础。

Morgan Stanley · p.5

#09 DRAM ASP 历史波动

帮助理解周期性；不能仅凭历史波形确定新一轮拐点。

Morgan Stanley · p.6

#10 各国存储收入份额历史

反映行业领导地位迁移，不意味着中国已完成全面替代。

Morgan Stanley · p.7

#11 产业领导权更替的原因

商业化、质量、规模成本与战略目标是不同阶段的竞争基础。

Morgan Stanley · p.7

#12 资本开支与资本强度

价格推高营收，可使开支绝对额上升、开支率下降。

Morgan Stanley · p.8

#13 关键 DRAM 厂商数量

1996年24家、2010年9家、2020年3家、2026年4家。

Morgan Stanley · p.8

#14 回报率向常态回落

图用三星净经营资产回报作为行业代理，不是全行业直接统计。

Morgan Stanley · p.9

#15 市净率的周期范围

历史时点估值，不能与当前行情混用。

Morgan Stanley · p.9

#16 2026E DRAM 收入份额

三星37%、海力士31%、美光24%、长鑫6%、其他2%。

Morgan Stanley · p.9

#17 2026E NAND 收入份额

三星30%、美光15%、铠侠14%、闪迪12%、海力士12%、长存11%、Solidigm6%。

Morgan Stanley · p.9

#18 DRAM 工艺迁移里程碑

节点命名不能直接换算成本、可靠性或认证水平。

Morgan Stanley · p.11

FIGURE INDEX

Morgan Stanley 摩根士丹利 · 图表导读

#19 NAND 层数迁移里程碑

层数与架构、良率、单片位元和企业产品组合共同影响竞争力。

Morgan Stanley · p.11

#20 长江存储扩产与 AI SSD 需求矩阵

36个格点敏感性分析，固定非AI需求+5%，无概率和插值。

Morgan Stanley · p.12

PAGE GUIDE

把论点放回原文位置

Citi 花旗 / 2026-09-14

p.1 报告主张与预测总览 持续学习、HBM/eSSD需求、DRAM/NAND供需和受益链条构成全文骨架。摘要增速须与后文供需表核验。

p.2 持续学习的定义与难点 理解增量训练与灾难性遗忘，再读图1。报告把持续模型更新与历史数据访问联系起来。

p.3 AI工作负载演进 先看图2的时间轴，再看图3的token样本。两图分别属于需求路径判断和所跟踪平台数据。

p.4 HBM需求拆分 按客户、代际和占比阅读图4。注意Broadcom与其他客户分类沿用原模型。

p.5 完整客户路线图 包括Nvidia、AMD、AWS、Google、Microsoft、Meta等。未来代际、设计伙伴与TBD按原文保留。

p.6 服务器DRAM与新存储方向 服务器需求2027年预计超过50%增长，LTA延长至3—5年。报告也讨论XL Flash、HBF、CXL与近GPU QLC。

p.7 SSD需求与三类存储总结 图7是全部SSD，CAGR34%。总结HBM、服务器DRAM/SoCAMM2、eSSD各自需求驱动。

p.8 供给增长与长期缺口 四图同时观察位元/晶圆、晶圆总量和供需比。未来值为花旗模型。

p.9 HBM die数与DRAM需求逻辑 图12/13讨论制造与技术方向，下半页连接scale-out、服务器容量与供给限制。

p.10 DRAM应用、容量与产能 图14—17拆分需求和供给。边栏Capex与正文晶圆能力表述要按实际单位解读。

p.11 DRAM供需模型的核心页 图20提供季度和年度原始量。课程以该表复算供需比并标出摘要增速差异。

p.12 DRAM价格路径 现货/合约图与年度ASP预测并列。全年平均涨幅不是年末对年末涨幅。

p.13 HBM供给、需求和TSV 2027缺口约21%；不同厂商产能、产出和份额需分开。与大摩图8存在显著模型差异。

p.14 HBM表再列与NAND需求 图28再列HBM分项，下方切换到NAND与eSSD。重复表的个别历史数据也需核对。

p.15 NAND需求与晶圆产能 图29—32讨论产品密度及晶圆供给。需求水平、增速与Total产能柱存在口径/数值待核验。

p.16 NAND长期结构与供需 先进存储方向与SSD占比上升是主线。长期缺口仍依赖需求与投资假设。

p.17 NAND总表与价格 图36给出可复算绝对量；图37—39分别提供现货和ASP。保留正文与表内增速差异。

p.18 完整资本开支表 图40—42列到2031E。WFE/Total表述边界和分项尾差均保留提示。

p.19 投资语境中的公司名单 列出AMAT、Lam、存储厂商、澜起与韩国设备公司等。股价和评级均属于报告时点。

p.20—29：分析师认证、评级规则、披露与发行说明。完整页面均附于书后。

PAGE GUIDE

逐页阅读导航 · 续

Bernstein 伯恩斯坦 / 2026-09-03

p.1 8月价格与季度判断总览 常规DRAM和含SSD调整后的NAND接近20%，涨速减缓但报告仍认为2027年供应偏紧。

p.2 评级与估值总表 三星、海力士、美光、闪迪跑赢，铠侠跑输；币种、财年和目标价口径不能混用。

p.3 样本方法与DRAM观察 现货规模约一成、噪声较大；固定样本不能直接等同公司ASP。解释PC与服务器行情。

p.4 DRAM分化、LTA和降配 包含DDR4长尾、服务器LTA上限、移动减产和消费DRAM。供应紧张与涨幅放缓可并存。

p.5 NAND观察与权重方法 现货走强但模组厂谨慎；样本缺SSD导致与行业判断不同。中国竞争构成NAND长期风险。

p.6 价格原始总表 最适合练习单位、时间基准与复算。模型脚注明确其最后更新在2026年6月。

p.7 PC DDR5现货 价格水平图与日涨跌图并列，区分长期趋势和日度噪声。

p.8 PC DDR4现货 供给退出与长尾需求支持旧规格，日变化仍会波动。

p.9 PC DRAM模组合约 DDR5 16GB、DDR4 8GB分别显示月变化；不要直接比较不同容量的绝对价格。

p.10 PC价差与服务器DDR5现货 PC DDR5折价与服务器紧张处于不同市场层次。

p.11 服务器DDR4与DDR5 旧规格现货下跌、新规格合约小涨并存，不能用一个系列代表全部。

p.12 服务器价差与移动LPDDR5 服务器DDR5溢价16%，移动季度涨幅约11%，分母不同。

p.13 LPDDR4X、eMCP和消费DRAM 整包与单芯片计价不同，消费DDR4的涨价更强。

p.14 NAND现货与合约对照 512Gb和1Tb走势不同，1Tb混合TLC/QLC类需保留。

p.15 NAND晶圆合约 TLC与QLC晶圆合约涨幅有限，显示模组厂消费者端需求顾虑。

p.16 移动NAND合约 eMMC/UFS季度涨幅约20%，但生产缩减加大议价阻力。

p.17 权重图与延伸阅读 DRAM图另含HBM8%，NAND不含SSD。下半页为相关研究入口而非本材料已提供内容。

p.18 延伸研究清单 保留研报题名与原件中的链接，课程不假装已经获得清单内其他报告全文。

p.19 估值方法与主要风险 说明各公司使用的盈利倍数和供需、竞争、估值风险。属于研究内容，纳入进阶课程。

p.20 风险续页与评级规则 铠侠上行风险、闪迪下行风险与评级定义。余下规则和声明随原件完整保留。

p.21—30：分析师认证、评级规则、披露与发行说明。完整页面均附于书后。

PAGE GUIDE

逐页阅读导航 · 续

Morgan Stanley 摩根士丹利 / 2026-08-26

p.1 中国存储挑战的主张 规模、政策、设备本地化和国内需求共同扩张，技术差距与竞争影响并存。

p.2 短期替代与长期竞争 2028年前增量可能被AI与替代进口吸收，之后边际定价影响更明显。

p.3 产品地位与长鑫规模 定性竞争矩阵与晶圆/位元份额相结合，规模相近不等于位元或成本相同。

p.4 产能路径和HBM机会成本 全球/长鑫产能、HBM晶圆负担和先进产能占比构成资源配置逻辑。

p.5 HBM和DRAM供需模型 提供TSV、良率、利用率和供需表。短缺阶段新增供给与下一轮竞争需要分别讨论。

p.6 周期历史与边际定价 历史ASP波动、不同战略目标和市场分层解释长期回报压力。

p.7 国家领导地位变迁 图10/11解释美国、日本、韩国的竞争基础及中国正在形成的规模。

p.8 资本开支与长期回报 绝对开支上升、资本强度下降、四家竞争与归一化利润率并列。

p.9 回报、估值与收入份额 ROE、RNOA、利润率、P/B各有分母；份额图为收入口径。

p.10 技术壁垒与长鑫进展 多重图形化、成本良率、HBM封装与认证，防止把规格追平等同全面竞争力。

p.11 DRAM/NAND工艺与YMTC产品 节点和层数之外还需看Xtacking、系统整合、耐久和企业级资格。

p.12 设备短板与2028情景测试 本地化仍有薄弱点；YMTC产能与AI SSD需求共同决定NAND供需分界。

p.13—16：分析师认证、评级规则、披露与发行说明。完整页面均附于书后。

GLOSSARY

28个术语，统一阅读语言

持续学习 Continual learning

在增量学习新任务和知识时，保留既有知识；花旗长期需求论点的起点。

Citi · p.2

灾难性遗忘 Catastrophic forgetting

学习新信息时丢失旧知识的挑战，使过去的数据访问与保留更重要。

Citi · p.2

训练 / 推理 Training / Inference

前者更新模型，后者执行任务；持续学习可能让两类需求同时增强。

Citi · p.3

高带宽内存 HBM

以堆叠 DRAM 服务加速器的产品；须同时看位元、TSV、封装和客户资格。

Citi · p.13

通用 / 低功耗 DRAM 产品 DDR5 / LPDDR5X

分别出现在服务器、PC、移动等应用口径中。具体平台需求不能混用。

Morgan Stanley · p.3

报告所述服务器内存模组方向 SoCAMM2

花旗将其与服务器 DRAM 一起讨论；三份材料没有提供完整规格定义。

Citi · p.3

闪存介质 / 企业级固态硬盘 NAND / eSSD

NAND 是介质，eSSD 是含控制与系统能力的产品，二者增长口径不同。

Citi · p.14

管理型闪存产品 eMMC / UFS

NAND 与控制器组合；伯恩斯坦以其作为移动 NAND 价格样本。

Bernstein · p.6

包含 DRAM 的组合封装 eMCP / uMCP

结合 DRAM、NAND 和控制器；报告中的价格是整包价格。

Bernstein · p.6

GLOSSARY

术语手册 · 续

每单元存储3/4位的 NAND TLC / QLC

基本术语解释；报告关注 QLC 大容量在企业级与近 GPU 存储的应用。

Citi · p.16

KV 缓存卸载 KV cache offloading

把部分推理缓存工作负载转移到外部内存或存储层；效果依具体架构。

Citi · p.9

单系统扩展 / 更多系统扩展 Scale-up / Scale-out

花旗用来解释单颗容量变化与整体出货增长之间的关系。

Citi · p.4

硅通孔 / 堆叠层数 TSV / hi / H

HBM 制造和堆叠口径；12hi/12H 表示报告所用的12层堆叠写法。

Morgan Stanley · p.10

千片晶圆/月 KWPM

晶圆能力指标；年末、平均、前端、TSV 口径需要进一步区分。

Citi · p.13

位元增长 Bit growth

总存储位元产出的增长，不等于晶圆片数或设备台数增长。

Citi · p.8

供需比 / 充足率 S/D / Sufficiency

本课程供需表以(供给÷需求−1)计算，负值为不足、正值为过剩。

Citi · p.11

现货 / 合约 Spot / Contract

成交范围、价格周期和客户不同，不能把二者混为财报售价。

Bernstein · p.3

平均售价 / 混合平均售价 ASP / Blended ASP

混合 ASP 受产品与客户组合影响，未必等于固定样本指数。

Bernstein · p.3

GLOSSARY

术语手册 · 续

月环比 / 季环比 / 同比 / 季度迄今 MoM / QoQ / YoY / QTD

时间基准不同；8月数据不能当作完整第三季度实现值。

Bernstein · p.6

低个位数百分比 / 长期协议 LSD% / LTA

LSD用于描述涨幅；LTA可能约束量、价和期限，作用取决于条款。

Bernstein · p.1

降配 De-spec

单设备配置下降；总需求要与设备数量和其他产品增量共同计算。

Bernstein · p.4

晶圆厂设备 / 资本开支 WFE / Capex

须核对是否包含其他投资边界；高投入需要时间才能转成有效产出。

Citi · p.18

深紫外 / 极紫外光刻 DUV / EUV

大摩讨论设备获取与多重图形化对步骤、周期和成本的影响。

Morgan Stanley · p.10

长江存储架构 / 混合键合 Xtacking / Hybrid bonding

大摩将架构进展与NAND密度、产品整合能力联系起来。

Morgan Stanley · p.11

擦写次数 / 每日整盘写入次数 P/E cycles / DWPD

一个是循环指标，一个是整盘写入强度指标，不是同一个寿命数字。

Morgan Stanley · p.11

互连及新存储方向 CXL / HBF / XL Flash

花旗提及的AI存储技术方向；三份材料未给出完整商用规模。

Citi · p.16

权益回报 / 净经营资产回报 / 市净率 ROE / RNOA / P/B

利润率、回报率与估值倍数不能混算；大摩用它们讨论跨周期回报。

Morgan Stanley · p.9

GLOSSARY

术语手册 · 续

预测或估计 / 待定 E / TBD

原文路线图与财务预测常用标记，应一直保留不确定性。

Citi · p.5

REFLECTION

从“看懂”，走到“讲清楚”

建议先遮住解析独立回答，再回到原文核对。交互版中可以逐题展开答案。

01 Bernstein 的 NAND +9.4% 与“接近 +20%”为何不矛盾?

前者是晶圆与 eMMC/UFS 的直接样本，未覆盖 SSD；后者加入机构对 SSD 的判断，是更广口径估计。两者不是同一范围的已实现价格。

Bernstein · p.5

02 16Gb 芯片的容量是多少 GB?

2GB，因为 8bit=1byte。若价格标为美元/GB，不能把这一报价直接当作16Gb芯片的整颗价格。

Bernstein · p.6

03 2027 年花旗 DRAM 缺口如何复算?

465,560÷510,101−1≈−8.73%。供给、需求须同单位、同期间；增长率相减不能代替这个计算。

Citi · p.11

04 设备量 +30%，单机容量 −20%，总需求怎么变?

1.30×0.80−1=+4%。这是教学假设，不是报告给出的市场预测。

Bernstein · p.4

05 为什么产能份额13%、位元份额11%、营收份额6%可以同时出现?

晶圆投入、每片位元产出和价格组合不同。三者分别衡量生产能力、实物供给和收入，不是相互矛盾的同一指标。

Morgan Stanley · p.3

06 HBM 占先进晶圆34%，是否等于占全行业收入34%?

不等于。分母是先进DRAM晶圆，且计量为晶圆占用；收入还受产品售价影响。

Morgan Stanley · p.4

REFLECTION

练习与解析 · 续

07 Citi 与 MS 对 2027 年 HBM 分歧主要在哪里?

Citi需求75.157bn Gb、MS需求56.085bn Gb，而供给分别59.347和56.300bn Gb。需求差异明显大于供给差异，但仍需追溯各自模型。

Citi · p.13

08 AI SSD 增长50%、长江存储430KWPM，MS矩阵结果是什么?

过剩3%。这个结果以非AI需求+5%及其他厂商扩产沿用原预测为前提，没有给出发生概率。

Morgan Stanley · p.12

09 为什么 NAND 现货强，并不能推断模组厂补库很积极?

报告同时观察到合约弱、交易量低、模组厂谨慎及采购低等级芯片。现货样本与实际补库范围不同。

Bernstein · p.5

10 如何处理花旗摘要和表格的需求增速不一致?

分别保留摘要原值与按表内水平重算的值，注明页码和算法，不擅自用一项覆盖另一项。

Citi · p.17

11 为什么看到 HBM stack 产出，还不能认定全面商业替代?

还需综合良率、封装测试、可靠性、交付能力与加速器客户认证。功能可用与规模商业资格是不同阶段。

Morgan Stanley · p.10

12 用三份材料，如何形成一条可被推翻的采购判断?

例：DDR4长尾需求与退出使合格供给偏紧；若新供应商认证放量、交期恢复且库存连续累积，则应重估紧张程度。这是教学迁移，需用实际材料数据验证。

Bernstein · p.4

ORIGINAL REPORTS

三份原件，完整保留

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Citi 花旗 / 2026-09-14

Global Semiconductors: Continual Learning to Fuel HBM/eSSD Demand (AI Training/Inference)

29 页原文；42 幅编号图表。

Bernstein 伯恩斯坦 / 2026-09-03

Global Memory: MEMORY TRACKER (Aug): Conventional DRAM & NAND price to rise nearly 20% in 3Q26

30 页原文；27 幅编号图表。

Morgan Stanley 摩根士丹利 / 2026-08-26

China Memory – Reshaping Global Competition

16 页原文；20 幅编号图表。

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14 Sep 2026 04:15:22 ET | 29 pages

Global Semiconductors

Continual Learning to Fuel HBM/eSSD Demand (AI Training/Inference)

Deep Dive

CITI'S TAKE

Despite memory de-spec and recent AI safety concerns, we believe that continual learning in AI will reshape long-term memory demand. We see it driving memory demand across diverse products including HBM (AI training), server DDR5 and eSSDs (AI inference), leading to memory undersupply worsening throughout 2028E and persisting into 2031E. Amid the structural growth trajectory of the memory market, we highlight major potential beneficiaries.

Entering Era of Continual Learning — We expect continual learning to be a key theme in AI over the next five years, as it would maximize resource efficiency and make AI increasingly intelligent. Continual learning is an AI learning approach that reinforces a model with training on new tasks and knowledge. As a result, continual learning could create persistent need for both model updates and access to previously acquired data, entailing significant usage of memory.

HBM and eSSDs Demand to Increase Exponentially — We expect both training and inference demand to grow simultaneously from 2027E onward, as the continual learning approach will increasingly become sophisticated. This should drive DRAM demand for server DDR5 used in AI inference and HBM, which is needed to acquire new information continuously. In particular, we forecast HBM bit demand to surge +62% YoY to 75.2bn Gb in 2027E and +69% YoY to 127.0bn Gb despite concerns over HBM de-spec and recent AI safety issues. On NAND, we project eSSDs demand to expand as retaining previously learned data requires substantial storage capacity.

27E/28E DRAM S/D Ratio of -8.7%/-9.7% — We forecast global DRAM demand to grow +30%/+35% YoY in 27E/28E driven by continual learning, which should drive server DDR5, and HBM demand simultaneously. With personal/physical AI emerging from 2H28E onwards, we project overall DRAM market undersupply to intensify throughout 2028E, with DRAM S/D ratio projected at -8.7%/-9.7% in 27E/28E amid limited supply growth of +19%/+22% in 27E/28E.

27E/28E NAND S/D Ratio of -6.1%/-5.5% — We project 27E/28E NAND demand to grow +29%/+33% YoY, exceeding supply growth of +21%/+25%, as memory suppliers are expanding DRAM/HBM green-field capacity at the expense of NAND. We anticipate continual learning as well as Nvidia's KV cache offloading trend to drive high-density eSSDs demand where we model +53%/+41% YoY demand growth in 27E/28E. As a result, we see NAND supply as much tighter and expect NAND S/D ratio at -6.1%/-5.5% in 27E/28E. (vs. -0.8% in 2026E)

Memory Top Picks: Samsung Electronics, SK Hynix, Micron, Sandisk, Kioxia; SPE/Materials Top Picks: Montage, AMAT, Lam Research, TES, Eugene Tech, and TechWing — We expect global memory suppliers to strongly benefit from the persistent memory supply shortage driven by continual learning and the potential emergence of personal AI in the mid-to-long-term. As demand is structurally growing, we think memory SPEs and material companies will incrementally grow on the back of massive capex plans and elevated memory production run rates.

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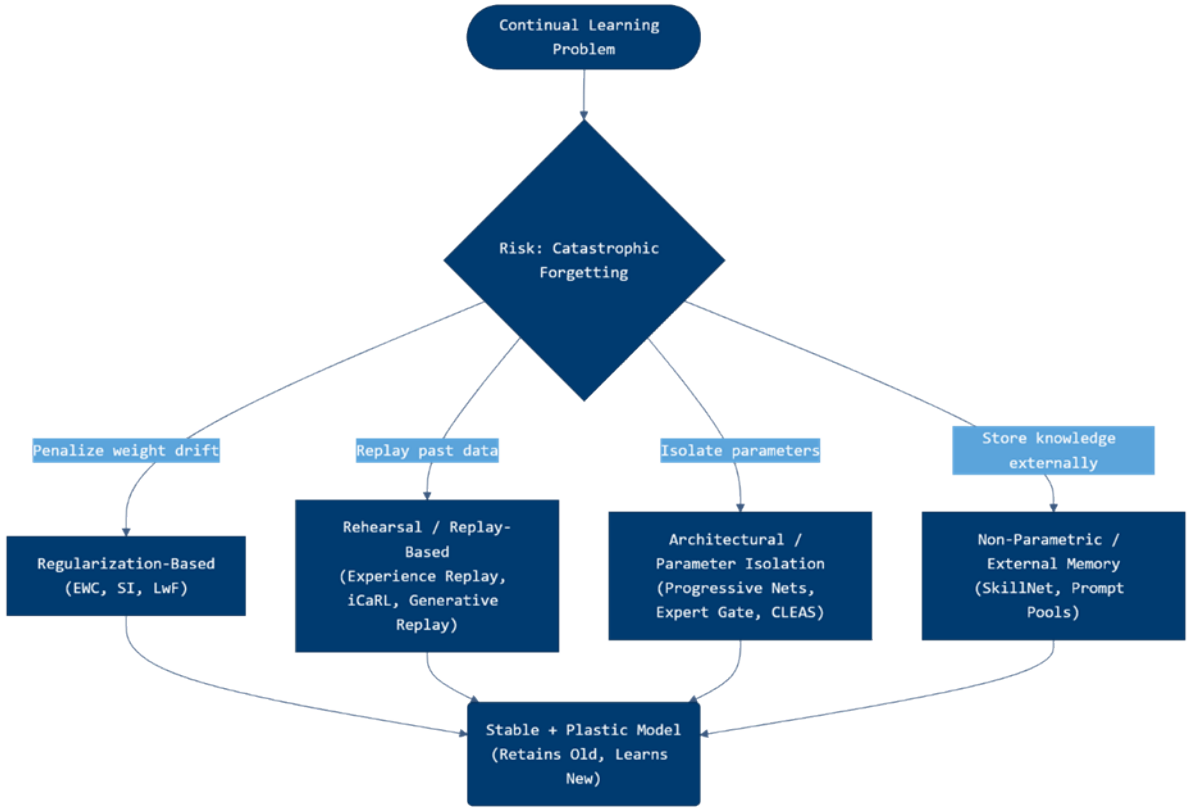
Continual Learning AI Era: HBM, Server DRAM and eSSDs Demand Looks Set to Exponentially Rise

Like the protagonist in the movie *Memento*, today's AI models retain what they have learned but do not automatically incorporate new interactions and data into their model parameters. We expect continual learning to be a key theme in AI over the next five years, as it would maximize resource efficiency and make AI increasingly intelligent.

Continual learning is an AI learning approach that reinforces a model by incrementally training on new tasks and knowledge. It involves training on new data while simultaneously preserving previously learned knowledge.

The key challenge is to balance learning new information with mitigating catastrophic forgetting. In particular, models are trained on a continuous stream of nonstationary data, with the total number of tasks to be learned not fixed in advance. As a result, continued learning could create persistent needs for both ongoing model updates and access to previously acquired data.

Figure 1. Continual Learning



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Source: Citi Research

In the memory market, HBM demand grew exponentially in 2022~2024, driven by increasingly complex and advanced training processes, while demand for conventional DRAM and NAND remained relatively soft. Since 2H25, however, demand for server DDR5 and eSSDs has surged as AI inference workloads have grown more complex, while market interest in HBM has cooled.

Figure 2. Evolution of AI Workloads and Memory Demand Implications

	22-1H25	2H25-26E	27E-1H28E	2H28E~
AI Era	AI Training Era	AI Inferencing Era (KV Cache)	Continual Learning Phase 1: AI Server	Continual Learning Phase 2: Physical AI/Personal AI
HBM Demand	Robust HBM Demand	HBM Demand relatively soft	Demand for HBM/Server DRAM/SoCAMM2/eSSDs to increase simultaneously	On-device/Edge Memory Demand to Emerge
DRAM/NAND Demand	Conventional DRAM/NAND relatively muted	Strong Server DRAM/eSSDs		

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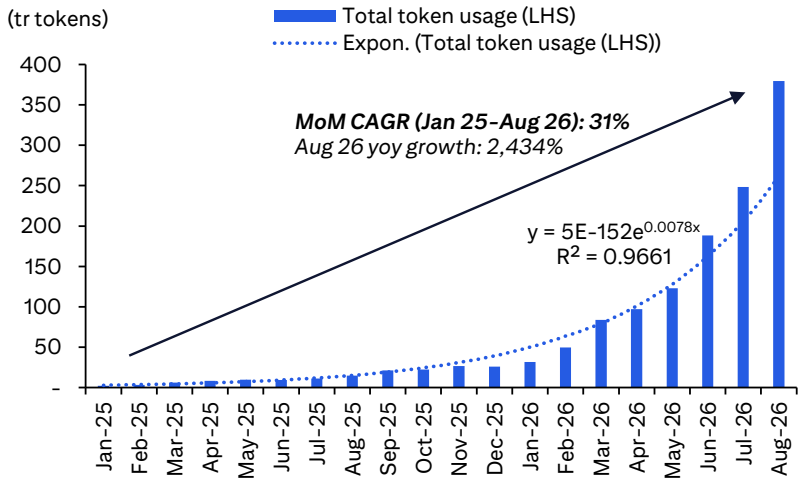
Source: Citi Research

On the other hand, we expect both training and inference demand to grow simultaneously after 2027E, as the continual learning approach itself increasingly becomes sophisticated. This should drive demand for server DDR5 and SoCAMM2 used in AI inference, as well as demand for HBM needed to continuously acquire new information. In addition, eSSDs demand will expand, on our analysis, as retaining previously learned data requires substantial storage capacity.

Beyond continual learning, we believe the next major shift in AI will lie in the era of personal AI/physical AI (See our previous note, Global Semiconductors - The AI Hardware Shift in IT Devices: The Era of Personal AI Server Begins). Specifically, we expect technology enabling edge AI execution and local data management to gain traction from 2H28E, allowing AI devices and robots to maintain persistent memory and adapt to their environments over time.

Meanwhile, we see AI chip architecture increasingly shifting from scale-up to scale-out. Additionally, continual learning should further raise memory intensity, as AI workloads increasingly incorporate continued training and model updates, which drives renewed focus on memory optimization. As shown in Figure 3, however, AI token demand has continued to grow exponentially.

Figure 3. Monthly AI Token Usage Trend



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Source: Citi Research, OpenRouter

We believe the recent de-specification issues indicate resource efficiency optimization amid memory shortage rather than demand weakness (See our previous note, Global Semiconductors - AI Chipmakers to Pivot from Scale-Up to Scale-Out Amid HBM Shortage). In particular, we expect HBM bit demand to surge 62% YoY to 75.2bn Gb in 2027E and 69% YoY to 127.0bn Gb in 2028E, double our previous 27E/28E estimates. This should be driven by robust demand growth not only from Nvidia but also from ASICs manufacturers such as Broadcom and Google.

Figure 4. HBM Demand Outlook

		2021	2022	2023	2024	2025	2026E	2027E	2028E
HBM Total(1Gb eq)		736	1,664	3,350	10,096	20,965	46,409	75,157	126,990
	Nvidia	512	1,024	1,728	4,160	8,400	22,003	32,503	40,939
	Broadcom	96	154	365	1,440	3,840	9,002	13,001	41,003
	Google	51	256	576	1,728	2,765	4,702	10,300	14,003
	AMD	64	128	384	832	2,160	4,200	7,000	8,200
	AWS	6	45	144	640	1,512	2,501	2,802	6,391
	Others	6	58	154	1,296	2,288	4,001	9,550	16,455
HBM DRAM (Mpcs, 1Gb eq)		736	1,664	3,350	10,096	20,965	46,409	75,157	126,990
Breakdown	HBM1(1Gb eq.)	368	832	-	-	-	-	-	-
	HBM2(1Gb eq.)	368	832	1,104	-	-	-	-	-
	HBM2E(1Gb eq.)	-	-	2,246	1,338	-	-	-	-
	HBM3(1Gb eq.)	-	-	-	6,560	6,598	1,601	1,364	-
	HBM3E 8hi(1Gb eq.)	-	-	-	1,680	3,840	6,104	1,005	-
	HBM3E 12hi(1Gb eq.)	-	-	-	518	10,526	25,313	14,153	12,628
	HBM4 8hi(1Gb eq.)	-	-	-	-	-	2,316	6,056	4,330
	HBM4 12hi(1Gb eq.)	-	-	-	-	-	11,075	32,107	47,379
	HBM4 16hi(1Gb eq.)	-	-	-	-	-	-	-	-
	HBM4e 8hi(1Gb eq.)	-	-	-	-	-	-	11,305	21,170
	HBM4e 12hi(1Gb eq.)	-	-	-	-	-	-	7,011	33,756
	HBM4e 16hi(1Gb eq.)	-	-	-	-	-	-	2,154	4,557
	HBM5 16hi(1Gb eq.)	-	-	-	-	-	-	-	3,169
Portion	HBM1(%)	50%	50%	0%	0%	0%	0%	0%	0%
	HBM2(%)	50%	50%	33%	0%	0%	0%	0%	0%
	HBM2E(%)	0%	0%	67%	13%	0%	0%	0%	0%
	HBM3(%)	0%	0%	0%	65%	31%	3%	2%	0%
	HBM3E 8hi(%)	0%	0%	0%	17%	18%	13%	1%	0%
	HBM3E 12hi(%)	0%	0%	0%	5%	50%	55%	19%	10%
	HBM4 8hi(%)	0%	0%	0%	0%	0%	5%	8%	3%
	HBM4 12hi(%)	0%	0%	0%	0%	0%	24%	43%	37%
	HBM4 16hi(%)	0%	0%	0%	0%	0%	0%	0%	0%
	HBM4e 8hi(%)	0%	0%	0%	0%	0%	0%	15%	17%
	HBM4e 12hi(%)	0%	0%	0%	0%	0%	0%	9%	27%
	HBM4e 16hi(%)	0%	0%	0%	0%	0%	0%	3%	4%
	HBM5 16hi(1Gb eq.)	0%	0%	0%	0%	0%	0%	0%	2%
Total DRAM Demand(Mpcs)		191,673	193,221	190,282	246,759	305,484	390,057	510,101	630,035
	HBM portion(%)	0.4%	0.9%	1.8%	4.1%	6.9%	11.9%	14.7%	20.2%

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Source: Citi Research Estimates, TrendForce

Figure 5. Customers' HBM Roadmap by GPU and ASIC product

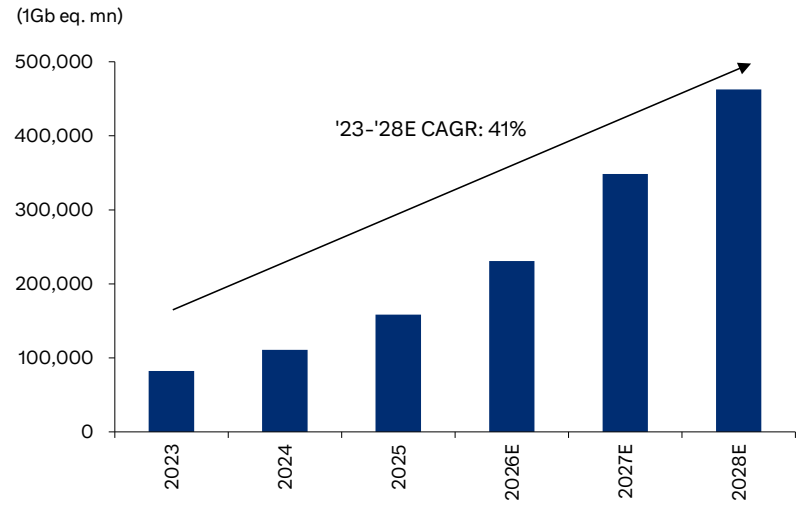
Company	AI Chips	External ASIC/ Design Partner	2024				2025				2026E				2027E				2028E			
			1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	1Q28E	2Q28E	3Q28E	4Q28E
Nvidia	H100	In-house	HBM3 8hi 80GB (16GB*5)																			
	GH200 (CPU+GPU)	In-house	HBM3e 8hi 144GB (24GB*6)																			
	H20	In-house	HBM3 8hi 96GB (16GB*6)																			
	H200	In-house	HBM3e 8hi 141GB (24GB*6)																			
	B100/ B200	In-house			HBM3e 8hi 192GB (24GB*8)																	
	GB200 (CPU+GPU)	In-house			HBM3e 8hi 192GB/ 384GB (24GB*8/ 192GB*2)																	
	B200A Ultra (B300A)	In-house					HBM3e 12hi 144GB (36GB*4)															
	Blackwell Ultra (B300)	In-house							HBM3e 12hi 288GB (36GB*8)													
	GB300 (CPU+GPU)	In-house							HBM3e 12hi 576GB (288GB*2)													
	Rubin	In-house									HBM4 12hi 288BG (36GB*8)											
	Rubin Ultra	In-house															HBM4 8hi 192GB (24GB*8)						
	Feynman	In-house																	custom HBM (possibly NVHBM)			
Company	**AI Chips**	**External ASIC/ Design Partner**	**2024**				**2025**				**2026**				**2027**				**2028**			
			1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26	1Q27	2Q27	3Q27	4Q27	1Q28	2Q28	3Q28	4Q28
AMD	MI200	In-house	HBM2e 8hi 128GB (16GB*8)																			
	MI300X	In-house	HBM3 12hi 192GB (24GB*8)																			
	MI300A (CPU+GPU)	In-house	HBM3 8hi 128GB (16GB*8)																			
	MI325X	In-house				HBM3e 8hi 256GB (32GB*8)																
	MI350	In-house						HBM3e 12hi 288GB (36GB*8)														
	MI400	In-house									HBM4 12hi 432GB (36GB*12)											
	MI500	In-house															HBM4e (TBD) (within 2027)						
	MI600	In-house																	HBM5 (TBD; within 2028)			
Company	**AI Chips**	**External ASIC/ Design Partner**	**2024**				**2025**				**2026**				**2027**				**2028**			
			1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26	1Q27	2Q27	3Q27	4Q27	1Q28	2Q28	3Q28	4Q28
AWS	Trainium 1	Alchip	HBM2e 32GB (16GB*2)																			
	Inferentia 2	Alchip	HBM2e 32GB (16GB*2)																			
	Trainium 2	Marvell				HBM3 96GB (24GB*4)																
	Trainium 2.5	Marvell							HBM3e 144GB (36GB*4)													
	Trainium 3	Alchip										HBM3e 12hi 144GB (36GB*4)										
	Trainium 4	Alchip, possibly Marvell																HBM4 12hi 288GB (36GB*8)				
Google	TPU v5i	Broadcom	HBM2e 16GB (8GB*2)																			
	TPU v5t	Broadcom	HBM2e 96GB (16GB*6)																			
	TPU v6i	Broadcom				HBM3 32GB (16GB*2)																
	TPU v7t	Broadcom							HBM3e 192GB (24GB*8)													
	TPU v8i	Broadcom											HBM3e 12hi 288GB (36GB*8)									
	TPU v8t	MediaTek											HBM3e 12hi 216GB (36GB*6)									
	TPU v9 Base (Humufish)	Marvell																	HBM4 (TBD)			
	TPU v9i (Triggerfish)	MediaTek																	HBM4e 8hi 384GB (32GB*12)			
	TPU v9t	MediaTek																	HBM4e 8hi 384GB (32GB*12)			
	TPU v10 (Icefish)	MediaTek / AMD																				TBD
	Merope (variant of TPU v10 or LPU)	Marvell / AMD																				
Microsoft	Maia 100	GUC	HBM2e 64GB (16GB*4)																			
	Maia 200	GUC									HBM3e 12hi 216GB (36GB*6)											
	Maia 300	Marvell															HBM4 12hi 288GB (36GB*8)						
Meta	MTIA 200	Broadcom			LPDDR5 128GB																	
	MTIA 300	Broadcom							HBM3e 216GB (36GB*6)													
	MTIA 400	Broadcom										HBM3e 12hi 288GB (36GB*8)										
	MTIA 450	Broadcom													HBM4 12hi 288GB (36GB*8)						
	MTIA 500	Broadcom																HBM4e 12hi 384GB (48GB*8)				
	MTIA 600	Broadcom; possibly Alchip / GUC / MediaTek																	TBD (2028E+)			
OpenAI	Jalapeño	Broadcom													HBM4 12hi 216 GB (36GB*6)							
	TBD	Broadcom																	TBD (2028E+)			
Apple	Baltra	Broadcom																TBD				
Anthropic	TBD	MatX																	TBD (2028E+)			

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Source: TrendForce, Companies, Citi Research Estimates

We forecast AI inference demand to rise continuously due to the impact of continual learning. Specifically, we expect server DRAM demand to grow more than 50% YoY in 2027E, with eSSDs demand rising ~50% YoY. Server DRAM demand should rise alongside growing CPU demand. We believe this improving multi-year demand visibility is also consistent with memory customers increasingly extending LTAs from 3 to 5 years.

Figure 6. Server DRAM Demand Outlook

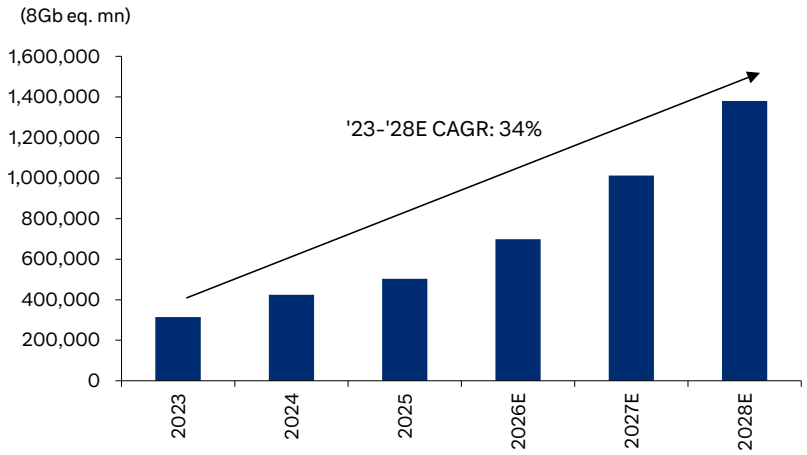


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Source: Citi Research Estimates; TrendForce

In terms of NAND, we expect growing demand for advanced solutions such as XL Flash and HBF with storage-complementary products such as CXL. We also expect near GPU storage solutions using QLC SSDs to gain traction as AI systems require larger pools of storage closer to accelerators to enhance memory retention.

By contrast, consumer NAND demand could soften amid weaker mobile and PC market conditions, though we expect this to be offset by rising AI inference-related NAND demand. In addition, AI data centers, particularly in China, are increasingly considering SSDs over HDDs, potentially creating further upside to eSSDs demand from 2027E.

Figure 7. SSD Demand Outlook



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Source: Citi Research Estimates; TrendForce

In conclusion, we believe continual learning will drive simultaneous growth in both training and inference related memory demand from 2027E. HBM should benefit from ongoing training and AI model updates, server DRAM/SoCAMM2 from rising inference demand and CPU workloads, and eSSDs from persistent storage requirements. As a result, we expect AI memory demand to strengthen beyond current market expectations, with demand for server DRAM/SoCAMM2/eSSDs particularly robust in 2027E.

Memory Supply/Demand Outlook

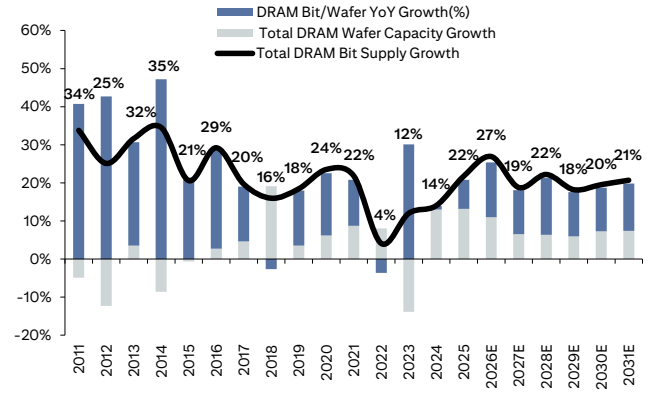
Sustained demand and constrained supply to drive an undersupplied memory market in 27E

We expect memory market tightness to intensify in 2027E, driven by limited supply growth given higher HBM production volume allocation and increasing difficulty in tech migration.

In terms of DRAM, we expect DRAM supply growth of ~19% in 2027E, constrained by higher HBM capacity allocation and slowing node migration. On the demand side, despite HBM de-specification issues, we forecast ~30% growth, driven by continued HBM demand from AI chipmakers' shift toward scale-out from scale-up and robust server DRAM demand.

Regarding NAND, we expect NAND demand to grow ~29% in 2027E amid constrained industry capacity. This is supported by rising AI inference demand and incremental eSSDs demand from offloading as lower HBM content shifts more memory workloads to external storage.

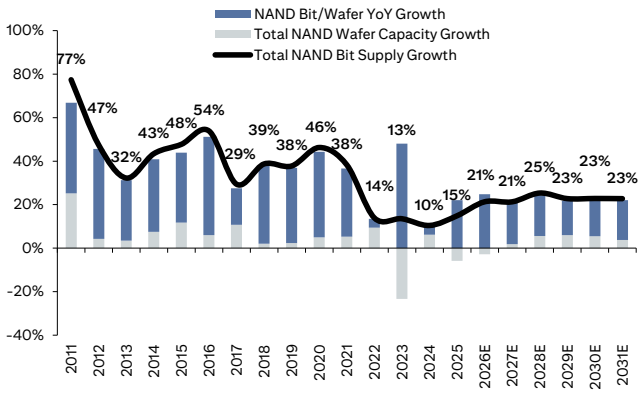
Figure 8. DRAM – Bit Supply Growth Outlook



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Source: DRAMeXchange, Citi Research estimates

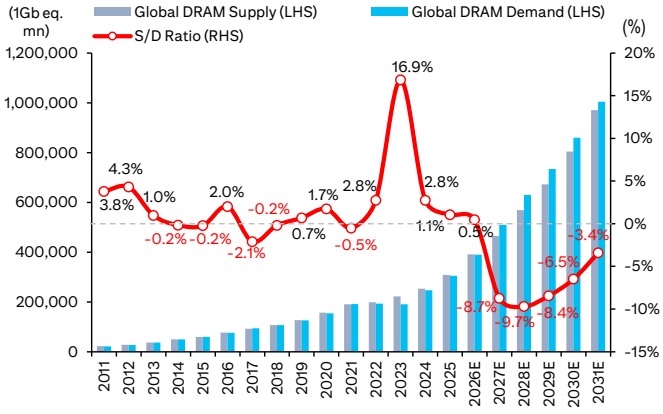
Figure 9. NAND – Bit Supply Growth Outlook



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Source: DRAMeXchange, Citi Research estimates

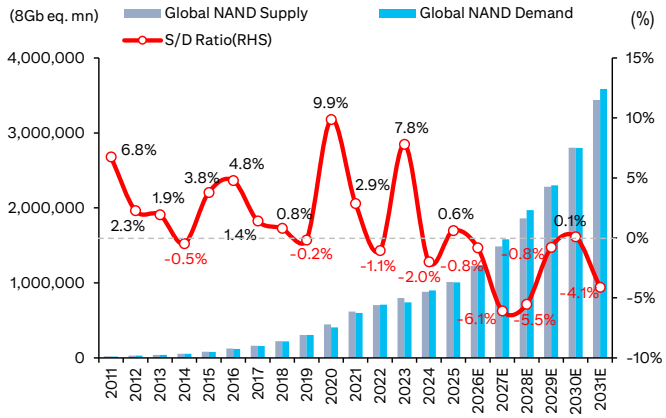
Figure 10. DRAM - Supply/Demand Outlook (2027E S/D Ratio: -8.7%)



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Source: Citi Research Estimates, DrameXchange

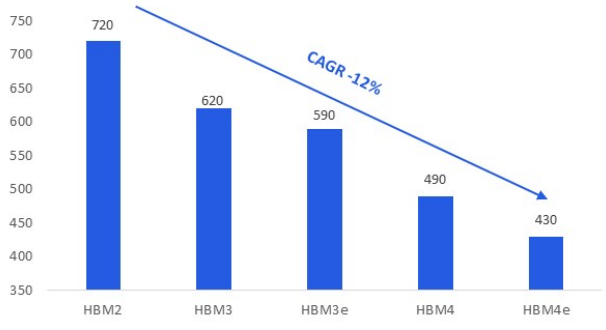
Figure 11. NAND - Supply/Demand Outlook (2027E S/D Ratio: -6.1%)



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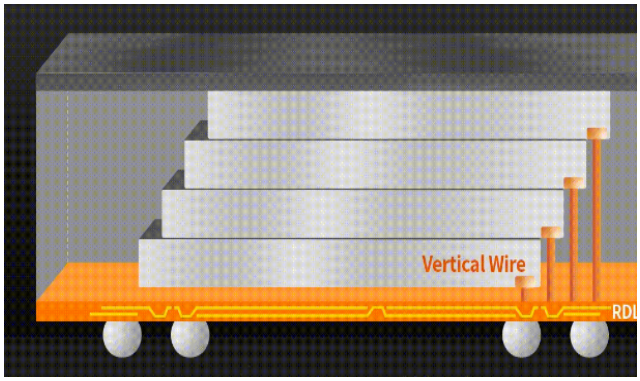
Source: Citi Research Estimates, DrameXchange

Figure 12. # of Dies per HBM Wafer



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Source: Citi Research estimates; TrendForce

Figure 13. Wide I/O (Mini HBM) for Mobile with VFO



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Source: SK Hynix, Citi Research

DRAM Supply/Demand Outlook

DRAM to be undersupplied in 27E

In 2027E, we forecast overall DRAM S/D to be undersupplied driven by [1] continued HBM demand from AI chipmakers' shift toward scale-out, [2] incremental demand for server DRAM, [3] constrained supply growth due to higher HBM capacity allocation, and [4] slowing node migration.

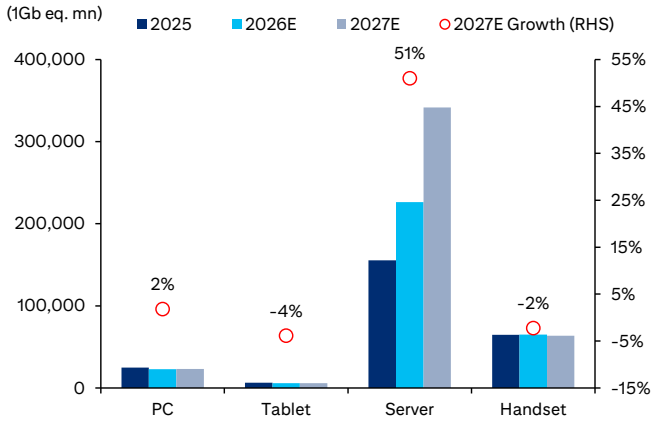
DRAM demand outlook: +30.2% YoY in 27E

We expect 27E DRAM bit demand to grow +30.2% YoY, significantly outpacing bit supply growth of +18.8% YoY, driving the industry S/D ratio to -8.7% from +0.5% in 26E. On the demand side, we expect robust growth to be driven by continued HBM demand as AI chipmakers shift toward scale-out, together with strong server DRAM demand, while PC demand remains relatively soft and mobile demand sees a modest recovery.

We view recent HBM de-specification issues as a response to constrained HBM availability rather than weaker underlying demand. Given AI chipmakers are pursuing a larger number of accelerator units amid limited HBM supply, we believe it will support continued HBM demand at a system level.

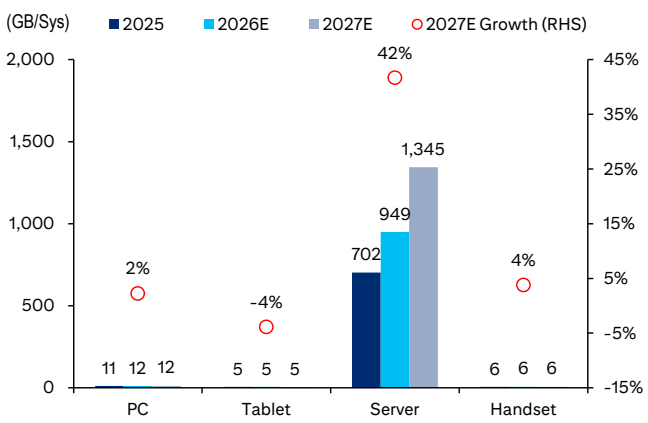
At the same time, server DRAM will benefit from memory workload offloading as lower HBM content should increase KV-cache and provide incremental demand outside HBM. Specifically, we forecast server DRAM demand to grow from 226.3bn 1Gb equivalent units in 2026E to 341.7bn units in 2027E (+51% YoY). In 2027E, we expect server will continue to account for the largest DRAM demand at 67%.

Figure 14. DRAM Set Application Forecasts



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Source: Citi Research Estimates, DRAMeXchange

Figure 15. DRAM Avg. Content Adoption Forecasts

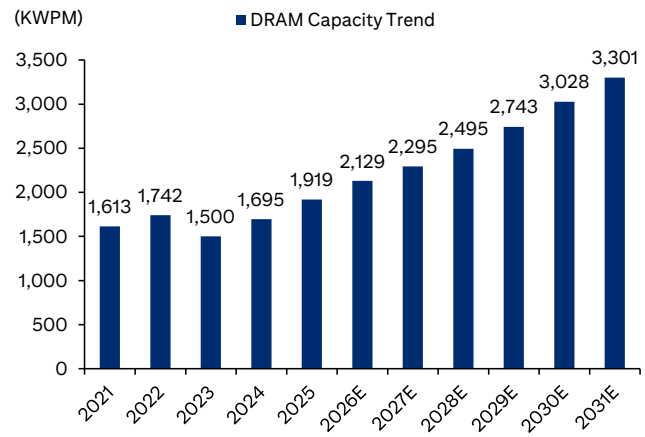


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Source: Citi Research Estimates, DRAMeXchange

DRAM Capex outlook: +8% YoY in 27E

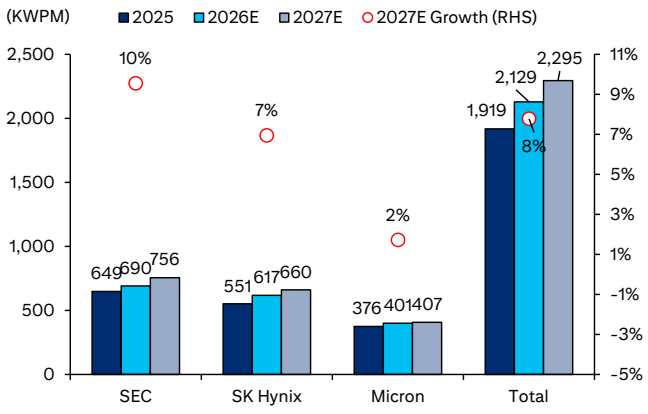
On the supply side, we expect average DRAM industry wafer capacity to increase only ~8% YoY to 2,295KWPM in 2027E from 2,129KWPM in 2026E. By supplier, we expect Samsung Electronics' DRAM greenfield capacity to increase by ~70KWPM in end 4Q27E vs. end-4Q26E. For SK Hynix, we project its greenfield capacity to increase by ~70KWPM in end 4Q27E vs. end-4Q26E

Figure 16. Average DRAM Industry Capacity Trend



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Source: Citi Research Estimates, TrendForce

Figure 17. Average DRAM Capacity Change by Company

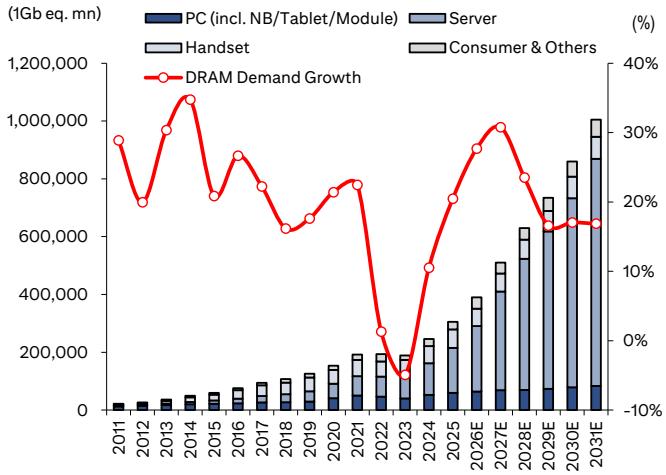


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Source: Citi Research Estimates, TrendForce

DRAM supply growth: +18.8% in 27E

We forecast DRAM industry bit supply growth of +18.8% YoY in 2027E. By company, we forecast Samsung to realize YoY DRAM supply B/G of +17.6% in 2027E, Hynix to increase its DRAM B/G by +20.6% and expect Micron to increase its supply B/G by +21.1% in 2027E. While industry DRAM Capex rises, a large portion of incremental investment could continue to be directed toward HBM. Additionally, slower technology migration limits bit growth from existing wafer capacity. Moreover, the long lead time for greenfield additions should prevent the sharp increase in bit supply. As a result, we expect DRAM supply to remain constrained throughout 2027E.

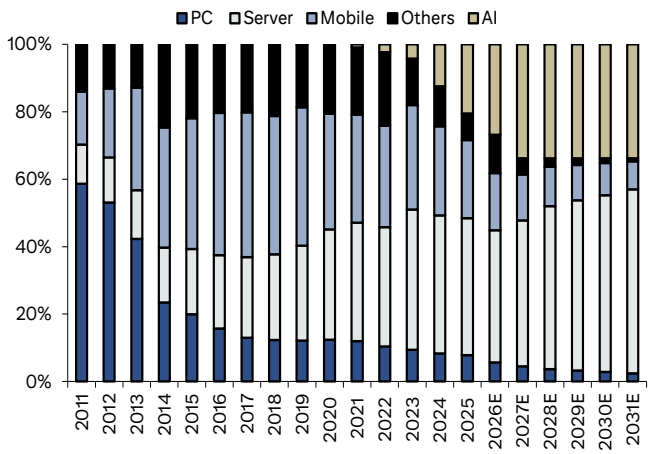
Figure 18. DRAM Demand Forecast by Application



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Source: Citi Research Estimates, TrendForce

Figure 19. DRAM Demand Portion by Application



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Source: Citi Research Estimates, TrendForce

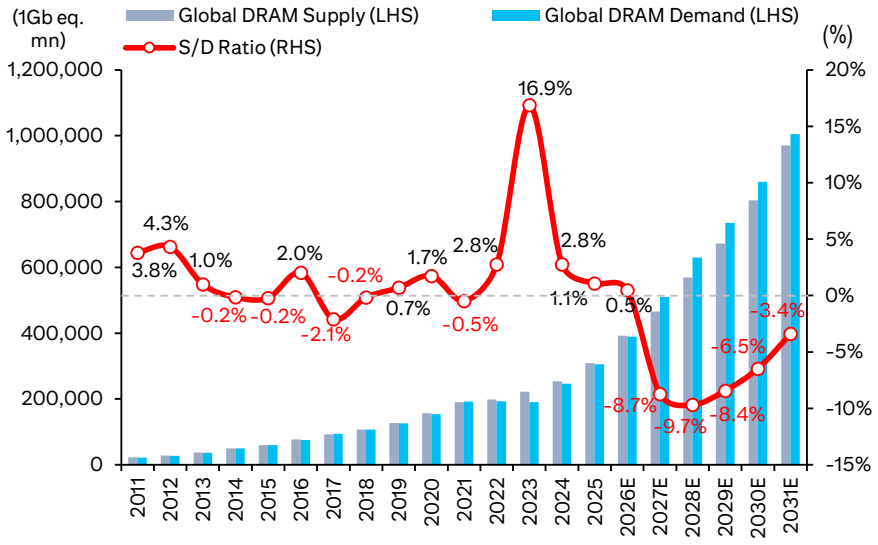
Figure 20. Quarterly/Annual DRAM Supply / Demand Forecasts

(1Gb eqiv. mn units)	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	1Q28E	2Q28E	3Q28E	4Q28E	2026E	2027E	2028E
DRAM Supply	88,212	96,103	102,411	105,166	105,274	111,354	121,625	127,307	126,993	134,665	150,071	157,240	391,891	465,560	568,969
DRAM Demand	88,049	95,749	102,294	103,966	108,824	118,010	135,373	147,894	142,892	149,641	165,370	172,132	390,057	510,101	630,035
DRAM S/D Ratio (%)	0.2%	0.4%	0.1%	1.2%	-3.3%	-5.6%	-10.2%	-13.9%	-11.1%	-10.0%	-9.3%	-8.7%	0.5%	-8.7%	-9.7%

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Source: DRAMeXchange, Citi Research estimates

Figure 21. DRAM Supply / Demand Forecasts



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Source: DRAMeXchange, Citi Research Estimates

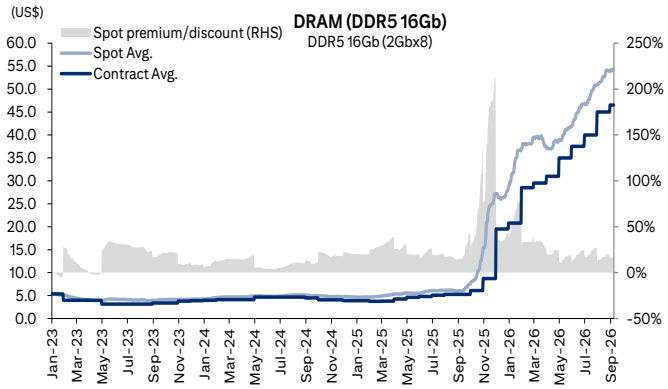
DRAM price outlook

DRAM ASP outlook: +23.1% YoY in 27E

Given the sharp deterioration in the DRAM S/D balance, we expect favorable pricing conditions to persist through 2027E. We forecast blended DRAM ASP to increase +23.1% YoY in 2027E, following an outsized +242.4% YoY increase in 2026E. While the magnitude of price increases should moderate from the exceptional 2026E level, we expect persistent supply tightness in both HBM and conventional DRAM to support further annual pricing upside.

By quarter, we expect blended DRAM ASP to move +20% QoQ in 3Q26E, and +13% QoQ in 4Q26E. On an annual basis, we expect annual average DRAM ASP to rise +242.4%/+23.1% in 2026E/2027E.

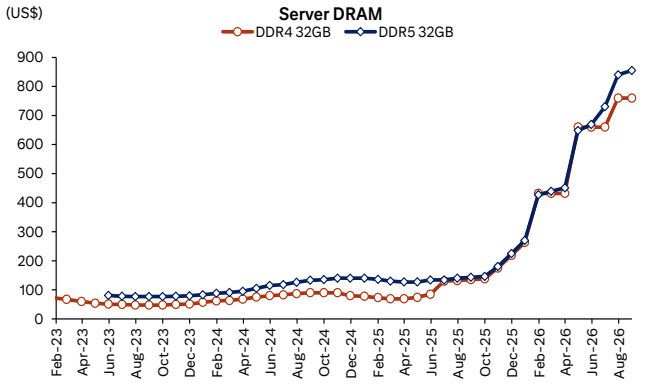
Figure 22. DDR5 PC DRAM Price Trend



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Source: DRAMeXchange, Citi Research estimates

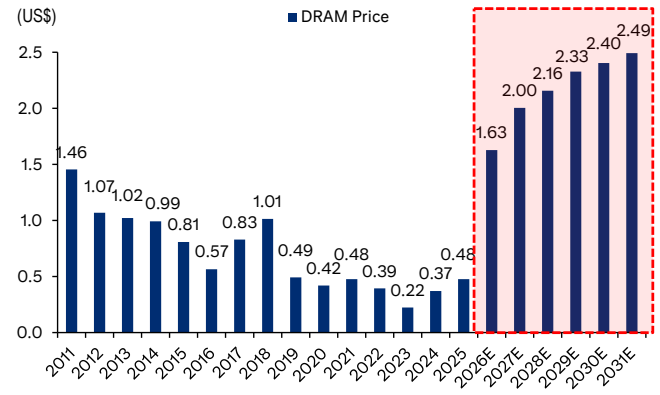
Figure 23. Server DRAM Price Trend



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Source: DRAMeXchange, Citi Research estimates

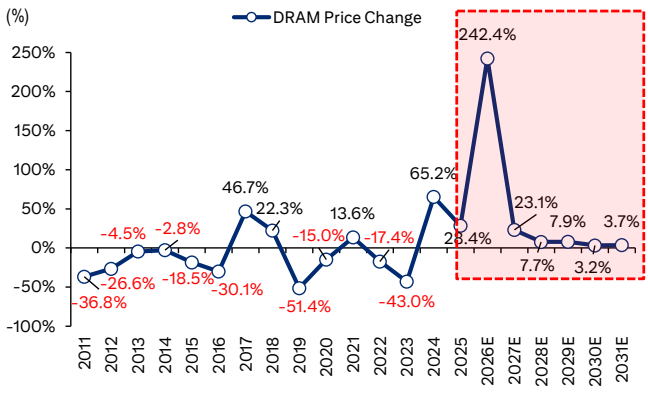
Figure 24. DRAM Price Outlook



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Source: DRAMeXchange, Citi Research estimates

Figure 25. DRAM Price Change Outlook



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Source: DRAMeXchange, Citi Research estimates

HBM supply/demand outlook

HBM S/D: -21% in 27E

We expect the HBM market to remain significantly undersupplied in 2027E despite the continued capacity expansion by memory suppliers. Our HBM demand forecast rises +62% YoY from 46.4bn 1Gb-equivalent units in 2026E to 75.2bn in 2027E, while supply increases from 36.1bn to 59.3bn, +64% YoY. As a result, we forecast the HBM S/D ratio to stay negative at -21% in 2027E, broadly unchanged vs -22% in 2026E. As the HBM demand from accelerated ASIC shipments is expected to grow further in 2028E, we project the 28E HBM S/D to worsen at -36%.

At the system level, total HBM unit demand will increase from 178.5mn units in 2026E to 269.1mn units in 2027E, we estimate, supported by continued growth from Nvidia as well as expanding ASIC demand. Therefore, we continue to view recent HBM de-specification issues as a practical way of maximizing accelerator shipments amid constrained supply rather than a sign of weaker HBM demand.

Figure 26. HBM Supply and Demand Outlook

		2021	2022	2023	2024	2025	2026E	2027E	2028E
HBM	Demand(Mpcs, 1Gb eq)	736	1,664	3,350	10,096	20,965	46,409	75,157	126,990
	Supply(Mpcs, 1Gb eq)	752	1,699	3,104	9,200	17,068	36,137	59,347	80,936
	S/D Ratio	2%	2%	-7%	-9%	-19%	-22%	-21%	-36%

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Source: Citi Research Estimates; TrendForce

Figure 27. HBM Supply Outlook

		2021	2022	2023	2024	2025	2026E	2027E	2028E
Year-End TSV Capacity(Kwafer/month)		32	53	106	180	280	420	700	870
	Samsung	20	28	49	70	110	155	240	300
	SK Hynix	8	20	50	75	110	150	270	290
	Micron	4	6	7	35	55	95	140	200
	CXMT	-	-	-	-	5	20	50	80
HBM Supply(Mpcs, 1Gb eq)		752	1,699	3,104	9,200	17,068	36,137	59,347	80,936
	Samsung	483	896	1,424	3,027	5,788	11,000	25,000	33,006
	SK Hynix	206	661	1,481	4,887	8,543	16,999	21,002	27,025
	Micron	63	142	200	1,285	2,719	8,034	13,009	20,001
	CXMT	-	-	-	-	19	104	336	904
M/S		100%	100%	100%	100%	100%	100%	100%	100%
	Samsung	64%	53%	46%	33%	34%	30%	42%	41%
	SK Hynix	27%	39%	48%	53%	50%	47%	35%	33%
	Micron	8%	8%	6%	14%	16%	22%	22%	25%
	CXMT	0%	0%	0%	0%	0%	0%	1%	1%

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Source: Citi Research Estimates; TrendForce

Figure 28. HBM Demand Outlook

		2021	2022	2023	2024	2025	2026E	2027E	2028E
HBM Total(1Gb eq)		736	1,664	3,350	10,096	20,965	46,409	75,157	126,990
	Nvidia	512	1,024	1,728	4,160	8,400	22,003	32,503	40,939
	Broadcom	96	154	365	1,440	3,840	9,002	13,001	41,003
	Google	51	256	576	1,728	2,765	4,702	10,300	14,003
	AMD	64	128	384	832	2,160	4,200	7,000	8,200
	AWS	6	45	144	640	1,512	2,501	2,802	6,391
	Others	6	58	154	1,296	2,288	4,001	9,550	16,455
HBM DRAM (Mpcs, 1Gb eq)		736	1,664	3,350	10,096	20,965	46,409	75,157	126,990
Breakdown	HBM1(1Gb eq.)	368	832	-	-	-	-	-	-
	HBM2(1Gb eq.)	368	832	1,104	-	-	-	-	-
	HBM2E(1Gb eq.)	-	-	2,246	1,338	-	-	-	-
	HBM3(1Gb eq.)	-	-	-	6,560	6,598	1,601	1,364	-
	HBM3E 8hi(1Gb eq.)	-	-	-	1,680	3,840	6,104	1,005	-
	HBM3E 12hi(1Gb eq.)	-	-	-	518	10,526	25,313	14,153	12,628
	HBM4 8hi(1Gb eq.)	-	-	-	-	-	2,316	6,056	4,330
	HBM4 12hi(1Gb eq.)	-	-	-	-	-	11,075	32,107	47,379
	HBM4 16hi(1Gb eq.)	-	-	-	-	-	-	-	-
	HBM4e 8hi(1Gb eq.)	-	-	-	-	-	-	11,305	21,170
	HBM4e 12hi(1Gb eq.)	-	-	-	-	-	-	7,011	33,756
	HBM4e 16hi(1Gb eq.)	-	-	-	-	-	-	2,154	4,557
	HBM5 16hi(1Gb eq.)	-	-	-	-	-	-	-	3,169
Portion	HBM1(%)	50%	50%	0%	0%	0%	0%	0%	0%
	HBM2(%)	50%	50%	33%	0%	0%	0%	0%	0%
	HBM2E(%)	0%	0%	67%	13%	0%	0%	0%	0%
	HBM3(%)	0%	0%	0%	65%	31%	3%	2%	0%
	HBM3E 8hi(%)	0%	0%	0%	17%	18%	13%	1%	0%
	HBM3E 12hi(%)	0%	0%	0%	5%	50%	55%	19%	10%
	HBM4 8hi(%)	0%	0%	0%	0%	0%	5%	8%	3%
	HBM4 12hi(%)	0%	0%	0%	0%	0%	24%	43%	37%
	HBM4 16hi(%)	0%	0%	0%	0%	0%	0%	0%	0%
	HBM4e 8hi(%)	0%	0%	0%	0%	0%	0%	15%	17%
	HBM4e 12hi(%)	0%	0%	0%	0%	0%	0%	9%	27%
	HBM4e 16hi(%)	0%	0%	0%	0%	0%	0%	3%	4%
	HBM5 16hi(1Gb eq.)	0%	0%	0%	0%	0%	0%	0%	2%
Total DRAM Demand(Mpcs)		191,673	193,221	190,282	246,759	305,484	390,057	510,101	630,035
	HBM portion(%)	0.4%	0.9%	1.8%	4.1%	6.9%	11.9%	14.7%	20.2%

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Source: Citi Research Estimates; TrendForce

NAND Supply/Demand Outlook

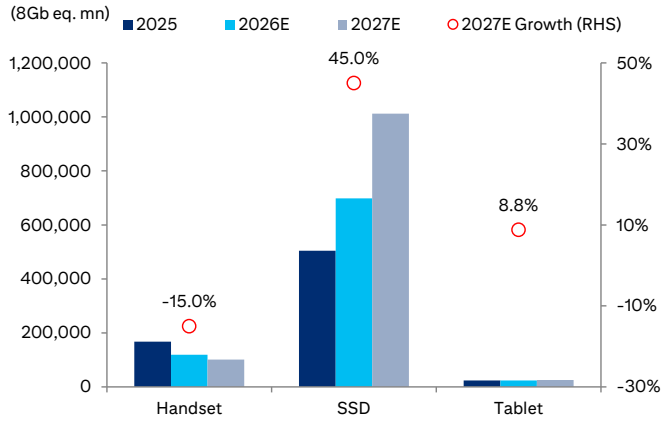
NAND S/D ratio: -6.1% in 27E

We forecast NAND demand to grow +29.1% YoY in 2027E, ahead of supply growth of +21.2% YoY, tightening the NAND S/D ratio to -6.1% from -0.8% in 2026E. We expect demand growth to be increasingly driven by AI inference and eSSDs applications, offsetting relatively softer consumer NAND demand.

NAND demand mainly driven by eSSDs

In particular, we expect eSSDs demand to grow at +52.9% YoY (total SSD demand growth: +45.0% YoY) in 2027E, given strong demand from HBM offloading, as lower HBM content per accelerator shifts a greater portion of KV-cache and other memory workloads toward external storage. Stronger AI inference demand should also continue to support higher-capacity QLC eSSDs adoption at AI servers. This is consistent with recent memory suppliers plans that NAND production is increasingly being allocated toward higher-value eSSDs products despite softer consumer demand.

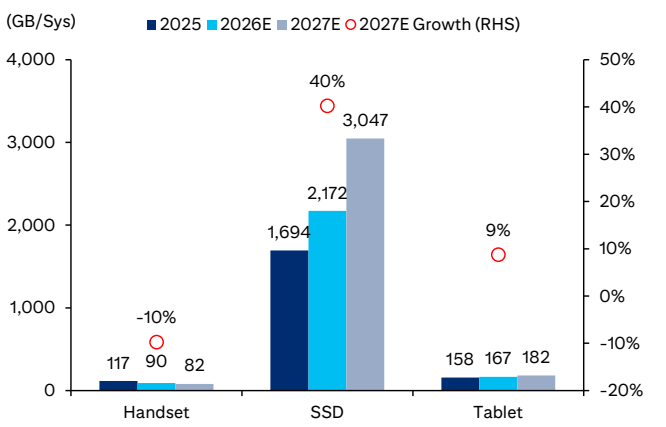
Figure 29. NAND Set Application Forecasts



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Source: DRAMeXchange, Citi Research estimates

Figure 30. NAND Avg. Content Adoption Forecasts



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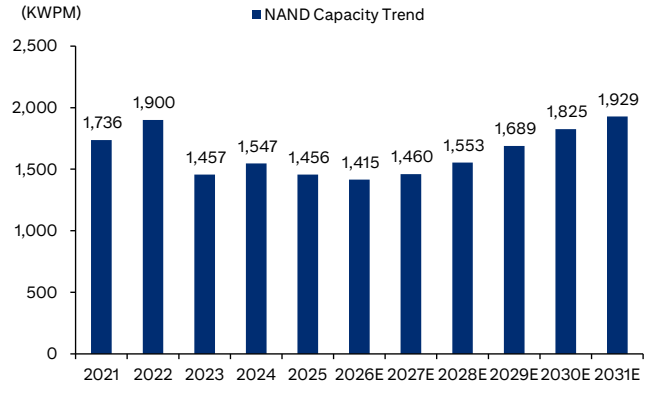
Source: DRAMeXchange, Citi Research estimates

NAND capacity outlook: +3.2% YoY in 27E

Meanwhile, NAND capacity expansion remains relatively disciplined. We forecast average industry average NAND wafer capacity to increase only ~3.2% YoY to 1,460KWPM in 2027E from 1,415KWPM in 2026E. By supplier, we project Samsung Electronics' average NAND capacity to decrease by -4.7% to 331KWPM in 2027E. We forecast average NAND capacity of both Hynix and Micron to remain unchanged in 2027E.

Despite stronger pricing and AI-related demand, we expect suppliers to remain cautious on incremental greenfield NAND capacity, supporting a sustained supply deficit in 2027E.

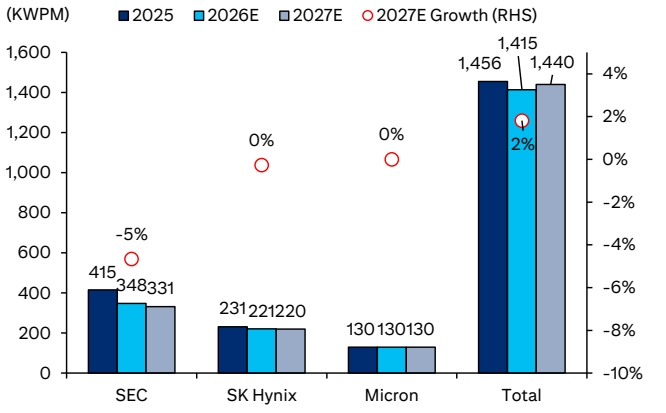
Figure 31. Average NAND Industry Capacity Trend



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Source: DRAMeXchange, Citi Research estimates

Figure 32. NAND Capacity Change by Company



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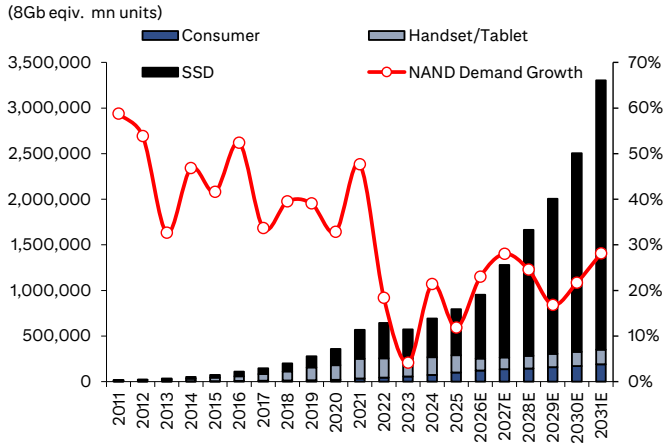
Source: DRAMeXchange, Citi Research estimates

NAND demand growth: +29.1% YoY in 27E

We forecast NAND industry supply/demand growth at +21.2% YoY/+29.1% YoY in 2027E, driven by strong eSSDs demand and limited capex investment. In bit terms, we expect NAND demand to reach 1,486bn units of 8Gb equivalent in 2027E, driven by incremental eSSDs demand on the back of strong AI inference demand and incremental eSSDs demand from KV Cache offloading.

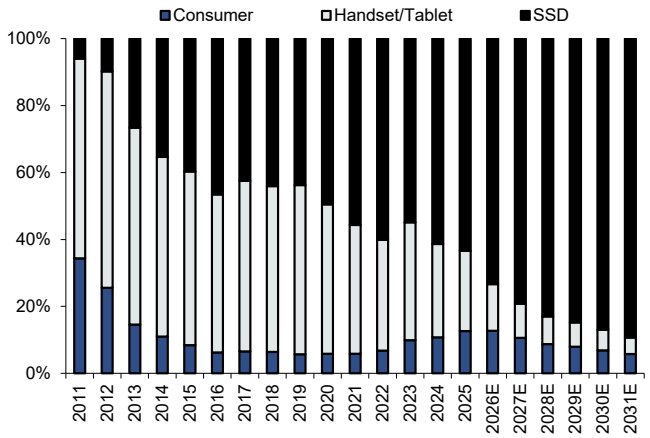
Looking ahead, we expect NAND demand to increasingly favor high-bandwidth, high-density, and AI-optimized solutions. In terms of NAND, we expect growing demand for advanced solutions such as XL Flash and HBF, alongside storage-complementary products such as CXL, as major suppliers position these technologies to utilize for AI workloads. As AI systems require larger pools of storage closer to accelerators to enhance memory retention, we expect near-GPU storage solutions using QLC SSDs to gain traction, reinforcing NAND's expanding role across in AI memory architectures.

Figure 33. NAND Demand Forecast by Application



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Source: DRAMeXchange, Citi Research estimates

Figure 34. NAND Demand Portion by Application

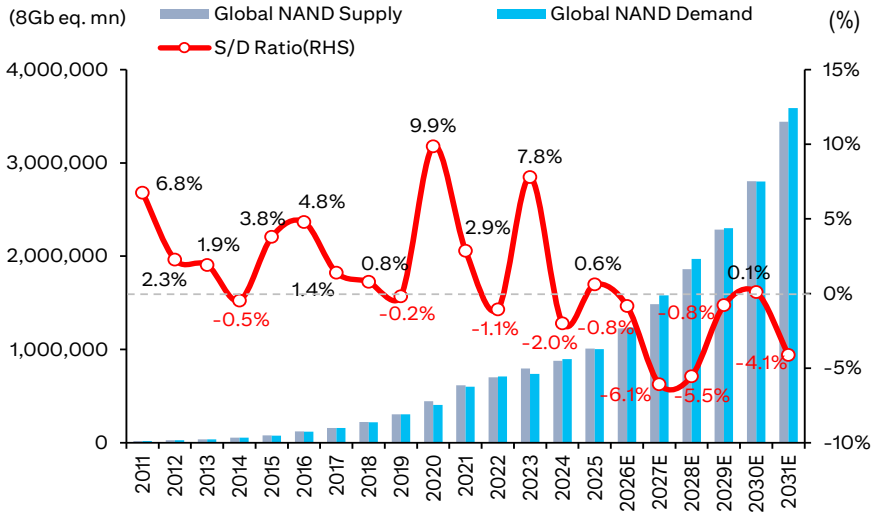


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Source: Citi Research; TrendForce

NAND S/D Ratio -6.1% in 2027E

We forecast the NAND S/D ratio to become more undersupplied, widening to -6.1% in 2027E from -0.8% in 2026E, driven by robust eSSDs demand against limited NAND supply growth on constrained capex.

Figure 35. NAND Supply/Demand Outlook



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Source: DRAMeXchange, Citi Research Estimates

Figure 36. NAND Supply / Demand Forecasts

(8Gb eqiv. mn units)	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	1Q28E	2Q28E	3Q28E	4Q28E	2026E	2027E	2028E
NAND Supply	279,454	297,395	318,953	329,554	325,920	354,904	394,187	410,625	411,362	445,526	493,595	510,598	1,225,356	1,485,636	1,861,081
NAND Demand	285,896	299,454	319,392	330,808	342,803	381,628	416,256	441,104	448,510	485,207	518,871	517,540	1,235,550	1,581,792	1,970,128
NAND S/D Ratio	-2.3%	-0.7%	-0.1%	-0.4%	-4.9%	-7.0%	-5.3%	-6.9%	-8.3%	-8.2%	-4.9%	-1.3%	-0.8%	-6.1%	-5.5%

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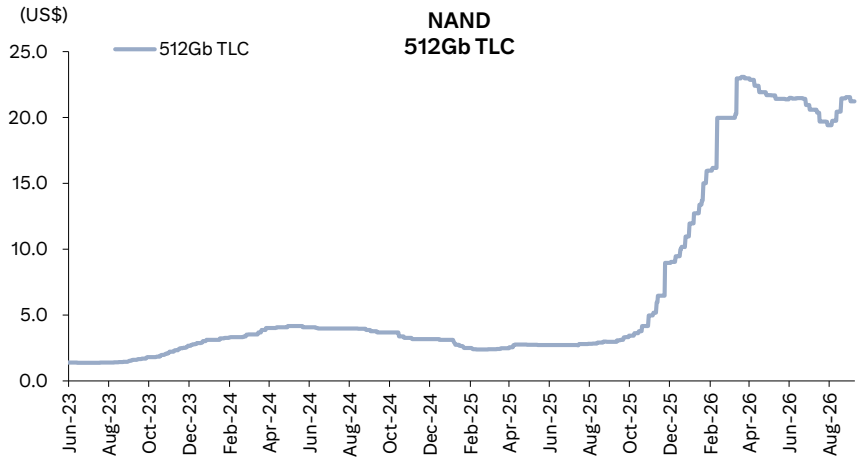
Source: DRAMeXchange, Citi Research Estimates

NAND price outlook

NAND ASP to rise +45.3% YoY in 27E

We also expect NAND pricing to remain constructive as AI-led eSSDs demand strengthens while supply expansion remains limited. By quarter in 2026E, we project blended NAND ASP to grow +34%/ +15% in 3Q/4Q26E. On an annual basis, NAND ASP is expected by us to rise +45.3% YoY in 2027E, following +236.2% YoY in 2026E. Despite weaker consumer NAND demand, sustained eSSDs growth and a tighter industry S/D balance should support continued pricing strength.

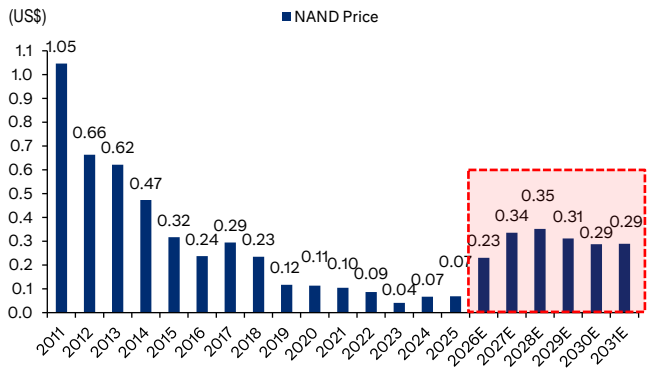
Figure 37. NAND Wafer (512Gb TLC) Spot Price Trend



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Source: Bloomberg, Citi Research; Note: Based on Spectrum's 512Gb spot price index

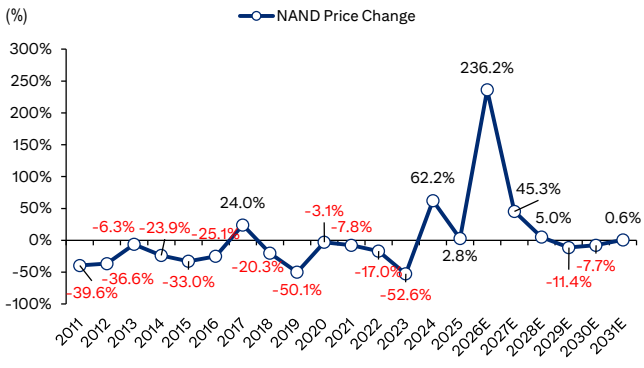
Figure 38. NAND ASP Outlook



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Source: Citi Research; TrendForce

Figure 39. NAND ASP Change



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Source: Citi Research; TrendForce

Total Memory Capex to Increase +46.5% YoY to US$80.4bn in 2027E

Memory WFE Capex to rise +46.5% YoY in 27E

We forecast global memory Capex to increase +46.5% YoY to US$80.4bn in 2027E, accelerating from +39.5% YoY growth to US$54.9bn in 2026E, as memory suppliers respond to structurally stronger AI-driven demand.

DRAM WFE Capex +51.6% YoY in 27E

NAND WFE Capex +34.2% YoY in 27E

We expect DRAM Capex to rise +51.6% YoY to US$58.6bn in 2027E, led by Samsung Electronics at US$20.6bn, SK Hynix at US$17.5bn, and Micron at US$15.8bn. We expect a meaningful portion of incremental DRAM investment to remain directed toward HBM capacity and advanced DRAM production, limiting the supply response in conventional DRAM.

On the NAND side, we forecast Capex to increase +34.2% YoY to US$21.8bn in 2027E, versus US$16.3bn in 2026E. We forecast Samsung/SK Hynix/ Kioxia/Sandisk/Micron NAND Capex at US$7.9bn/US$3.1bn/US$3.7bn/ US$2.9bn/US$2.1bn, respectively. Despite the sizable headline Capex increase, we believe the long lead time for additional capacity means that higher spending should not be sufficient to close the 2027E supply-demand gap.

Figure 40. Total DRAM+NAND CAPEX

Total DRAM+NAND Capex (US$mn)	2025	2026E	2027E	2028E	2029E	2030E	2031E
Total DRAM+NAND Capex	**39,364**	**54,905**	**80,417**	**118,622**	**168,795**	**234,043**	**322,531**
DRAM+NAND Capex YoY %	7.5%	39.5%	46.5%	47.5%	42.3%	38.7%	37.8%

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Source: Companies, Citi Research estimates

Figure 41. DRAM CAPEX by Player

DRAM Capex by Player (US$mn)	2025	2026E	2027E	2028E	2029E	2030E	2031E
Samsung	9,595	13,625	20,574	32,918	48,389	66,293	90,822
SK Hynix	7,211	10,889	17,532	27,174	39,402	54,769	76,129
Micron	7,100	10,630	15,828	22,317	32,360	47,570	66,598
Total DRAM Capex (US$mn)	**26,875**	**38,640**	**58,594**	**89,150**	**129,910**	**182,766**	**254,031**
DRAM Capex YoY %	15.2%	43.8%	51.6%	52.1%	45.7%	40.7%	39.0%

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Source: Companies, Citi Research estimates

Figure 42. NAND CAPEX by Player

NAND Capex by Player (US$mn)	2025	2026E	2027E	2028E	2029E	2030E	2031E
Samsung	4,672	5,466	7,926	11,651	17,477	24,468	34,989
SK Hynix	1,987	2,384	3,099	4,432	6,205	8,997	12,596
Kioxia	2,069	3,200	3,700	4,200	4,500	4,800	5,000
Sandisk	1,324	2,202	2,900	2,999	3,485	4,008	4,609
Micron	1,500	1,650	2,100	2,940	3,969	5,755	8,057
Total NAND Capex (US$mn)	**12,488**	**16,264**	**21,823**	**29,471**	**38,885**	**51,277**	**68,500**
NAND Capex YoY %	-5.9%	30.2%	34.2%	35.0%	31.9%	31.9%	33.6%

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Source: Companies, Citi Research estimates

Companies mentioned in an investment context:

Applied Materials Inc (AMAT.O; US$456.49; 1; 11 Sep 26; 16:00) | EugeneTechnology (084370.KQ; W131500.0; 1; 11 Sep 26; 15:45) | Kioxia Holdings (285A.T; ¥54030.0; 1H; 11 Sep 26; 15:30) | Lam Research Corp (LRCX.O; US$298.22; 1; 11 Sep 26; 16:00) | Micron Technology Inc (MU.O; US$975.26; 1; 11 Sep 26; 16:00) | Montage Technology (6809.HK; HK$258.0; 1; 11 Sep 26; 16:10) | Samsung Electronics (005930.KS; W259500.0; 1; 11 Sep 26; 15:45) | Sandisk Corp (SNDK.O; US$1633.35; 1; 11 Sep 26; 16:00) | SK Hynix (000660.KS; W1812000.0; 1; 11 Sep 26; 15:45) | TechWing (089030.KQ; W46350.0; 1H; 11 Sep 26; 15:45) | TES (095610.KQ; W143500.0; 1; 11 Sep 26; 15:45)

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Appendix A-1

ANALYST CERTIFICATION

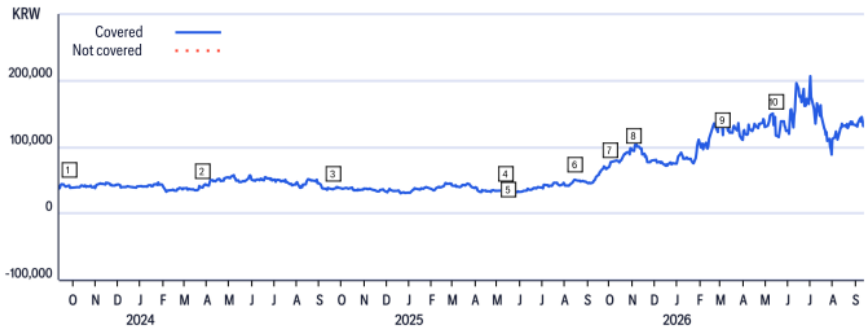
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IMPORTANT DISCLOSURES

EugeneTechnology (084370.KQ)

Ratings and Target Price History
Fundamental Research

Analyst: Peter Lee



	Date	Rating	Target Price	Closing Price
1	26-Sep-23 10:43:43	*3	*35,000.00	42,050.00
2	26-Mar-24 10:32:16	*1	*55,000.00	39,500.00
3	19-Sep-24 18:15:47	1	*60,000.00	36,350.00
4	12-May-25 06:52:03	1	*50,000.00	35,950.00
5	15-May-25 06:21:53	1	*47,000.00	33,350.00
6	14-Aug-25 04:20:21	1	*62,000.00	50,800.00
7	01-Oct-25 10:01:49	1	*90,000.00	72,500.00
8	03-Nov-25 01:31:16	1	*132,000.00	94,100.00
9	04-Mar-26 06:12:02	1	*163,000.00	117,900.00
10	14-May-26 07:37:22	1	*193,000.00	145,200.00

*Indicates Change

Rating/target price changes above reflect Eastern Time

TechWing (089030.KQ)

Ratings and Target Price History
Fundamental Research

Analyst: Jayden Oh



	Date	Rating	Target Price	Closing Price
1	11-Mar-24 17:00:45	*1H	*41,000.00	23,900.00
2	14-Apr-24 18:50:06	1H	*49,000.00	38,600.00
3	14-Jul-24 16:00:12	1H	*90,000.00	65,700.00
4	14-Jul-25 06:42:58	1H	*70,000.00	29,000.00
5	21-Oct-25 10:29:44	1H	*80,000.00	63,400.00

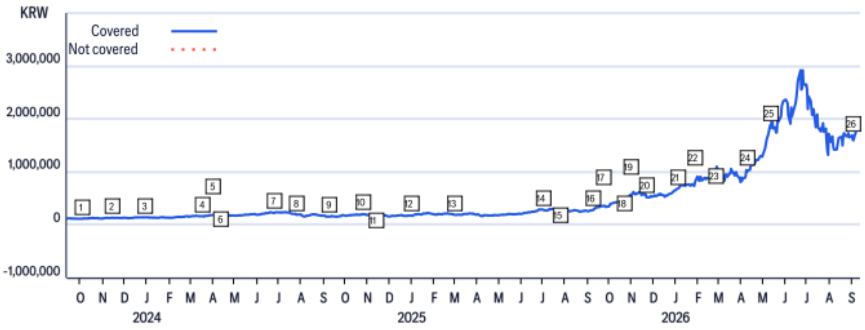
*Indicates Change

Rating/target price changes above reflect Eastern Time

SK Hynix (000660.KS)

Ratings and Target Price History
Fundamental Research

Analyst: Peter Lee



	Date	Rating	Target Price	Closing Price
1	04-Oct-23 06:14:16	1	*185,000.00	115,400.00
2	13-Nov-23 09:52:50	1	*190,000.00	131,800.00
3	01-Jan-24 18:20:58	1	*230,000.00	141,500.00
4	18-Mar-24 10:47:11	1	*234,000.00	164,300.00
5	01-Apr-24 10:34:31	1	*238,000.00	185,500.00
6	12-Apr-24 08:17:40	1	*310,000.00	187,400.00
7	25-Jun-24 06:27:18	1	*350,000.00	225,000.00
8	25-Jul-24 03:44:27	1	*337,000.00	190,000.00
9	09-Sep-24 02:31:32	1	*310,000.00	157,000.00
10	24-Oct-24 06:29:30	1	*330,000.00	198,200.00
11	12-Nov-24 10:31:16	1	*350,000.00	185,800.00
12	31-Dec-24 06:45:21	1	*340,000.00	173,900.00
13	03-Mar-25 15:00:11	1	*350,000.00	190,200.00
14	01-Jul-25 03:45:10	1	*430,000.00	285,500.00
15	24-Jul-25 04:40:40	1	*380,000.00	269,500.00
16	08-Sep-25 10:00:23	1	*430,000.00	277,000.00
17	22-Sep-25 06:19:28	1	*480,000.00	351,000.00
18	20-Oct-25 04:22:50	1	*640,000.00	485,500.00
19	29-Oct-25 02:38:54	1	*770,000.00	558,000.00
20	23-Nov-25 10:20:04	1	*830,000.00	521,000.00
21	02-Jan-26 03:22:42	1	*900,000.00	677,000.00
22	26-Jan-26 07:57:46	1	↑1,400,000.00	736,000.00
23	24-Feb-26 08:08:02	1	↑1,550,000.00	1,005,000.00
24	08-Apr-26 07:47:52	1	↑1,700,000.00	1,033,000.00
25	11-May-26 07:42:14	1	↑3,100,000.00	1,880,000.00
26	01-Sep-26 04:40:27	1	↑3,000,000.00	1,693,000.00

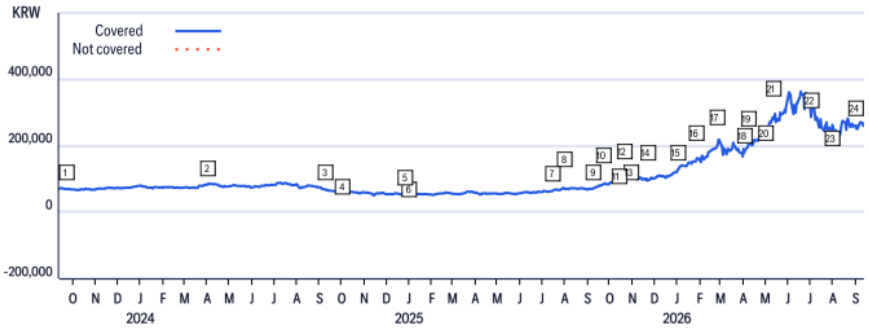
*Indicates Change

Rating/target price changes above reflect Eastern Time

Samsung Electronics (005930.KS)

Ratings and Target Price History
Fundamental Research

Analyst: Peter Lee



	Date	Rating	Target Price	Closing Price
1	22-Sep-23 09:53:22	1	*110,000.00	68,800.00
2	01-Apr-24 05:23:20	1	*120,000.00	82,000.00
3	09-Sep-24 02:31:35	1	*110,000.00	67,500.00
4	02-Oct-24 05:16:51	1	*97,000.00	61,300.00
5	26-Dec-24 07:42:50	1	*87,000.00	53,600.00
6	31-Dec-24 06:21:22	1	*83,000.00	53,200.00
7	16-Jul-25 05:08:04	1	*90,000.00	64,700.00
8	31-Jul-25 02:49:57	1	*100,000.00	71,400.00
9	08-Sep-25 09:30:31	1	*110,000.00	70,100.00
10	22-Sep-25 05:57:03	1	*120,000.00	83,500.00
11	14-Oct-25 07:05:40	1	*133,000.00	91,600.00
12	20-Oct-25 06:25:10	1	*145,000.00	98,100.00
13	30-Oct-25 11:44:04	1	*150,000.00	104,100.00
14	23-Nov-25 10:07:18	1	*170,000.00	94,800.00
15	02-Jan-26 02:41:08	1	*200,000.00	128,500.00
16	26-Jan-26 09:32:20	1	*240,000.00	152,100.00
17	24-Feb-26 05:11:17	1	*280,000.00	200,000.00
18	02-Apr-26 07:54:27	1	*300,000.00	178,400.00
19	07-Apr-26 02:31:16	1	*320,000.00	196,500.00
20	30-Apr-26 03:49:59	1	*300,000.00	220,500.00
21	11-May-26 07:42:55	1	*460,000.00	285,500.00
22	02-Jul-26 03:16:27	1	*530,000.00	286,000.00
23	30-Jul-26 04:00:39	1	*450,000.00	207,000.00
24	31-Aug-26 08:46:42	1	*430,000.00	260,000.00

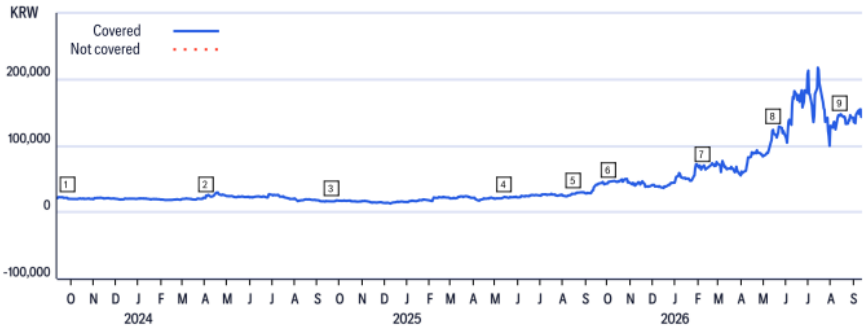
*Indicates Change

Rating/target price changes above reflect Eastern Time

TES (095610.KQ)

Ratings and Target Price History
Fundamental Research

Analyst: Peter Lee



	Date	Rating	Target Price	Closing Price
1	26-Sep-23 10:43:43	*2	*23,000.00	21,700.00
2	01-Apr-24 07:15:32	*1	*35,000.00	21,400.00
3	19-Sep-24 18:15:47	1	*31,500.00	16,400.00
4	12-May-25 07:04:44	1	*32,000.00	22,200.00
5	14-Aug-25 10:38:10	1	*37,000.00	27,550.00
6	01-Oct-25 10:29:58	1	*57,000.00	43,600.00
7	08-Feb-26 16:10:33	1	*100,000.00	67,400.00
8	14-May-26 06:35:59	1	*160,000.00	123,900.00
9	13-Aug-26 08:15:21	1	*180,000.00	145,000.00

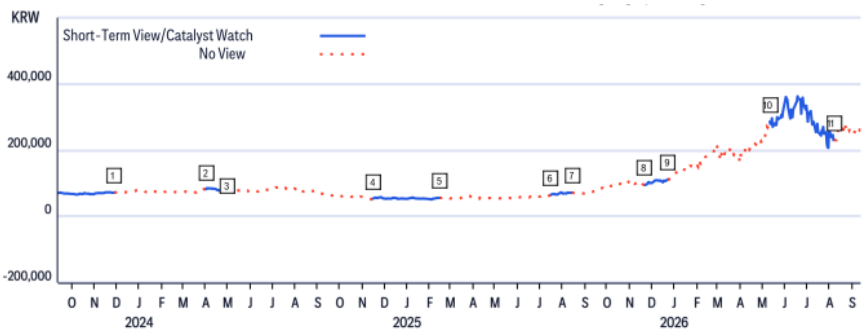
*Indicates Change

Rating/target price changes above reflect Eastern Time

Samsung Electronics (005930.KS)

Short-Term View/Catalyst Watch Research

Analyst: Peter Lee



	Date	Action	Expected Direction	Duration	Closing Price
1	28-Nov-23 21:19:48	Remove CW	Upside	90 Days	72,700.00
2	01-Apr-24 01:23:20	Add CW	Upside	30 Days	82,000.00
3	30-Apr-24 22:52:33	Remove CW	Upside	30 Days	77,500.00
4	17-Nov-24 11:09:02	Add CW	Upside	90 Days	53,500.00
5	14-Feb-25 12:17:17	Remove CW	Upside	90 Days	56,000.00
6	16-Jul-25 01:08:04	Add CW	Upside	30 Days	64,700.00
7	15-Aug-25 14:07:20	Remove CW	Upside	30 Days	71,600.00
8	23-Nov-25 05:07:18	Add CW	Upside	30 Days	94,800.00
9	23-Dec-25 20:51:38	Remove CW	Upside	30 Days	111,500.00
10	11-May-26 03:42:55	Add CW	Upside	90 Days	285,500.00
11	09-Aug-26 23:05:45	Remove CW	Upside	90 Days	231,000.00

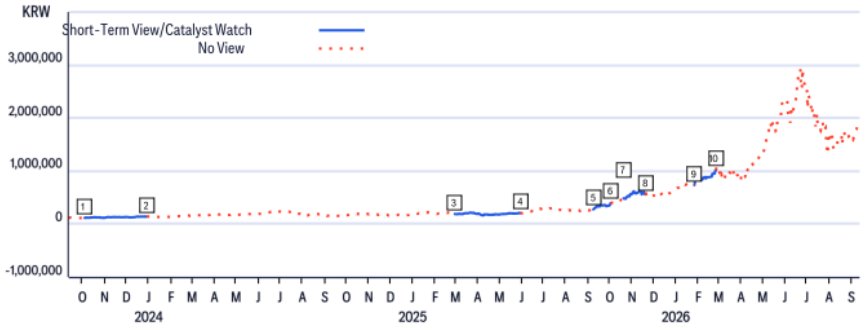
CW - Catalyst Watch , STV - Short-Term View

Rating/target price changes above reflect Eastern Time

SK Hynix (000660.KS)

Short-Term View/Catalyst Watch Research

Analyst: Peter Lee



	Date	Action	Expected Direction	Duration	Closing Price
1	04-Oct-23 02:14:16	Add CW	Upside	90 Days	115,400.00
2	01-Jan-24 21:19:41	Remove CW	Upside	90 Days	141,500.00
3	03-Mar-25 10:00:11	Add CW	Upside	90 Days	190,200.00
4	01-Jun-25 22:55:32	Remove CW	Upside	90 Days	204,500.00
5	08-Sep-25 06:00:23	Add CW	Upside	30 Days	277,000.00
6	08-Oct-25 22:50:00	Remove CW	Upside	30 Days	395,500.00
7	20-Oct-25 00:22:50	Add CW	Upside	30 Days	485,500.00
8	19-Nov-25 21:15:03	Remove CW	Upside	30 Days	562,000.00
9	26-Jan-26 02:57:46	Add CW	Upside	30 Days	736,000.00
10	25-Feb-26 20:57:35	Remove CW	Upside	30 Days	1,018,000.00

CW - Catalyst Watch , STV - Short-Term View

Rating/target price changes above reflect Eastern Time

The Firm has made a market in the publicly traded equity securities of Montage Technology Co., Ltd. on at least one occasion since 1 Jan 2025.

Citigroup Global Markets Inc. or its affiliates has a net short position of 0.5% or more of any class of common equity securities of Kioxia Holdings.

Within the past 12 months, Citigroup Global Markets Inc. or its affiliates has acted as manager or co-manager of an offering of securities of Applied Materials Inc,SK Hynix,Samsung Electronics,Sandisk Corp.

Citigroup Global Markets Inc. or its affiliates has received compensation for investment banking services provided within the past 12 months from Applied Materials Inc,Micron Technology Inc,SK Hynix,Samsung Electronics,Sandisk Corp.

Citigroup Global Markets Inc. or its affiliates expects to receive or intends to seek, within the next three months, compensation for investment banking services from SK Hynix,Samsung Electronics,Sandisk Corp.

Citigroup Global Markets Inc. or its affiliates received compensation for products and services other than investment banking services from Applied Materials Inc,Kioxia Holdings,Lam Research Corp,Micron Technology Inc,Montage Technology,SK Hynix,Samsung Electronics,Sandisk Corp,TechWing in the past 12 months.

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ADDITIONAL INFORMATION IS AVAILABLE UPON REQUEST

BERNSTEIN
SOCIETE GENERALE GROUP

3 September 2026

Global Memory

MEMORY TRACKER (Aug): Conventional DRAM & NAND price to rise nearly 20% in 3Q26


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*This monthly tracker (download dataset **here**) summarizes contract/spot price data released by TrendForce/DRAMeXchange & compares that with our model & investor expectations.*

DRAM: Aug contracts was up LSD% MoM and suggested19% conventional DRAM price increase QoQ in 3QCY26.

Spot price: DRAM spot price remained robust in Aug. PC DRAM spot price rose by 5-6% MoM while server DDR5 was also up 3.9% MoM.

Contract price: DRAM contract prices in Aug rose moderately MoM & indicated 3QCY26 contract price to be 19% higher than 2QCY26, with 22% for PC, 17% for Server, 11% for Mobile & ~33% for Consumer. The size of the price increase is ahead of our model, but likely remained a bit weaker than Street expectations. It was partially due to LTA price ceilings, which in some cases memory vendors were trying to renegotiate. Overall server demand remained very robust despite moderation in PC and mobile demand. There were some signs of de-specing in server DRAM, but that should be offset by higher unit shipment, and should not impact the memory shortage and pricing.

NAND: wafer contract price remained sluggish despite some strength in spot price market

Spot price: NAND wafer spot price was quite strong with 1Tb rising by 23% MoM. 512Gb also rebounded by 7.6% MoM.

Contract price: Despite the strength in spot price, wafer contract price was up only slightly in Aug, as module makers remained very concerned about consumer demand. Transaction volume was also extremely low. And they were buying lower grade chips to save on costs. Mobile NAND price increase was more solid at ~20% QoQ in 3Q26, but the size of the increase also substantially smaller vs. 2Q as OEMs cut back on handset production & thus memory usage. Most importantly we believe client & enterprise SSD price is rising ~20% QoQ, making the overall NAND price hike close to 20% in 3Q. That however could be still a bit below the latest expectation.

In summary, recent pullback should offer a short-term technical rebound. Contract price data suggests conventional DRAM & NAND contract prices to rise by close to 20% QoQ in 3QCY26. The size of the increase narrowed notably vs. 2Q, and may be a bit below the latest Street expectation, but still very strong on a longer term historical context. Shortage should persist into CY27 to keep prices elevated, especially for DRAM, but the pace of further price rises will be certainly much smaller, given memory is becoming a burden for both AI & non-AI applications & price caps set in some LTAs. That said the recent pullback still opens an opportunity of a near-term rebound. See **Global Memory: Memory becoming a burden of AI too?** & **Memory: What do New Memory LTAs mean for sustainability of earnings and multiples? Raising SNDK TP to $3000** & recent earnings review (**Samsung**, **SK hynix**, **Micron**, **KIOXIA**, **SNDK**) for details.

See the Disclosure Appendix of this report for required disclosures, analyst certifications and other important information. Alternatively, visit our Global Research Disclosure Website.

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BERNSTEIN TICKER TABLE

			2 Sep 2026		TTM	Reported EPS				Reported P/E (x)	
Ticker	Rating	Cur	Closing Price	Price Target	Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
005930.KS (SEC- Samsung)	O	KRW	248,500	440,000	226.3%	KRW	6,611.53	48,393	77,273	37.6	5.1	3.2
005935.KS (SEC-Pref - Samsung)	O	KRW	185,500	374,000	192.1%	KRW	6,611.53	48,393	77,273	28.1	3.8	2.4
SMSN.LI (Samsung)	O	USD	4,636.00	7,350.00	238.4%	USD	116.15	812.39	1,290.80	39.9	5.7	3.6
000660.KS (SK hynix)	O	KRW	1,597,000	3,300,000	477.5%	KRW	60,341	395,677	568,862	26.5	4.0	2.8
MU (Micron)	O	USD	956.08	1,300.00	685.8%	USD	8.29	67.39	158.99	115.4	14.2	6.0
285A.JP (KIOXIA)	U	JPY	51,670	40,000	1840.2%	JPY	1,014.00	10,013	9,656.84	51.0	5.2	5.4
SNDK (SanDisk)	O	USD	1,553.40	3,000.00	2810.9%	USD	70.08	238.36	272.09	22.2	6.5	5.7
ASIAX			1,949.13									
EM			1,815.57									
SPX			7,666.60									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

MU, SNDK estimate is Adjusted EPS; MU, SNDK valuation is Adjusted P/E (x); SNDK base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

Samsung Electronics: We rate Samsung Electronics Outperform with price target of KRW 440,000.

SK hynix : We rate SK hynix Outperform with price target of KRW 3,300,000.

Micron: We rate Micron Outperform with price target of US$1,300.00.

KIOXIA: We rate KIOXIA Underperform with price target of JPY 40,000.

SanDisk: We rate SNDK Outperform with a price target of $3,000.00

DETAILS

*TrendForce (who publishes research on memory market & also maintains DRAMeXchange for memory spot market) has released the Aug 2026 contract prices for DRAM & NAND covering different applications (dataset can be downloaded at **Memory Price Tracker**). We select mainstream products from each segment, apply different weights on them to derive an industry average price change. SSD & HBM are not in the samples but will be additionally considered so that we can compare the overall ASP with our projection. Spot market is probably only 10% of the overall volume and price is also noisy, but we also summarize spot price data as it can be a leading indicator of contract price. Please feel free to find our **most recent market review** and download our industry and company models from the links below.*

- ***CoWoS/HBM Model***
- ***DRAM Industry Model***
- ***NAND Industry Model***
- ***Samsung Electronics Model***
- ***SK hynix Model***
- ***Micron Model***
- ***KIOXIA Model***
- ***SanDisk Model***

Aug contract prices went up LSD% MoM and suggested ~19% QoQ rise for 3QCY26 DRAM contract price, slightly more than indicated by Jul contract price.

- **Spot price:**
  - **PC DRAM**: PC DRAM chip spot price continued to be robust in Jul, rising by 5.8% MoM for DDR5 and 5.4% for DDR4 (Exhibit 1, Exhibit 3, Exhibit 4, Exhibit 5, Exhibit 6). As DRAM shortage is widely expected to continue into 2027 and may even worsen, module makers therefore generally were not worried about inventory risk and continued to buy DRAM.
  - **Server DRAM**: Server DRAM module spot price was also largely solid, with DDR5 rising by another 3.9% MoM, even though DDR4 fell slightly by 1% MoM (Exhibit 1, Exhibit 11, Exhibit 12). DDR5 module spot price remained significantly above contract price by about 100%, suggesting server DRAM shortage continued to be very acute.
- **Contract price**:
  - **Average**: Overall, taking a weighted average of multiple samples, we find contracts set in Aug indicated conventional DRAM prices to rise by 19% QoQ in 3QCY26. This is better than indicated by Jul contract price and a tad ahead of our model, but likely still somewhat below latest Street expectation. The rate of the increase also narrowed notably vs. 2Q. Please note this is only for conventional DRAM, and the actual company reported ASP may differ due to changing mix among PC, Mobile, server, etc., and also most importantly the balance from steadier HBM price that is largely set by annual contracts (Exhibit 1). Thus, the weighted average contract price increase here may not match the increase reported in company earnings releases but does serve as an important indication for where the market is heading.
  - **PC DRAM**: Contract price in Aug was up by 2-4% MoM, and indicated 3QCY26 contract price to be 22% higher QoQ (Exhibit 1, Exhibit 3, Exhibit 7). Price negotiation for 3QCY26 largely concluded in Aug. PC OEMs slightly increased production as CPU supply shortage eased. Though overall units were still expected to see meaningful QoQ and YoY decline in 3QCY26, due to memory cost. Going forward, PC OEMs are aware of potentially worsening DRAM shortage in

CY27. But they are unlikely to continue bidding for more supply from vendors or module makers, due cost pressure, and instead is likely to be more focused on conservation by lowering production of low-end models.

  - **PC DDR4:** PC DDR4 prices rose a bit less than DDR5 in Aug and DDR5's discount vs DDR4 narrowed slightly to 9% (Exhibit 5,Exhibit 8,Exhibit 9,Exhibit 10). DDR4 price has been even more robust than DDR5 as supply base largely shifts to Taiwanese makers led by Nanya (2408 TT, not covered). On the other hand demand remains quite resilient from long-tailed customers from consumer segments, such as low-end PC and eSSD.

- **Server DRAM:** Aug server DRAM contract price was up slightly by 1-2% MoM, and still suggested 3QCY26 price to be 17% higher than 2QCY26 (Exhibit 1, Exhibit 11, Exhibit 13). Memory vendors finished supply contract discussions with more OEM customers, and prices continued to be higher vs LTA covered CSP customers. And to boost pricing, vendors were trying to restart negotiation with some LTA customers to increase the ceiling price or charge higher prices on add-on volumes not under LTA. As of Aug these discussions were still ongoing. Another noteworthy observation from TrendForce is that CSPs were also "optimizing" memory usage given the supply constraints, by opting for lower capacity modules (32/64GB), which potentially can support more server units, and also lead to better TCO and ROI. This has led to shrinking of price premium of 128GB modules. But due to the overall bit supply constraint, this likely will not impact overall DRAM pricing environment.

  - **Server DDR4:** Server DDR4 contract price was unchanged (Exhibit 12,Exhibit 14). DDR5's premium over DDR4 went up slightly to 16% (Exhibit 15). Server DDR4 likely will remain in shortage with supply EOL from large suppliers while some legacy applications continue to require it. However, DDR5 was likely already over 90% of the market into 3QCY26 and hence should be the focus.

- **Mobile DRAM:** TrendForce updates Mobile DRAM contract price data only in the first month of a quarter. By the end of Jul, 3QCY26 contract price negotiation was still ongoing, but TrendForce estimates mobile DRAM contract price to increase by ~10% QoQ in 3QCY26, with SK hynix and CXMT potentially seeing more increases over Samsung, as their prices were behind in 2Q (Exhibit 1, Exhibit 16, Exhibit 17, Exhibit 18). Mobile OEM customers were notably more reluctant than before to accept price hikes, after they meaningfully reduced their smartphone production plans. As a result similar to PC DRAM, mobile DRAM will see the size of price hike notably more moderate in 3Q vs. 2Q. Having said that OEMs were still keen to have supply contracts in place to secure memory supply, at least for the next year. Into 4QCY26, TrendForce expects DRAM demand from smartphone OEMs to remain subdued due to reduced production, and as a result price increase to further narrow to single digit % QoQ.

- **Specialty DRAM:** Specialty DRAM contract price moved up by ~5% MoM in Aug, and Jul contracts suggested 3QCY26 to be 30-36% higher vs. 2QCY26 (Exhibit 1, Exhibit 19). No sign of easing of supply shortage in the consumer market either. TrendForce notes that some customers were also starting to enter LTAs hoping to secure supply, and for Nanya, LTA coverage had already reached 50% in 2Q. TrendForce also notes that SSDs are also growing as a main application of consumer DRAM, and along with server and networking, have taken over traditional applications like TV and set-top box as the main driver of consumer DRAM demand.

- **In summary, Aug contract prices went up LSD% MoM and suggested ~19% QoQ rise for 3QCY26 DRAM contract price, slightly more than indicated by Jul contract price. The rate of price increase is slowing down but there is no sign of easing shortage at all. On the contrary the supply shortage is widely expected to become more severe in CY27. Besides PC and mobile, there were also some signs of optimization or despecing of memory in server DRAM. But likely it will be offset by higher unit shipment and hence likely won't impact the memory shortage. Overall we expect the shortage situation to last into CY27 and pricing to continue growing, though at a more moderate pace vs before. See Global Memory: Memory becoming a burden of AI too? & our earnings reviews (Samsung, SK hynix, Micron & KIOXIA) for details.**

In Aug, NAND wafer contract price remained sluggish despite some strength in spot price market.

- **Spot price**:

  - **NAND Wafer:** Aug NAND wafer spot price saw robust rebound. 512Gb wafer price bounced back by 7.6% MoM while 1Tb wafer soared by 23% MoM (Exhibit 2, Exhibit 20, Exhibit 21). 1Tb wafer spot price also surpassed contract price, which was a first since early 2025. We wonder if the strength from 1Tb wafer was due to more eSSD demand flowing to module houses.

- **Contract price:**

  - **Average**: Taking the average of multiple samples of NAND wafer and eMMC/UFS, we find contracts set in Aug indicated just 9.4% QoQ NAND price increase from 2QCY26 to 3QCY26 (Exhibit 2). We highlight this is an underestimate due to over representation of wafers and omission of SSDs in our sample. While wafer price is slowing, we believe price hike of SSDs in 3QCY26 likely will around 20% for both client and enterprise, and therefore the blended NAND contract price increase likely should approach 20% as well. And this is mostly in line with our model, though possibly below some latest expectations. Still actual company reported 3QCY26 ASP may differ as product mix changes and timing difference between contract and shipment can have a significant impact on blended ASP.

  - **NAND Wafer:** NAND wafer contract price was up slightly by 1-3% MoM in Aug. Compared to the average in 2QCY26, Aug contract price was also only mildly higher by 1-5% (Exhibit 20, Exhibit 21, Exhibit 22, Exhibit 23). Despite the strength in spot market, TrendForce observed no change in demand from module houses. TrendForce noted that module houses were purchasing more lower grade NAND chips, which is still considered adequate for consumer markets, mainly to save on costs. But overall they remained very cautious on their purchase on the soft consumer market demand, whereas wafer supply remained very tight as well.

  - **eMMC/UFS**: TrendForce expects 3QCY26 Mobile NAND contract price to rise by ~20% QoQ (Exhibit 24,Exhibit 25). Similar dynamics was in play as in PC & Mobile DRAM. Customers were fighting harder on price increase as they cut production volume & thus memory need. Suppliers still tried to get prices higher to match profitability of eSSD. Overall eMMC/UFS price grew a bit more than other memories mainly to catch up. TrendForce also observed that while global suppliers were more firm on price hikes, Chinese supplier (i.e. YMTC) was not asking as much price increase, and hence won more volume. Going forward, TrendForce expects price hikes will likely become more difficult and moderate in 4QCY26.

- **Overall NAND wafer spot price saw notable strength but contract price remained sluggish and module house generally remained very passive on buying NAND wafers. Mobile NAND price was relatively better but also at a decelerating pace vs. 2Q. Overall 3QCY26 NAND contract price increase likely is a bit weaker than hoped, even as the price rise of eSSD is better. We think price increase will continue to moderate going forward and perhaps reach a peak some time in CY27. Structurally we remain more cautious on NAND as well, due to more competition from China (details in our deep dive on YMTC).**

Our industry average is derived from an average of multiple memory products used in different applications, weighted by their contributions to the overall market, and with the weights on DDR5 & LPDDR5 regularly adjusted to reflect its penetration. HBM & SSD are not in our samples & are considered separately.

- Our average DRAM contract price change mentioned above is a weighted average of multiple applications, with Mobile, Server, PC and Specialty DRAM accounting for 34%, 34%, 9% and 15%, respectively. And within PC DRAM, we include both DRAM chips and DRAM module prices in our samples, assuming 10%/90% weight for chips/modules (Exhibit 26). Additionally, the weights assigned to DDR5 & LPDDR5 will be adjusted according to their adoptions. For NAND, the average is based on a weighted average of eMMC/UFC (40%) and NAND wafer (60%) (Exhibit 27). We then additionally discuss the influence of HBM & SSD as they're not in our samples.

EXHIBIT 1: **Aug DRAM contract price indicated 19% QoQ increase for conventional DRAM price in 3QCY26.**

		Price at End of Month						Quarterly Avg		
					MoM				3QCY26	
DRAM		Jun-26	Jul-26	Aug-26	Jun-26	Jul-26	Aug-26	2QCY26	QTD	QoQ
Spot Price (US$/GB)										
PC DRAM	DDR4 8Gb Chip	36.0	42.1	44.4	5.6%	17.0%	5.4%	33.4	41.6	24.5%
	DDR5 16Gb Chip	23.4	25.5	27.0	11.8%	8.8%	5.8%	20.1	25.6	27.3%
Server DRAM	DDR4 64GB RDIMM*	24.5	22.9	22.7	6.1%	-6.7%	-1.0%	23.1	23.2	0.5%
	DDR5 64GB RDIMM*	44.9	46.6	48.4	26.4%	3.8%	3.9%	38.1	47.2	24.0%

		Prior Price			Price for 3QCY26			3QCY26 Over 2QCY26		
DRAM		1QCY26	2QCY26	QoQ	Jul-26	Aug-26	Sep-26	Jul-26	Aug-26	Sep-26
Contract Price (US$/GB, except the price of uMCP & eMCP is module price)										
PC DRAM	DDR4 8Gb Chip	12.5	19.0	52.0%	24.0	25.0		26.3%	31.6%	
	DDR5 16Gb Chip	14.9	18.8	25.7%	22.5	23.3		20.0%	24.0%	
	DDR5 16GB UDIMM*	8.7	12.8	47.8%	14.9	15.4		16.1%	20.5%	
	DDR4 16GB UDIMM*	10.3	14.2	37.6%	16.6	16.9		16.7%	18.9%	
	DDR4 8GB UDIMM*	10.6	14.9	40.0%	17.4	17.8		16.8%	19.3%	
Server DRAM	DDR5 96GB RDIMM*	13.9	21.5	54.9%	24.8	25.1		15.7%	16.7%	
	DDR5 64GB RDIMM*	13.0	20.1	55.5%	23.2	23.4		15.2%	16.3%	
	DDR5 48GB RDIMM*	13.8	21.5	55.5%	24.8	25.0		15.2%	16.3%	
	DDR4 64GB RDIMM*	11.5	17.6	52.7%	20.2	20.2		14.9%	14.9%	
Mobile DRAM	LPDDR5 16GB Discrete	6.3	11.4	80.6%		12.7			10.7%	
	LPDDR5 12GB Discrete	6.3	11.4	80.2%		12.7			10.7%	
	LPDDR4X 8GB Discrete	6.2	10.9	76.9%		12.2			11.9%	
	LPDDR4X 4GB Discrete	6.2	10.9	77.4%		12.3			11.9%	
	uMCP (LPDDR4X 64Gb+NAND 128GB)**	65.7	119.4	81.9%		131.5			10.1%	
	eMCP (LPDDR4X 32Gb+NAND 64GB)**	34.8	76.6	120.0%		78.4			2.3%	
Specialty DRAM	DDR4 4Gb	12.7	23.3	84.2%	29.0	30.4		24.3%	30.3%	
	DDR3 4Gb	12.0	22.3	86.1%	29.0	30.4		29.9%	36.1%	
Weighted Average		**10.2**	**16.8**	**65.0%**	**19.6**	**20.0**		**17.0%**	**19.1%**	
Bernstein Estimate (Blended ASP, inc HBM)****										
Samsung		9.5	14.9	56.7%		16.5			10.3%	
SK hynix		8.3	12.5	50.7%		13.9			11.1%	
Micron***		7.1	10.7	50.4%		12.3			15.4%	
Weighted Average		**8.5**	**13.0**	**53.2%**		**14.5**			**11.6%**	

*: DIMM = Dual In-line Memory Module; UDIMM (Unbuffered DIMM) is the DIMM mostly for desktop PC; RDIMM (Registered DIMM) is mostly for server
**: eMCP (Embedded Multi-Chip Package) is mostly used in mobile applications & is a package standard that extends the eMMC standard (which consists of NAND & controller) to include DRAM too in the package. uMCP (UFS + MCP) is similar to eMCP but replace eMMC with UFS. Their prices include the DRAM, NAND & controller in the package but are classified as part of DRAM statistics as DRAM contributes more value.
***: Micron's quarter ends a month earlier.
****: Our models were last updated in Jun 2026.

Source: DRAMeXchange, Bernstein estimates and analysis

EXHIBIT 2: **Our sample for NAND only includes wafer and eMMC/UFS, which together rose by 9.4% QoQ in Aug. However we understand SSD contract price likely is also rising growing 20% QoQ, and with that included overall NAND contract price likely increased can also rise by close to 20% QoQ in 3QCY26.**

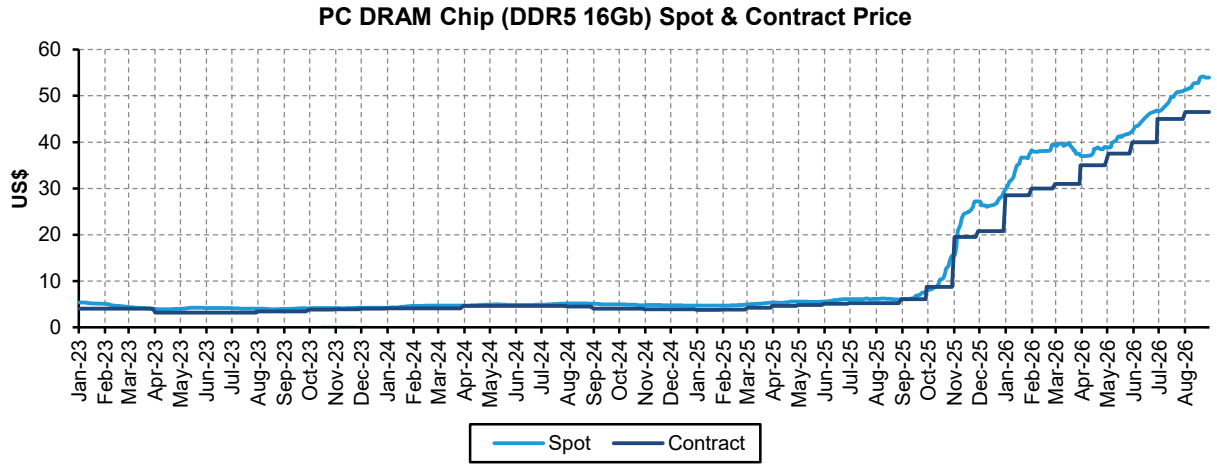
		Price at End of Month						Quarterly Avg		
					MoM					
NAND		Jun-26	Jul-26	Aug-26	Jun-26	Jul-26	Aug-26	2QCY26	3QCY26	QoQ
Spot Price (US$/GB)										
NAND Wafer	1Tb 3D TLC/QLC	0.190	0.206	0.254	-3.3%	8.2%	23.4%	0.197	0.222	12.7%
	512Gb 3D TLC	0.310	0.301	0.324	-3.8%	-3.1%	7.6%	0.322	0.314	-2.7%

		Prior Price			Price for 3QCY26			3QCY26 Over 2QCY26		
NAND		1QCY26	2QCY26	QoQ	Jul-26	Aug-26	Sep-26	Jul-26	Aug-26	Sep-26
Contract Price (US$/GB)										
NAND Wafer	1Tb 3D QLC	0.191	0.235	23.1%	0.237	0.237		0.8%	0.8%	
	1Tb 3D TLC	0.197	0.246	24.5%	0.254	0.257		3.3%	4.6%	
	512Gb 3D TLC	0.312	0.396	27.2%	0.403	0.413		1.6%	4.2%	
eMMC/UFS*	512GB UFS	0.132	0.238	80.0%		0.284			19.1%	
	256GB UFS	0.131	0.236	80.0%		0.286			21.3%	
	128GB TLC eMMC	0.131	0.231	75.4%		0.280			21.2%	
Weighted Average		**0.193**	**0.269**	**39.9%**	**0.292**	**0.295**		**8.4%**	**9.4%**	
Bernstein Estimate (Blended ASP, inc eSSD)****										
Samsung		0.156	0.259	66.0%		0.316			22.0%	
SK hynix		0.168	0.279	66.0%		0.341			22.0%	
Micron**		0.141	0.232	65.0%		0.302			30.0%	
KIOXIA		0.107	0.178	66.0%		0.217			22.0%	
SanDisk		0.242	0.314	29.9%		0.343			9.0%	
Weighted Average		**0.155**	**0.248**	**59.3%**		**0.300**			**21.1%**	

*: eMMC (Embedded Multi Media Card) is a standard that packages NAND dies & controller together. UFS (Universal Flash Storage) is an enhancement over eMMC.
**: Micron's quarter ends a month earlier.
****: Our models were last updated in Jun 2026.

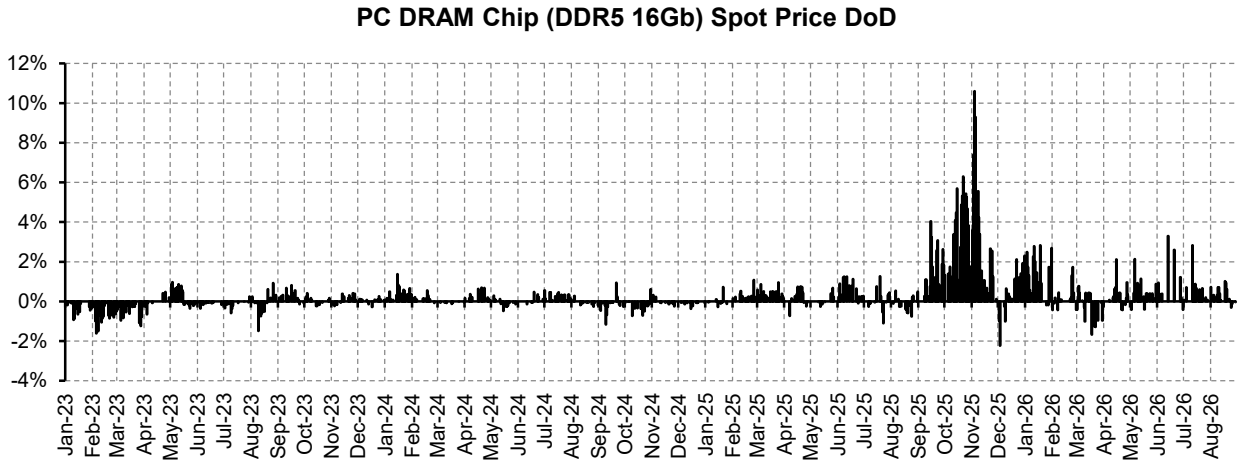
Source: DRAMeXchange, Bloomberg, Bernstein estimates and analysis

EXHIBIT 3: **PC DDR5 DRAM chip spot price continued rallying in Aug.**



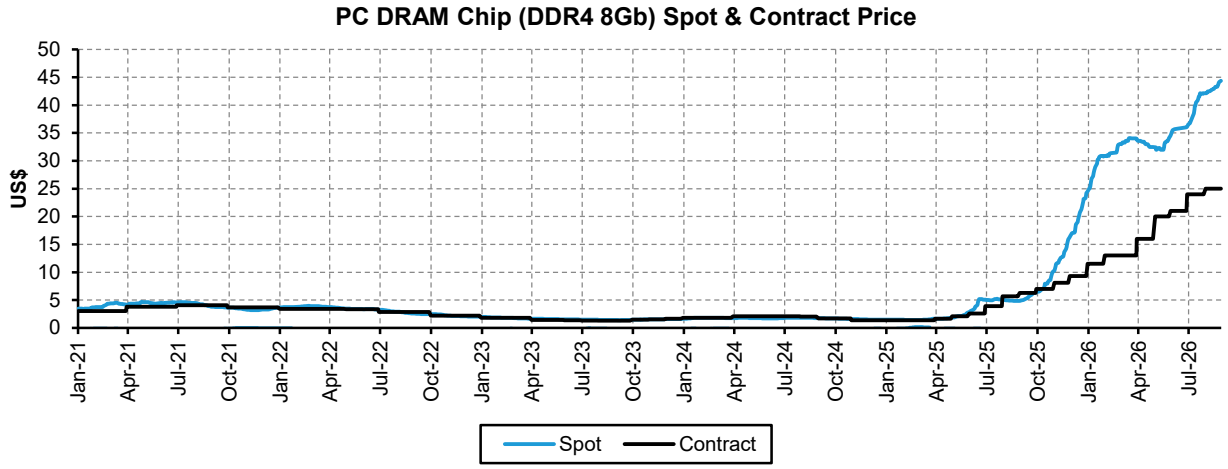
Source: DRAMeXchange and Bernstein analysis

EXHIBIT 4: **PC DDR5 chip spot price saw some correction in late Mar, but it has since been growing almost non-stop.**



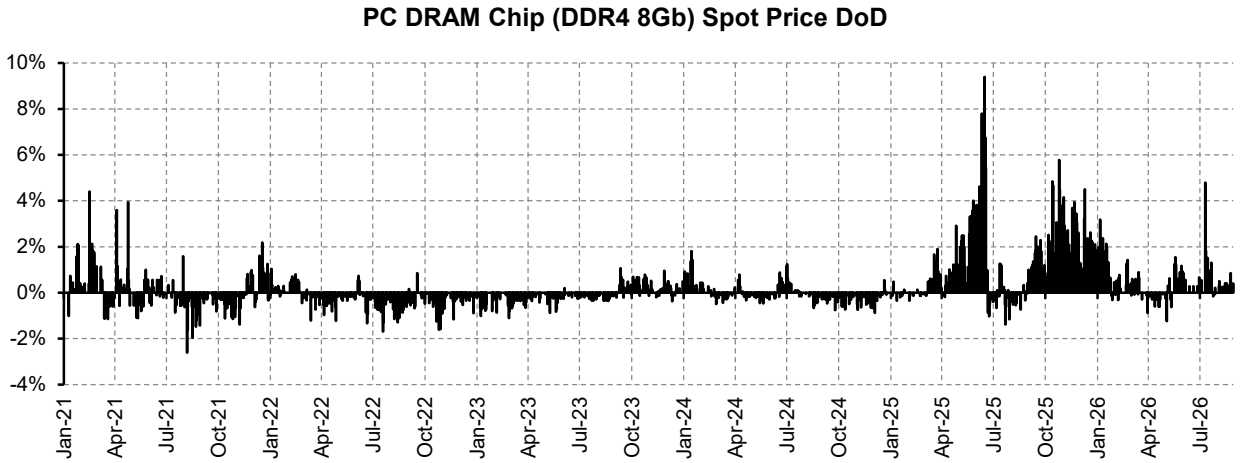
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 5: **PC DDR4 chip spot price also saw solid increase in Aug.**



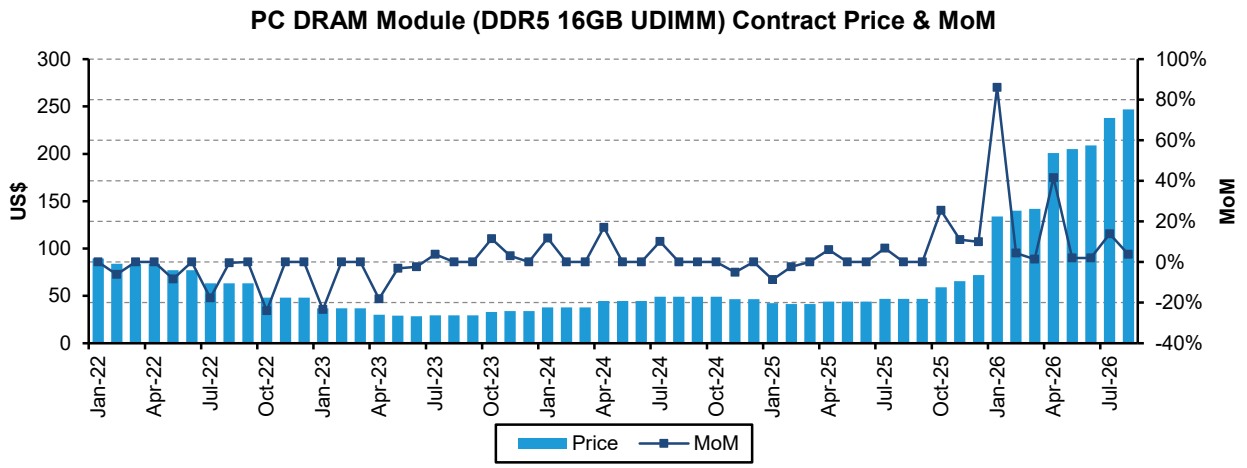
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 6: **Growth momentum slowed in Aug but remained quite robust for PC DDR4 chip spot price.**



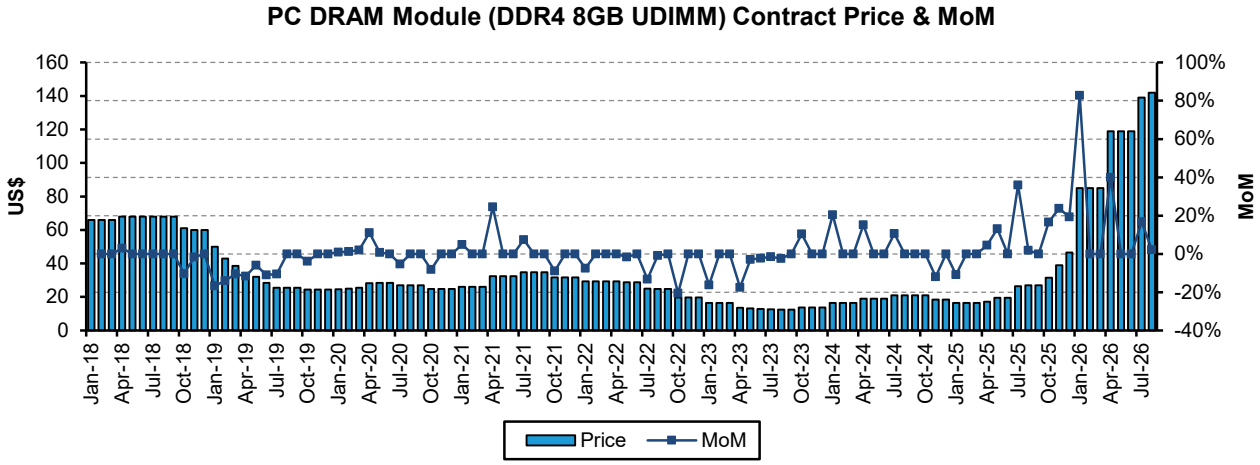
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 7: **PC DDR5 module contract price increased by 4% MoM in Aug as most of negotiations had wrapped up in Jul.**



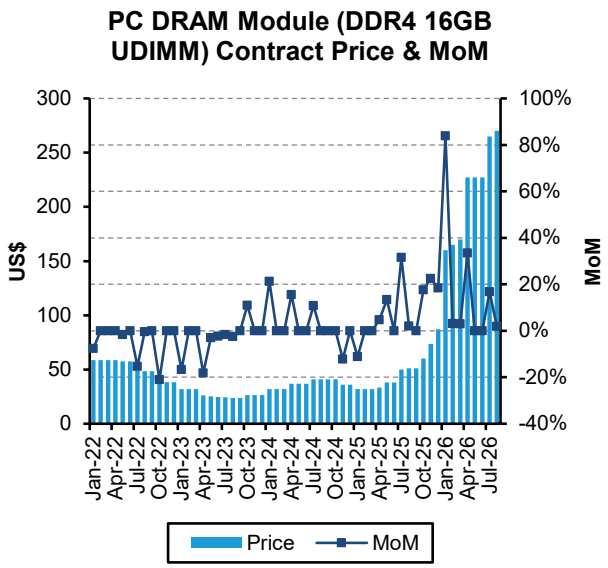
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 8: **PC DDR4 module contract price was up by 2% MoM in Aug.**



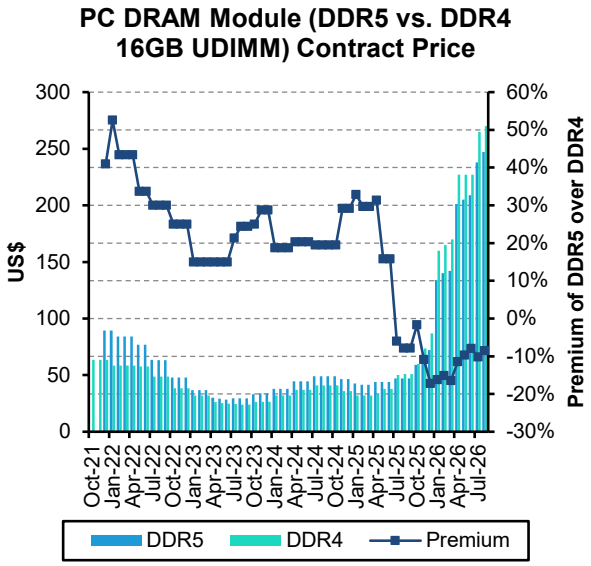
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 9: **16GB DDR4 modules contract price grew at similar rate to 8GB.**



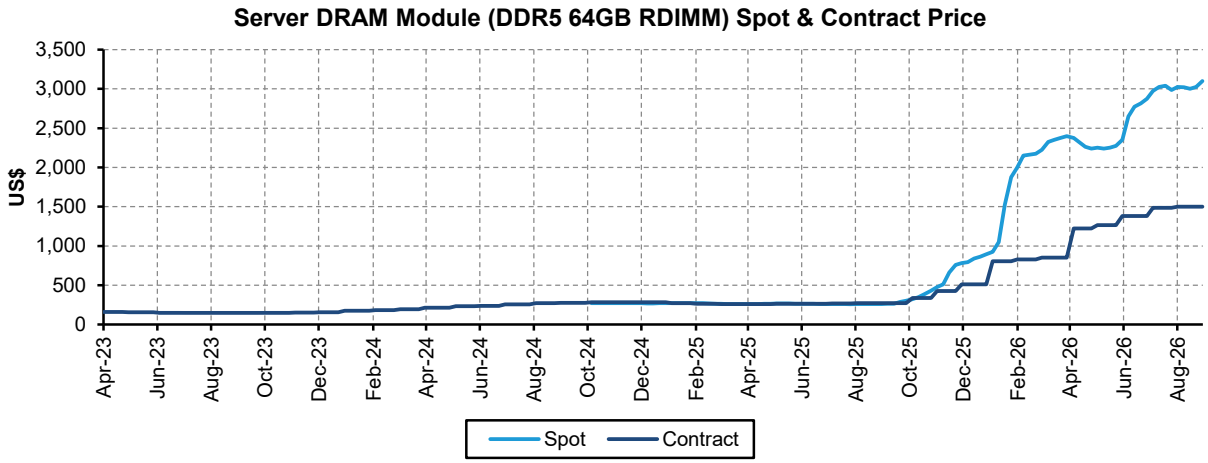
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 10: **PC DDR5 price discount vs. DDR4 was at 9%.**



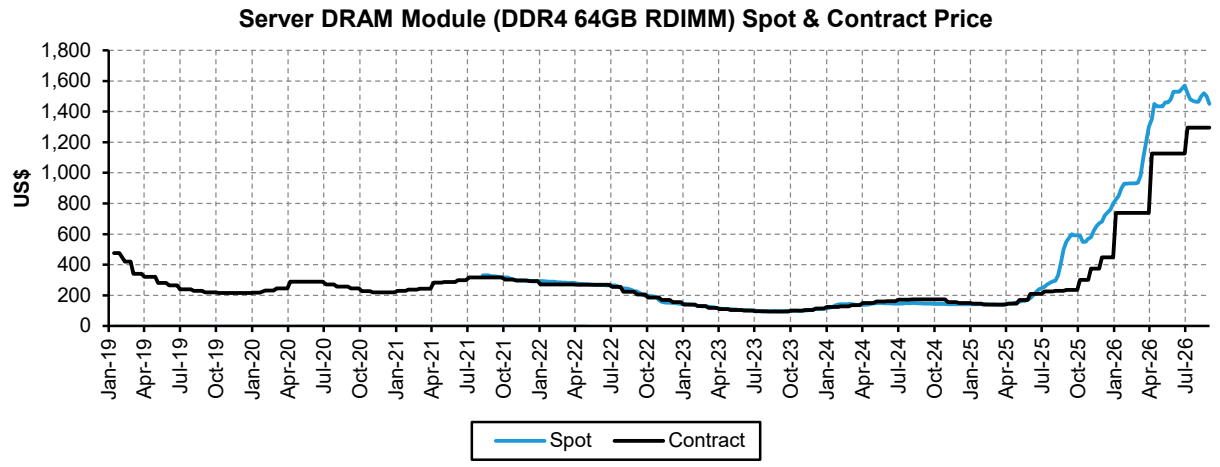
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 11: **Server DDR5 module spot price further grew in Aug and remained considerably above contract price.**



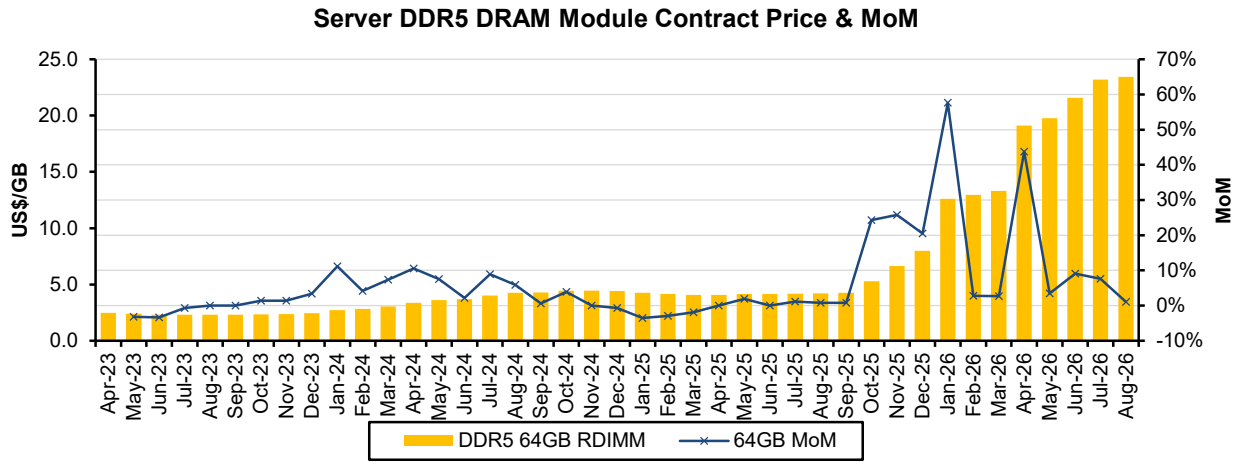
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 12: **Server DDR4 module spot price however fell further in Aug.**



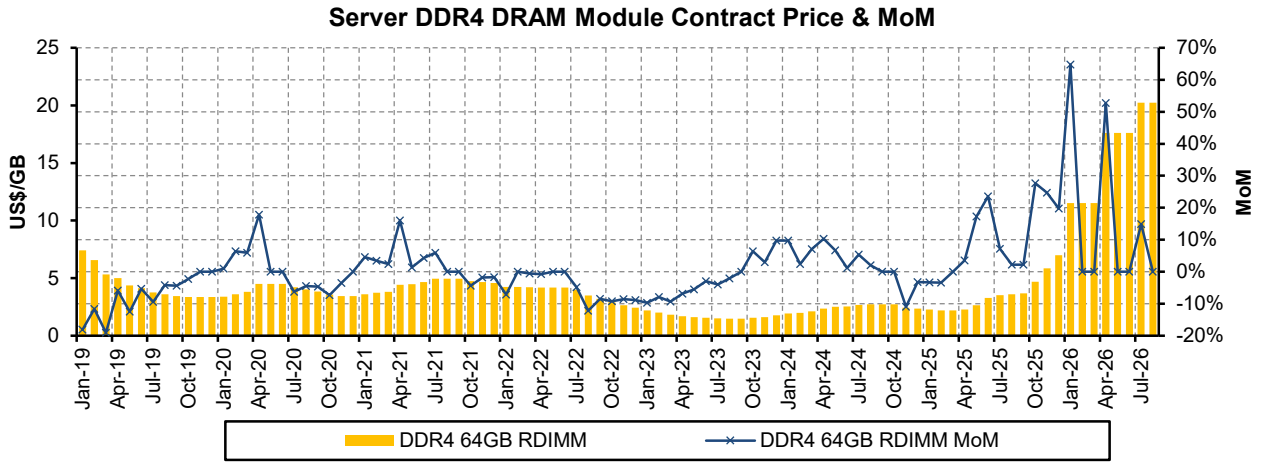
DRAMeXchange started to track server spot price in Aug 2021 as its trading volume became meaningful.
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 13: **Aug server DDR5 module contract price rose slightly by 1% MoM.**



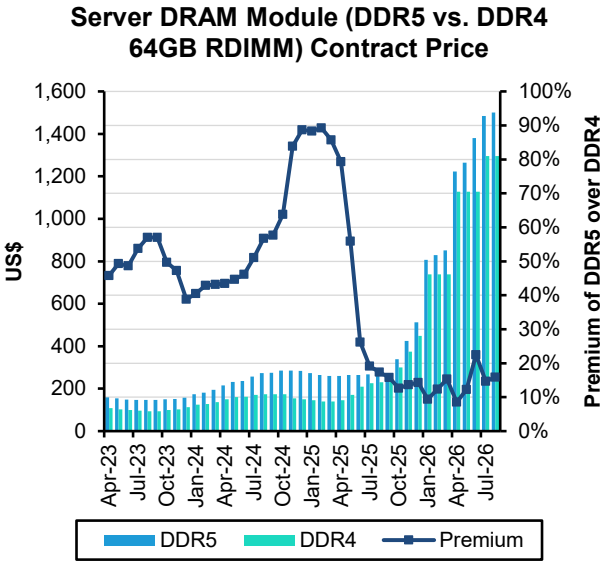
Source: DRAMeXchange and Bernstein analysis

EXHIBIT 14: **Server DDR4 module contract price was flat MoM.**



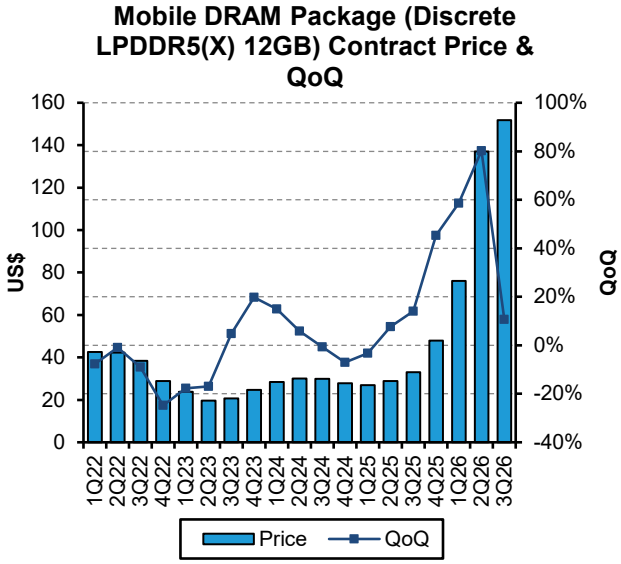
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 15: **Server DDR5 premium over DDR4 was at 16%.**



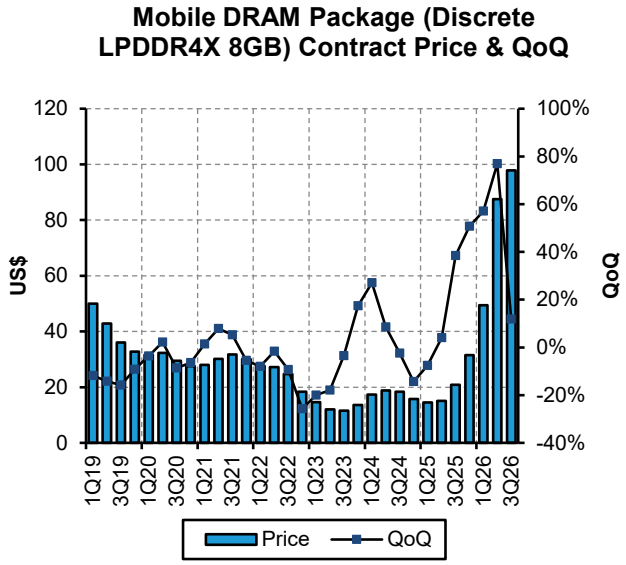
Source: DRAMeXchange and Bernstein analysis

EXHIBIT 16: **LPDDR5 contract price was estimated to rise by only 11% in 3QCY26.**



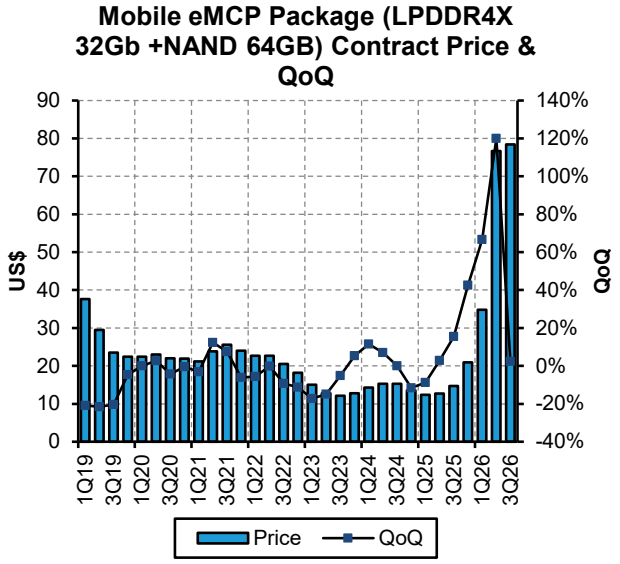
Source: DRAMeXchange and Bernstein analysis

EXHIBIT 17: **LPDDR4 price also estimated to rise by just 12% QoQ.**



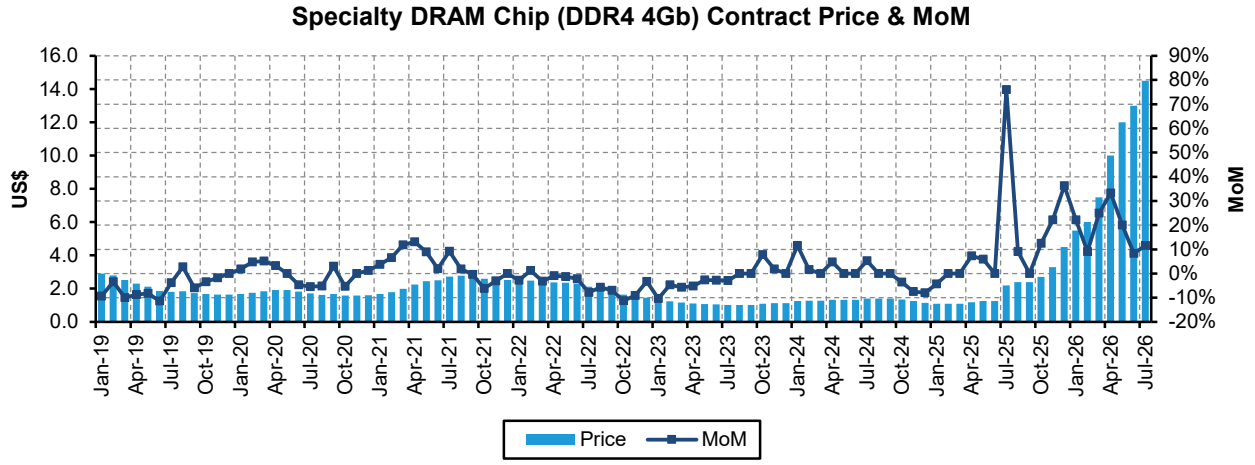
Source: DRAMeXchange and Bernstein analysis

EXHIBIT 18: **The price of low-end eMCP packages was largely flattish at 2.3% up QoQ for 3QCY26.**



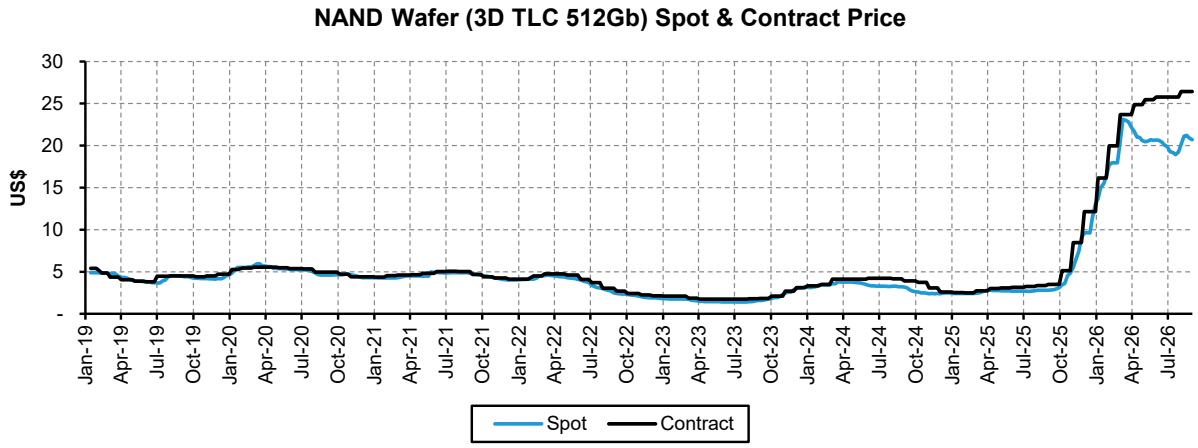
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 19: **Consumer DRAM shortage was likely more acute, with contract price rising 5% MoM.**



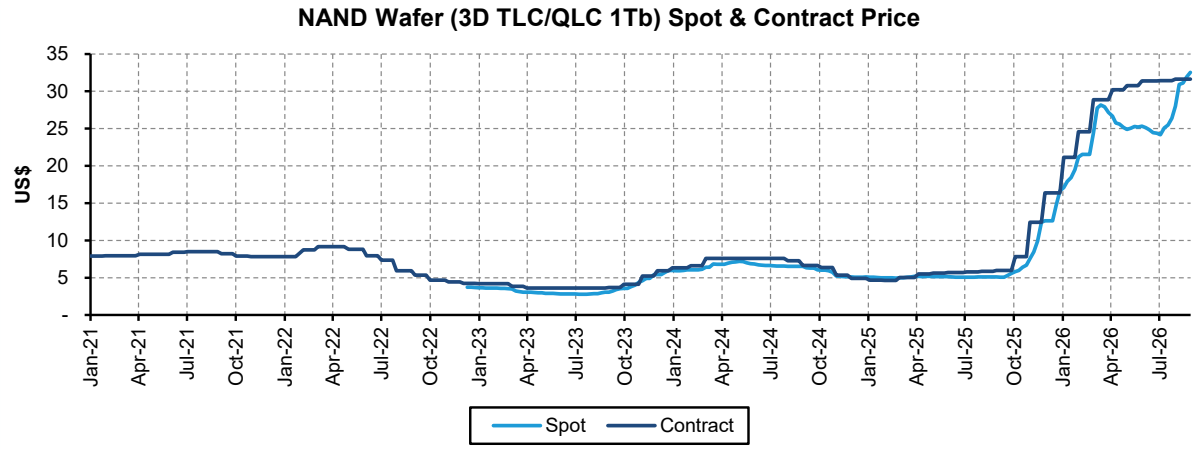
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 20: **NAND 512Gb spot price rebounded in Aug.**



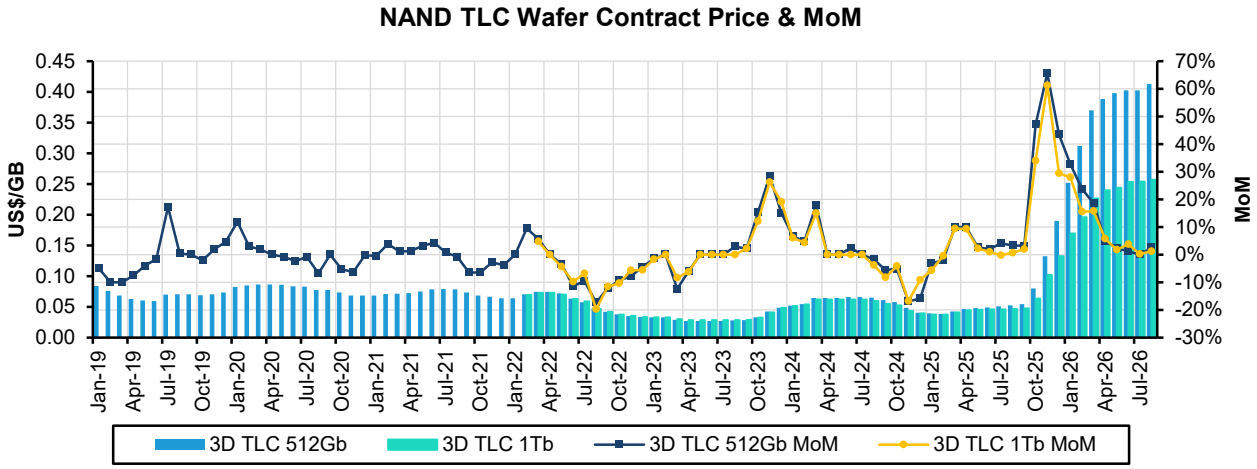
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 21: **1Tb wafer spot price soared in Aug and surpassed contract price too.**



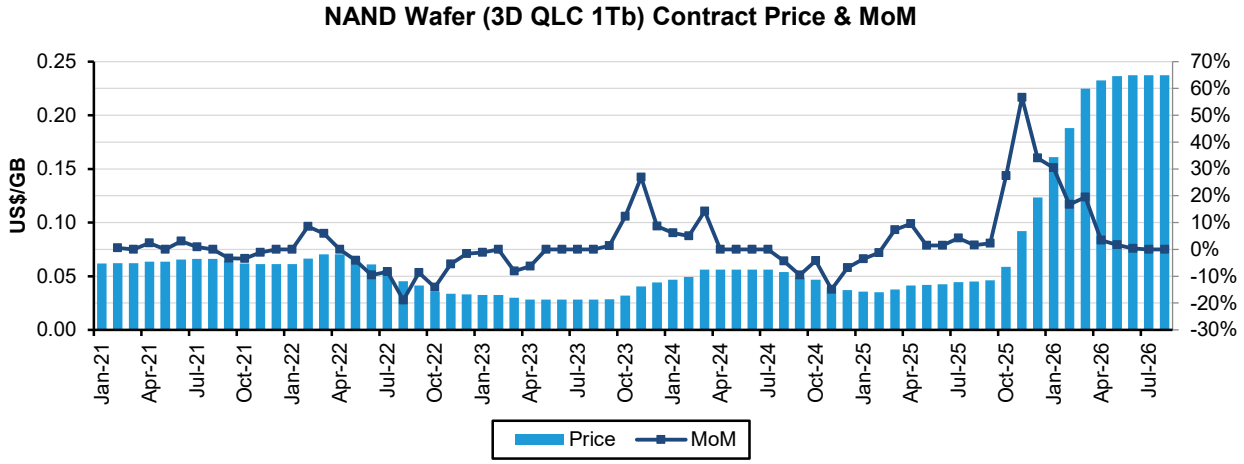
Source: DRAMeXchange and Bernstein analysis

EXHIBIT 22: **However, NAND wafer contract price remained largely sluggish as module houses stayed cautious.**



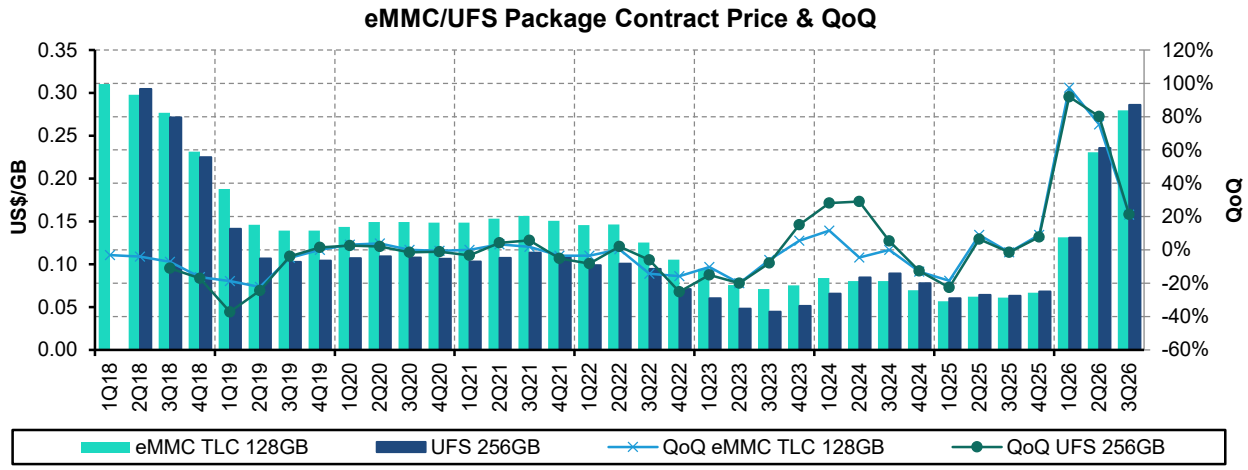
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 23: **QLC NAND wafer contract price was also flattish.**



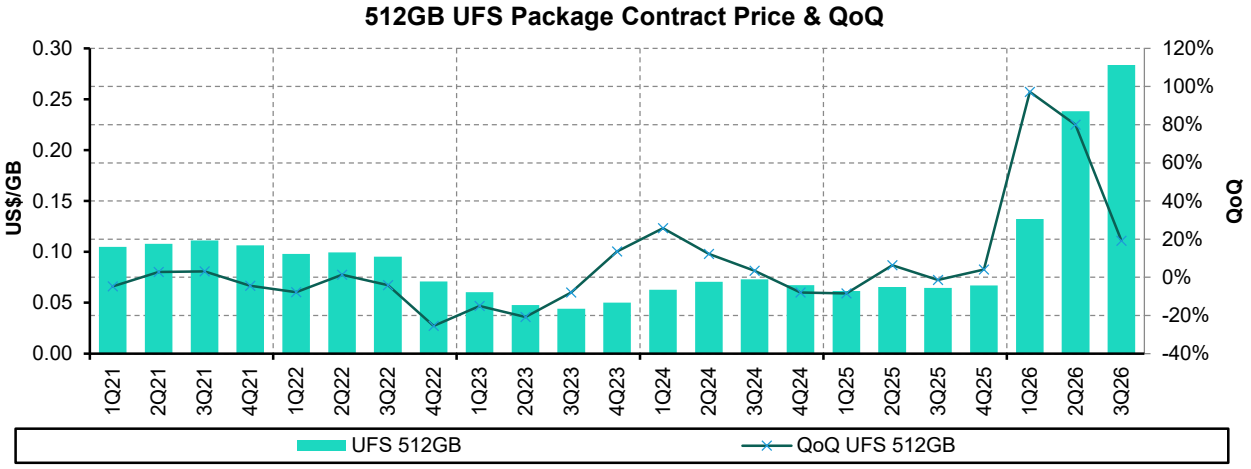
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 24: **Mobile NAND contract price for 3QCY26 is estimated to increase by 20% QoQ, notably slower vs 2Q as customers cut demand and push ed back on price hikes.**



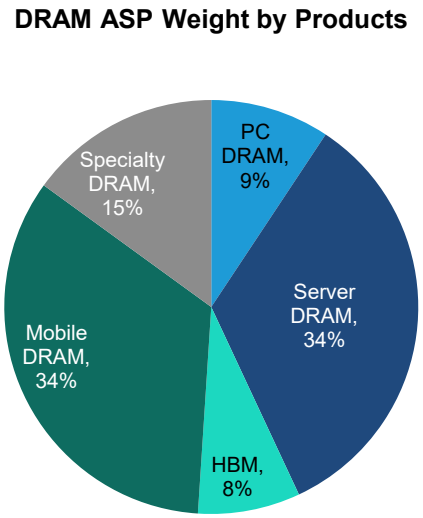
Source: DRAMeXchange & Bernstein analysis

EXHIBIT 25: **High-density 512GB UFS contract price is also expected to rise by ~20% in 3QCY26.**



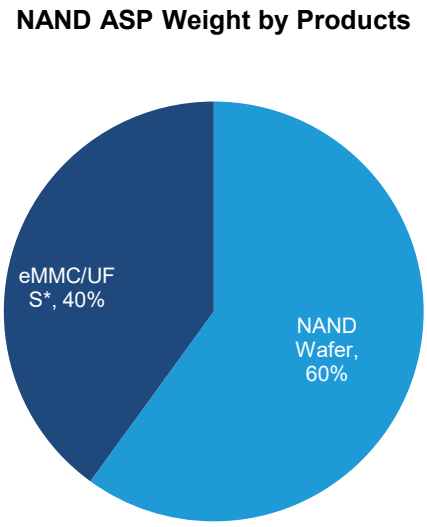
Source: DRAMeXchange and Bernstein analysis

EXHIBIT 26: **Our industry ASP is a weighted average of different products based on the size of their demand. We then further adjust it for HBM.**



Source: Bernstein analysis

EXHIBIT 27: **We allocate 40% & 60% to eMMC/UFS & wafer, respectively, to derive the industry NAND ASP. eSSD is then considered in addition.**



Source: Bernstein analysis

Further Reading Materials

- 28 Aug 2026 - **Global Memory: An AI memory primer**
- 21 Aug 2026 - **Korea Memory Export Tracker (Jul): Samsung continued ramping HBM4, but SK hynix delayed by Rubin?**
- 3 Aug 2026 - **2Q26 AI Server Pulse: Summertime Cool-Down**
- 31 Jul 2026 - **KIOXIA 1QFY26: Can buyback help KIOXIA avoid volatility?**
- 30 Jul 2026 - **Samsung 2Q26: Poised to Lead in HBM in 2H26?**
- 29 Jul 2026 - **SK hynix 2Q26: ASP should improve in 2H26. Overall thesis intact**
- 27 Jul 2026 - **Global Memory, NVIDIA & Broadcom: Quick thoughts on strategic partnerships**
- 25 Jun 2026 - **Micron 3QFY26: Strong near-term momentum but SCAs thus far still can't avoid cyclicality**
- 20 Jul 2026 - **Global Memory: New Memory LTAs Part 3 - Bull vs Bear Debate - Webinar slides, transcript and summary**
- 30 June 2026 - **Memory: What do New Memory LTAs mean for sustainability of earnings and multiples? Raising SNDK TP to $3000**
- 22 June 2026 - **Global Memory: Memory becoming a burden of AI too?**
- 15 June 2026 - **KIOXIA: All parties come to an end - Underperform**
- 2 June 2026 - **KIOXIA Investor Day: 22% bit CAGR good enough?**

- 1 May 2026 - **Sandisk FQ3'26: Nosebleed ASPs (now locked in); Raise estimates and TP to $1700, Reiterate Outperform (top ST pick)**
- 13 Apr 2026 - **Asia Semiconductors & Global Memory: AI, war, memory price - 1Q26 earnings preview**
- 9 Apr 2026 - **Sandisk: How high can estimates and the stock go? Raising estimates and TP to $1,250, with blue-sky scenario at $3000**
- 17 Mar 2026 - **Asia Semiconductors & Hardware: Stronger & longer - 2026 Tech Tour takeaways**
- 13 Mar 2026 - **Global Industrial Gases & Semis: Assessing the helium impact**
- 27 Jan 2026 - **AI Value Chain: Putting it all together - how much AI upside does each name really have, and who might be an underappreciated winner?**
- 2 Jan 2026 - **Asia Semiconductors & Global Memory: 2026 is still all about AI**
- 11 Dec 2025 - **Expert Insight: How much memory do AI Data Centers need?**
- 24 Nov 2025 - **Asian Semis & Global Memory: Two findings, one simulation & one model for AI**
- 27 Oct 2025 - **AI Value Chain: How much does a GW of data center capacity actually cost, and what goes into it?**
- 1 Sep 2025 - **Global Semicap and Memory - US revokes Samsung and SK hynix semicap license waivers for their China Fabs**
- 21 Aug 2025 - **KIOXIA: Decipher the complex JV with SanDisk**
- 29 Jul 2025 - **Samsung: Dimensioning the contribution of the Tesla project on Samsung & ASML**
- 13 Jun 2025 - **KIOXIA: The Last Samurai - initiating with Underperform**
- 6 Dec 2024 - **Global Memory: Decoupling in memory? A quantitative assessment**
- 21 Nov 2024 - **Global Memory: Will China/YMTC surprise us in NAND?**
- 16 Sep 2024 - **Global Memory: Is Samsung still the memory leader?**
- 8 Jul 2024 - **Global Semiconductors: AI semiconductor roadmap - 3 charts & 7 investment implications**
- 1 Jun 2023 - **10 FAQs on Investing in Semiconductors for Generative AI**
- 4 Mar 2022 - **Global Memory: Cornerstone of the Data Economy**
- 25 Oct 2021 **Global Memory: The cornerstone of the data economy but painful cyclicality - Initiating with Underperform**

I. REQUIRED DISCLOSURES

References to "Bernstein" or the “Firm” in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK（サンフォード・C・バーンスタイン株式会社）and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

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VALUATION METHODOLOGY

Samsung Electronics Co Ltd

We value Samsung Electronics with 6.2x of forward 5Q-8Q EPS of KRW71,172 and arrive at price target of KRW 440,000 for its common shares, KRW374,000 for preferred shares, and US$7,350 for GDR.

SK Hynix Inc

We value SK hynix with 6.2x of forward 5Q-8Q EPS of KRW536,158 and arrive at price target of KRW3,300,000.

Micron Technology Inc

We value Micron with 7.7x of forward 5Q-8Q EPS of US$169.8 and arrive at price target of US$1,300.

KIOXIA Holdings Corp

We value KIOXIA with 4.1x of 1-year forward EPS of ¥9,810 and arrive at price target of ¥40,000.

SanDisk Corp

We value SNDK at $3000 on 11x FY28 EPS, or 14x through cycle EPS (FY26-30 avg EPS).

RISKS

Samsung Electronics Co Ltd

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

SK Hynix Inc

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

Micron Technology Inc

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

KIOXIA Holdings Corp

Upside risk to our target price include 1) stronger than expected NAND demand growth, which likely would lead to better NAND pricing and hence company earnings, 2) favorable policy support from Japanese government such as capex subsidies, and 3) better than expected cost reduction

SanDisk Corp

The biggest risks to the downside on Sandisk and to our price target are that: 1) Near-term numbers look high and face cyclical downside from NAND; 2) SNDK's disclosures and broader investor communications have been confusing, potentially dissuading more quality-focused investors; and 3) NAND weakness could extend beyond the current cycle and be more structural, in which case Sandisk's DCF value could be structurally lower and the impaired value of its assets may be materially below replacement cost.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

EQUITY RATINGS DEFINITIONS

Bernstein brand

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAX) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/-15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp

- Neutral (N): Stock will perform in line with the market index to within +/-10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

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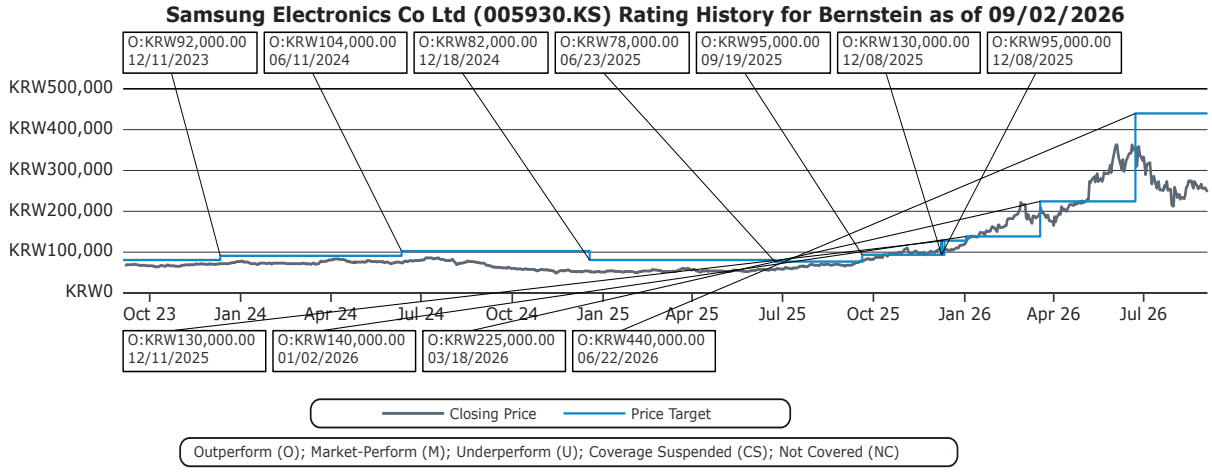
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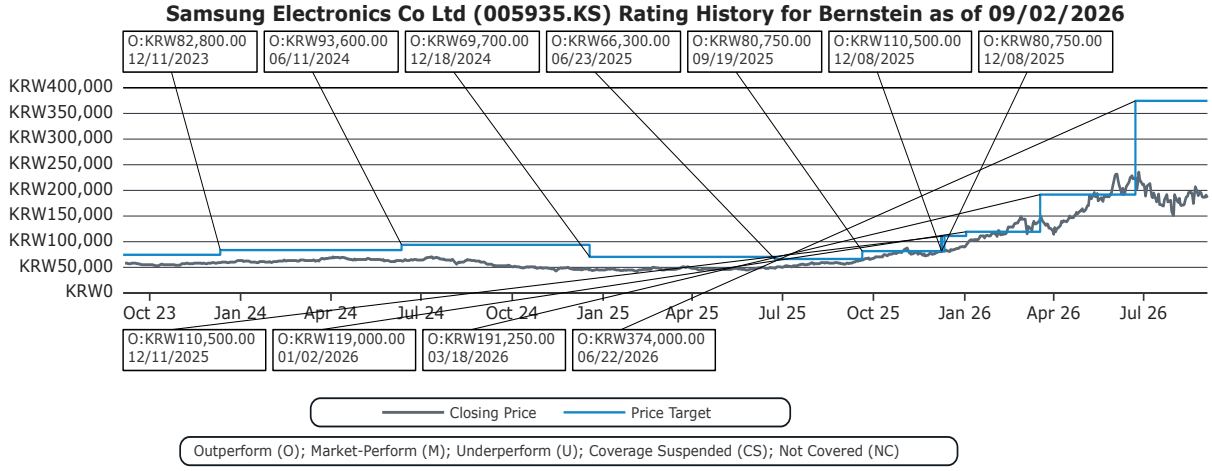
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Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
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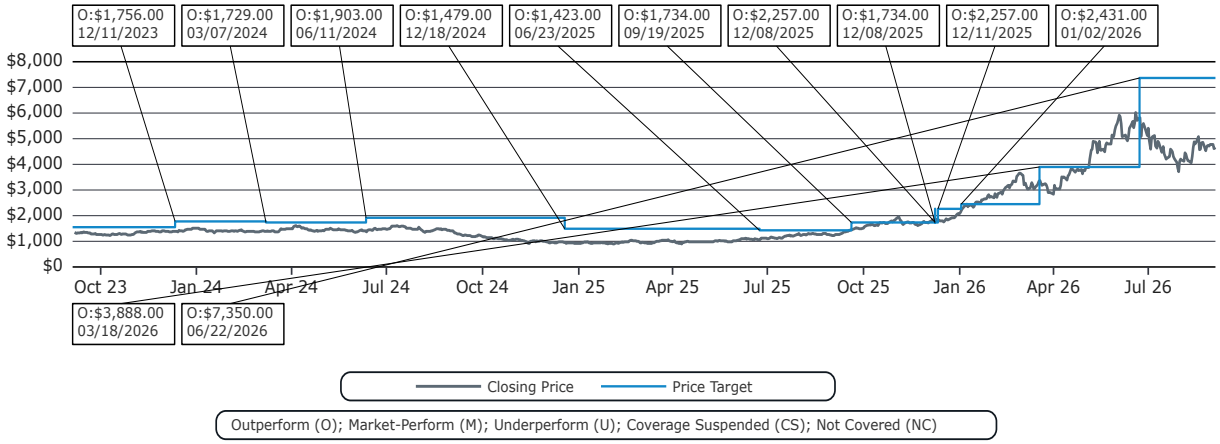
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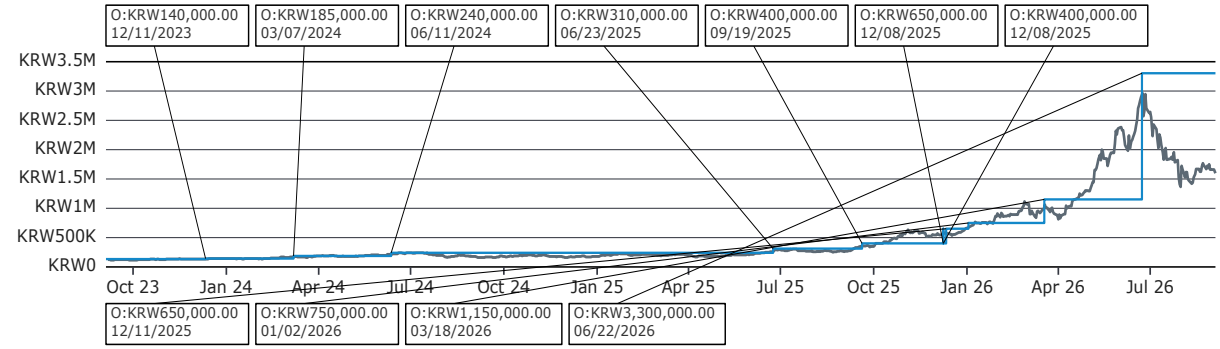


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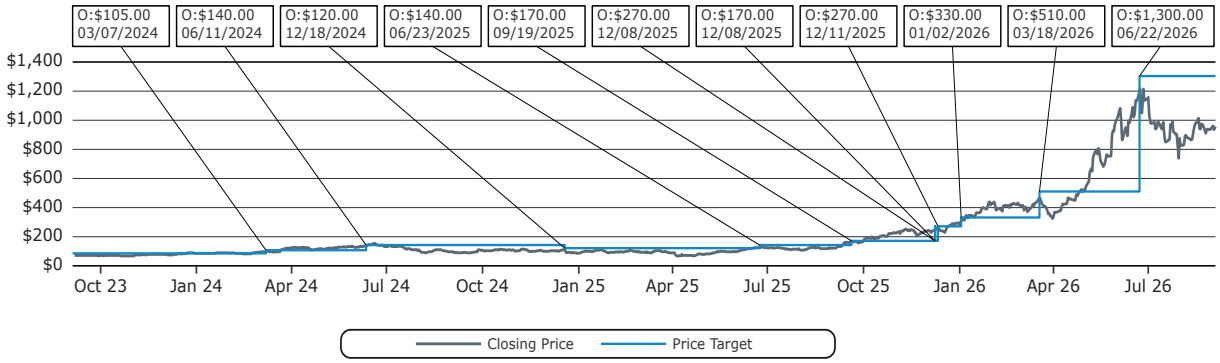
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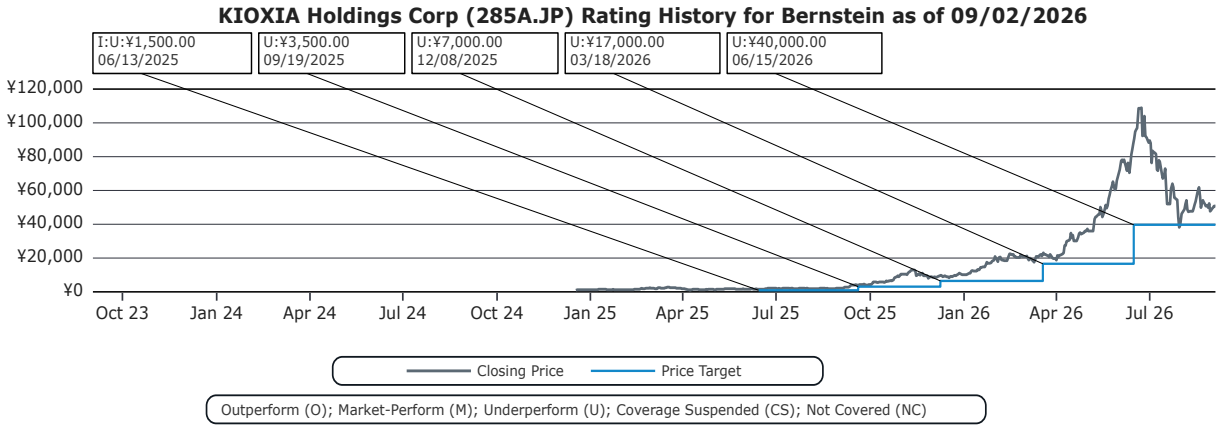


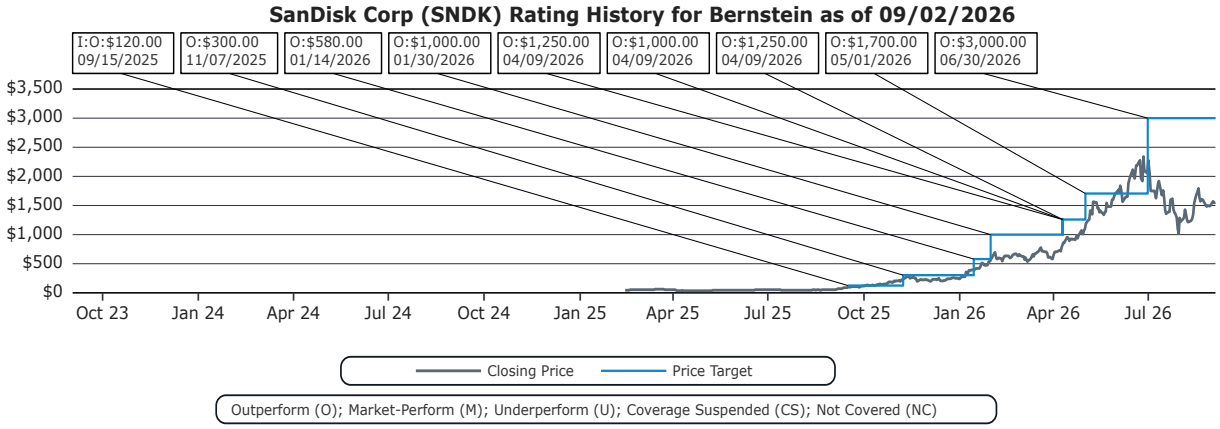
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Tech Bytes | Asia Pacific

China Memory – Reshaping Global Competition

China's memory makers are turning state support and domestic demand into manufacturing scale, reshaping industry capex, growth and returns. The technology gap remains, but the race is shifting from building the fastest DRAM to building enough capacity to power the next decade of AI.

China's turn to challenge. The US invention and dominance in DRAM during the 70's shifted to Japan in the 80's, then Korea out-investing in the 90's and taking dominant share since 1998, and may now be challenged by China in the coming decade. It is brutal business competition and only strong players survive, but Chinese companies like CXMT and YMTC are rejoining competition with government's endorsement, innovation and capital. CXMT is likely to surpass #3 Micron by 2028 and potentially 800k wafers per month by 2031. YMTC is adding about 100k wafers capacity per month each year and likely to be the global #2 NAND player by 2028. These expansions are supported by increased localization efforts, with YMTC's localization penetration reaching near 60% for new fabs.

Competition or substitution? There is still a long way to go before CXMT matches the technology depth, customer trust and global reach of the market leaders. China's memory market is experiencing significant growth driven by strong AI and memory investment cycles. AI datacenters don't just need HBM (exclusive to GPU and ASIC chips), they need commodity DRAM too. As AI demand and the migration of leading-edge capacity toward HBM tighten conventional-memory supply, both CXMT and YMTC can relieve shortages and support import substitution in selected product categories. Over time, however, their expansion could erode the supply discipline that has underpinned incumbent returns.

Overcoming geopolitical tensions and export restrictions. US semi equipment export restrictions have impacted node migration processes, however Chinese companies are increasingly collaborating with local WFE vendors to mitigate these challenges. Overall, China's memory market is poised for continued growth, supported by domestic investments and strategic expansions, with geopolitical factors increasingly driving efficiency and innovation.

Implication. Global memory leaders should be able to compete on cost, but others face challenges given high cost structures and limited scale, and the biggest negative initially should come from higher volume markets with lower barriers to entry. Competition impacts market share, returns over a cycle via reduced l-t sustainable margins and high capital spend, which lower ability to sustain capital returns and hence ROE. China's expansion of domestic memory production creates sustained demand for locally made semiconductor fabrication equipment, with competitive pressure increasing across deposition, cleaning, inspection and polishing systems.

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China is introducing a new variable into the memory cycle

Altering the supply side of the equation. For the past decade, the DRAM industry's improved economics have depended on a rational three-player oligopoly. While CXMT does not challenge that by overtaking Samsung, SK hynix or Micron in HBM, it challenges it by scaling mainstream DRAM against a different objective. The Big Three are directing more wafers, R&D and packaging resources toward HBM and advanced server memory. CXMT, meanwhile, is expanding conventional DRAM capacity to capture domestic demand, improve manufacturing yields and establish greater scale. This influx of Chinese memory has forced foreign giants like Samsung, SK Hynix, and Micron to abandon the lower end of the market such as DDR4 DRAM or legacy MLC NAND. With larger share of the legacy chip market, China effectively sets the price floor for older memory tech while accelerating the global pivot toward advanced AI components.

Near-term relief could become a longer-term threat. AI is creating enormous demand for memory, excess returns and China clearly wants a much larger piece of that market to support its AI ambitions and reduce exposure to US restrictions. AI datacenters don't just need HBM (which is exclusive to GPU and ASIC chips), they also need commodity DRAM too. As AI demand and the migration of leading-edge capacity toward HBM are tightening conventional-memory supply, both CXMT and YMTC can immediately relieve shortages and support import substitution in selected product categories. Over time, however, their expansion could weaken the supply discipline that has underpinned incumbent returns.

A cyclical inflection from 2028. Before then, strong AI demand, HBM-related capacity crowd-out, domestic import substitution and product-qualification time lines should absorb much of China's incremental output. From 2028 onward, however, CXMT and YMTC could become sufficiently large, along with incumbent unprecedented capacity expansions, and could influence global pricing and the industry's response to a downturn.

Exhibit 1: Supplier competitive position, by memory type

Supplier position by memory type

Market position by product pool: HBM remains Big 3-led; China emerges in commodity DRAM/NAND

Market leader Strong Present / growing Limited / emerging Not present

	HBM AI accelerator	Server DRAM DDR5	PC & mobile LPDDR	Enterprise NAND	SSD	Consumer NAND
SK hynix South Korea	**Recovery candidate** Ceded early HBM4 lead to Samsung's 1c/4nm ramp; recovery hinges on next-gen Nvidia qualification.	**Strong** DDR5 ahead of peers. Wafer shift to HBM tightens conventional supply.	**Strong** LPDDR5X for premium smartphones. Losing low-end to CXMT.	**Present** Enterprise SSD via Solidigm.		**Present** Lower priority -- HBM and server absorb most capacity.
Samsung South Korea	**Leader** Early mass production of HBM4 and began shipping HBM4E sample	**Leader** Largest conventional DRAM capacity. Broadest customer base.	**Leader** Largest LPDDR supplier. Inside most premium Android phones.	**Leader** Largest NAND capacity globally. V-NAND technology advantage.		**Leader** eMMC, UFS, client SSDs across all tiers. Dominant share.
Micron USA	**Growing fast** HBM4E qualified for Vera Rubin; only US-headquartered HBM supplier.	**Strong** DDR5 competitive. US manufacture differentiates for government buyers.	**Strong** LPDDR5X competitive. Apple historically significant customer.	**Strong** QLC NAND competitive for hyperscale SSDs. US manufacture a differentiator.		**Present** Consumer flash. Lower priority versus HBM/server mix shift.
Kioxia / SanDisk Japan	NAND only -- no DRAM	--	--	**Present** 2nd-largest NAND capacity globally. BiCS technology.		**Strong** USB, client SSDs, mobile UFS. Financially more fragile in downturns.
CXMT China, state-backed	**Attempting** HBM3 development / sampling, but not a near-term global HBM relief valve.	**Limited** Emerging in DDR4/DDR5; not yet qualified at scale for mainstream global hyperscale server DRAM	**Growing** Mid/low LPDDR5 for Chinese Android market. Supporting Huawei.	DRAM only -- no NAND		--
YMTC China, state-backed	NAND only -- no DRAM	--	--	**Emerging** Growing in Chinese market. Export restrictions limit global access.		**Growing fast** Domestic NAND ramp; China self-sufficiency ambition; export restrictions limit global access

Source: Company commentary, Morgan Stanley Research estimates. Note: Market positions are illustrative and intended to frame supplier participation by memory type.

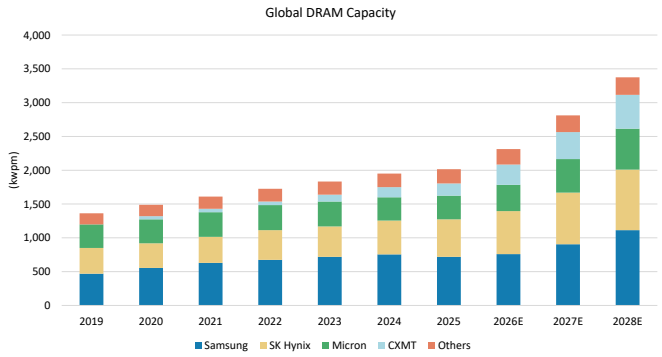
China's memory strategy – scale before technology leadership

CXMT is rapidly shifting from a technology follower to a supplier at scale... The company offers mainstream DDR5, LPDDR5/5X and selected server DRAM, while new funding provides additional resources for capacity expansion and process development. We estimate CXMT will reach 300k wafers per month in 2026, equivalent to 13% of global DRAM wafer capacity and 11% of global bit shipments. We forecast capacity rising to 500k wafers per month in 2028 and 800k by 2031, with global bit-shipment share reaching approximately 15% by 2030.

... however, wafer parity does not equal bit parity... We estimate that CXMT has transitioned to its Gen 4B (approximately 16nm-class process) with yields above 90%. Nevertheless, its reliance on immersion DUV and multi-patterning requires more masks and process steps, lengthens cycle times and creates a structural cost-per-bit disadvantage relative to the leaders. Even when visible DDR5 specifications are comparable, the incumbents retain advantages in yield, bits per wafer, premium-grade output and qualification breadth.

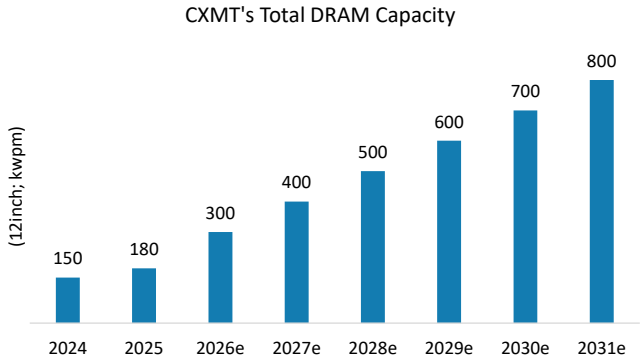
... but technology parity is not required for pricing influence. In commodity memory, a supplier becomes strategically relevant once its incremental output is large enough to affect the marginal price. This is particularly important because HBM is ultimately built from DRAM dies and competes for the same leading-edge DRAM wafer capacity. As CXMT approaches a mid-teens share of global capacity, broad adoption in the US or Europe is unnecessary, given that domestic Chinese output can displace imports and force incumbent suppliers to redirect production into other markets.

Exhibit 2: Global DRAM wafer capacity to reach 3,374kwpm in 2028e, while CXMT may account for 15%



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 3: CXMT to expand capacity from 180kwpm in 2025 to 300kwpm in 2026, followed by annual capacity additions of approximately 100k wpm, reaching 800k wpm by 2031

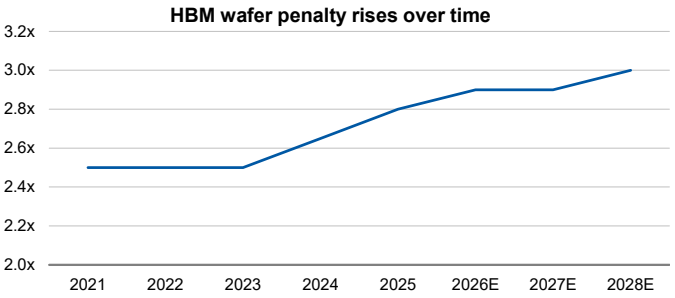


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

The AI paradox – HBM creates the opening

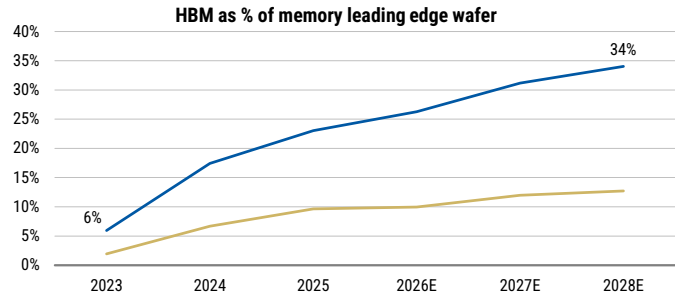
HBM is increasing the opportunity cost of conventional DRAM. Compared with conventional DRAM, HBM uses larger dies and requires stacking, TSV integration, advanced packaging, extensive testing and customer qualification. We estimate the HBM bit-output penalty will rise from approximately 2.5x conventional DRAM wafers in 2021–23 to approximately 3x by 2028E. HBM's share of leading-edge DRAM wafer consumption is also expected to increase from approximately 6% in 2023 to 34% by 2028E. This draws incumbent capital and leading-edge capacity toward premium AI memory, creating more room for CXMT in mainstream DDR5 and LPDDR5X. At the same time, AI demand absorbs new supply and should delay a broad China-led pricing impact until 2028.

Exhibit 4: HBM die penalty – bit-output is reduced due to much larger DRAM dies



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 5: HBM cannibalization: Share of advanced DRAM wafer is rising sharply

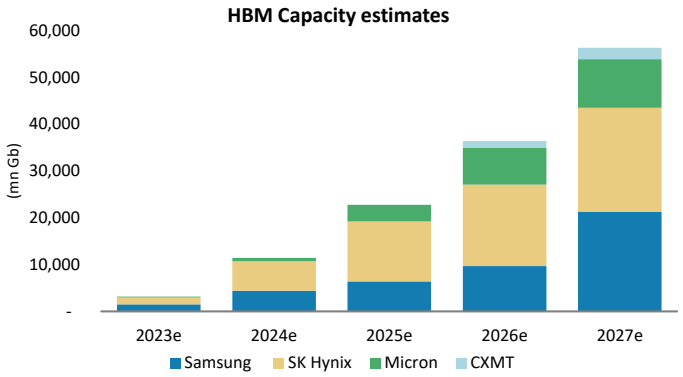


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

China's domestic HBM requirement is becoming more immediate. HBM3E is likely to be a major bottleneck for Chinese AI GPUs in 2027, particularly as global vendors convert more capacity toward HBM4. Our checks suggest CXMT could produce approximately 3–4mn HBM3E stacks in 2027 despite low yields, sufficient to support around 0.8–1.0mn domestic AI GPUs. This would make CXMT strategically relevant to China's AI supply chain, although yield, packaging, testing and accelerator qualification mean it is not yet a global substitute for the Big Three. After incorporating CXMT's planned HBM ramp, our combined DRAM model remains approximately 17% undersupplied in 2026E and 15%

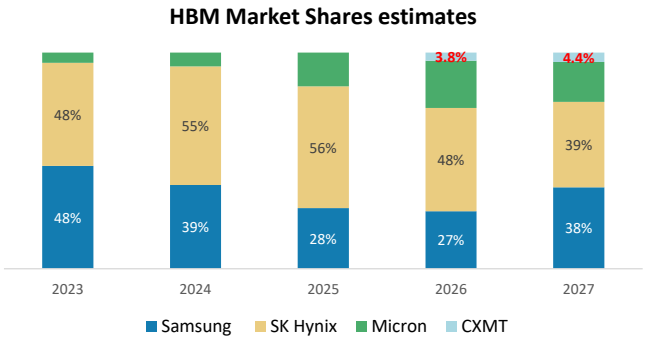
undersupplied in 2027E.

Exhibit 6: Global HBM capacity build



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 7: CXMT's HBM to account for 3.8%/4.4% if the total HBM market in 2026e and 2027e in terms of bit shipment



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 8:

The DRAM market forecast to remain undersupplied in 2027 despite CXMT's capacity

Sufficiency estimates	2023	2024	2025	2026e	2027e
HBM TSV Capacity (K wpm)					
Samsung	45	130	150	180	250
SK Hynix	45	120	150	200	250
Micron	3	20	60	100	110
CXMT				20	40
Total	**93**	**270**	**360**	**500**	**650**
Yield rate assumptions				**38.9%**	**30.0%**
Samsung		*50%*	*60%*	*55%*	*70%*
SK Hynix		*60%*	*75%*	*65%*	*70%*
Micron		*50%*	*70%*	*65%*	*70%*
CXMT				*40%*	*50%*
UTR assumptions					
Samsung		*80%*	*60%*	*70%*	*100%*
SK Hynix		*100%*	*100%*	*100%*	*100%*
Micron		*100%*	*100%*	*100%*	*100%*
CXMT				*100%*	*100%*
Implied HBM production (mn Gb)					
Samsung	1,500	4,435	6,387	9,696	21,239
SK Hynix	1,500	6,273	12,830	17,363	22,226
Micron	150	729	3,548	7,937	10,372
CXMT				1,382	2,462
Total	**3,150**	**11,436**	**22,765**	**36,379**	**56,300**
Sufficiency Ratio				**59.8%**	**54.8%**
HBM Demand (mn Gb)	1,866	9,059	19,516	34,205	56,085
Commodity DRAM market demand (mn Gb)	209,527	243,423	316,640	469,675	619,970
Total HBM+DRAM demand (mn Gb)	211,393	252,481	336,156	503,879	676,055
Commodity DRAM market supply (mn Gb)	200,214	228,086	301,257	382,107	519,666
Total HBM+DRAM supply (mn Gb)	203,364	239,523	324,022	418,486	575,965
Total DRAM sufficiency	***-4%***	***-5%***	***-4%***	***-17%***	***-15%***

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

The next cycle is the real test

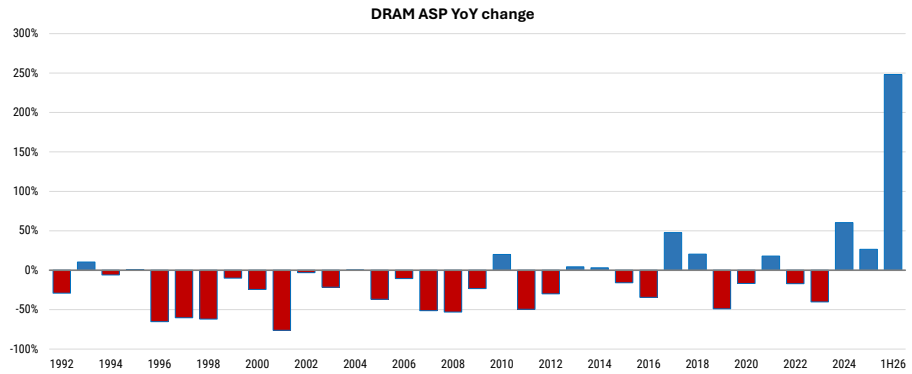
CXMT may be relatively benign during a shortage. Its incremental output can meet demand in areas the incumbents increasingly prefer not to prioritise, particularly mainstream DDR, LPDDR and China-specific applications. The risk grows once its installed capacity becomes large enough to influence pricing during a weaker part of the cycle.

The key issue is how CXMT acts when demand slows. Samsung, SK hynix and Micron are ultimately profit-driven suppliers and have become more willing to reduce capex, utilization and bit growth when market conditions weaken. CXMT's objectives may be different and include domestic substitution, technology learning curve, supply-chain resilience and semiconductor sovereignty.

A different objective may change industry's behavior. If CXMT continues expanding through the next downturn, the Big Three may have to shoulder a greater share of the production cuts required to restore market balance. The risk is therefore not simply

"cheap Chinese DRAM," but a shift from a three-player market with broadly aligned profit incentives to a four-player market with asymmetric strategic objectives.

Exhibit 9: DRAM ASP YoY change



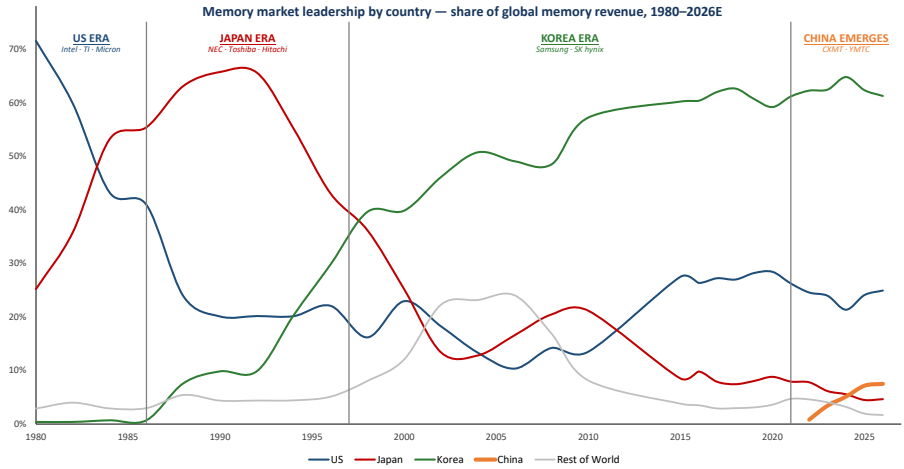
Source: WSTS

What history tells us

Memory leadership has historically shifted when a new entrant combined scale with a change in the basis of competition. US companies led through technology innovation and the early commercialization of DRAM. Japan overtook the US in the early 1980s by turning memory into a high-quality, high-yield manufacturing business, while Korea rose to leadership through the 1990s by combining aggressive investment, cost-per-bit leadership and rapid process migration, and has retained leadership since then. China appears to be following a different version of the same playbook: CXMT and YMTC are building scale first in mainstream memory, supported by a large domestic market, while the incumbents increasingly direct capital toward HBM, advanced server DRAM and enterprise storage. China therefore does not need to displace the Big Three to alter industry economics or achieve broad adoption in the US and Europe. Once Chinese output is sufficiently large, domestic production can displace imports, redirect incumbent supply into other markets and influence the marginal price of mainstream memory.

The end game may therefore be bifurcation before displacement. China could increasingly set the marginal price in mainstream memory, while the incumbents defend returns through HBM, advanced server DRAM and enterprise storage. This may preserve a differentiated premium profit pool even as conventional-memory returns become more competitive. The longer-term risk is what happens when China follows the incumbents up the value chain: if CXMT becomes credible in advanced server DRAM and HBM, or YMTC achieves broad enterprise-SSD qualification, China would move from competing for the industry's volume pool to challenging its profit pool.

Exhibit 10: Memory market leadership by country — share of global memory revenue, 1980–2026E



Source: TrendForce, Morgan Stanley Research

Exhibit 11: Memory leadership eras: who led, why they won, and what changed

ERA	PERIOD	KEY PLAYERS	WHY THEY WON	WHY LEADERSHIP SHIFTED / INVESTOR LESSON
US ERA	1970s–early 1980s	Intel, Mostek, Texas Instruments Also: Fairchild, Motorola, National Semiconductor, IBM	Technology invention and commercialization. US companies created the early commercial DRAM market, supported by Silicon Valley, computing customers and strong device-design capability. Intel's 1103 was an early commercially successful DRAM; Mostek and TI were important scale players.	As DRAM standardized, competition shifted from invention toward yield, quality, scale and cost per bit. The 1985 downturn accelerated the exit of most US merchant DRAM producers, with TI and Micron among the few survivors. **Lesson: inventing a technology does not guarantee manufacturing leadership.**
JAPAN ERA	Early 1980s–early 1990s	NEC, Toshiba, Hitachi, Fujitsu, Mitsubishi Electric Also: Oki, Sharp	Manufacturing quality and coordinated scale. Japanese producers combined high yields and reliability with aggressive investment, strong domestic electronics demand and coordinated VLSI research, jointly establishing dominance at the 64K DRAM generation and beyond.	Leadership weakened as demand shifted toward PCs, DRAM prices collapsed in 1995–96, and manufacturing complexity rose across 200mm fabs and 64Mb processes. Japanese suppliers responded less systematically than Korean rivals. **Lesson: manufacturing excellence can still be disrupted when market economics and process complexity change.**
KOREA ERA	1990s–present	Samsung Electronics, Hyundai Electronics/Hynix, LG Semicon Current leaders: Samsung, SK hynix; Micron the main US competitor	Scale, cost leadership and investment through the cycle. Korean producers invested aggressively across successive technology generations, expanded during downturns and converted scale into yield learning and lower cost per bit. Samsung became the largest DRAM supplier in 1992; Hyundai acquired LG Semicon in 1999, creating the business that became Hynix and then SK hynix.	Korea consolidated leadership as Japanese suppliers cut investment or exited commodity DRAM, and the industry concentrated around Samsung, SK hynix and Micron. **Lesson: memory rewards suppliers that can fund the next node and keep investing when weaker competitors retrench.**
CHINA EMERGING	2020s — not yet a completed leadership era	CXMT (DRAM), YMTC (NAND) Watch: JHICC, XMC	Scale before frontier leadership. China is combining domestic demand, import substitution, policy support and sustained capacity additions. CXMT is gaining share in conventional DDR and LPDDR while YMTC builds scale in NAND, so China can matter to marginal pricing before reaching technology or cost parity.	China is not yet replacing Korea: HBM and advanced server DRAM remain concentrated among Samsung, SK hynix and Micron, while China is more relevant in mainstream DRAM and NAND. **Lesson: the first impact falls on the volume pool and industry supply discipline; the larger risk is China subsequently entering the premium profit pools.**

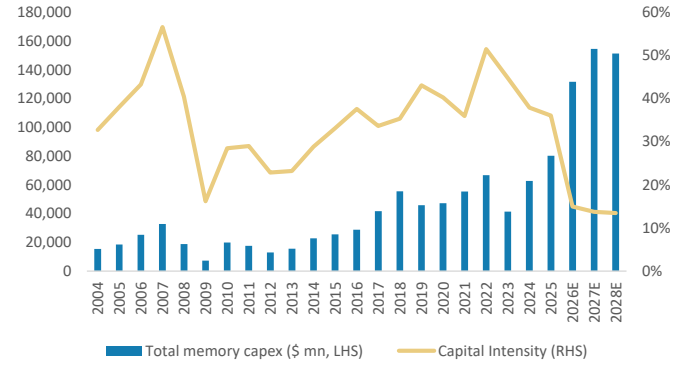
Source: Computer History Museum; National Academies; RIETI; academic literature on industry leadership transitions; Samsung and company disclosures; Morgan Stanley Research.

China Memory Ambitions – Industry Implications

Industry fragmentation. The number of key DRAM players declined from 24 in 1996 to three in 2020, before rising to four in 2026 with CXMT. Cost leadership remains critical, while process complexity, yield learning and customer qualification have become increasingly important barriers to entry. We see several key implications for the memory industry, including: 1) higher absolute capex requirements, 2) lower normalized returns over time, and 3) greater valuation dispersion. Strategic memory, including HBM and advanced server DRAM, remains concentrated among Samsung, SK hynix and Micron, while China is becoming more relevant in conventional DRAM and NAND.

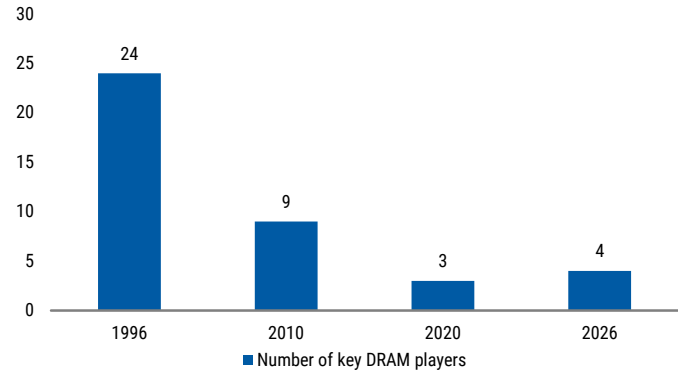
Higher absolute capex, despite moderating capital intensity. We expect total memory capex to increase sharply through 2026–28E, driven by AI-related memory demand and capacity expansion by Chinese memory companies. However, as shown in Exhibit 12, capital intensity as a percentage of industry revenue is expected to decline from its 2022 peak toward the low teens by 2028E mainly driven by significant price inflation. Samsung, SK hynix and Micron will need to continue investing to maintain their technology, cost and product-mix advantages across EUV-enabled DRAM, HBM, advanced packaging and 3D NAND scaling. HBM is particularly capital- and capacity-intensive: its larger die sizes, stacking, TSV integration, packaging and testing requirements reduce effective bit output per wafer and require additional investment beyond front-end wafer capacity. CXMT's expansion adds competitive pressure, but the near-term impact should be more visible in investment and market-share dynamics than in global pricing. We expect CXMT to continue adding capacity, while restrictions on EUV/DUV, servicing and advanced process-control tools remain important constraints.

Exhibit 12: Total memory capex is rising to record levels



Source: Company data, Morgan Stanley Research estimates

Exhibit 13: CXMT turns the DRAM Big Three into a Big Four, but the industry remains consolidated

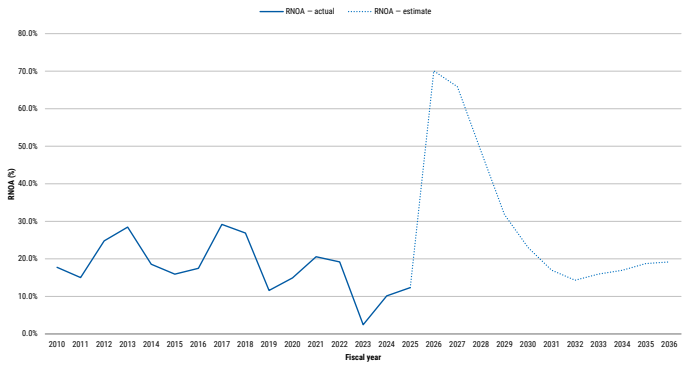


Source: Morgan Stanley Research

Reduced sustainable returns. With aggressive inroads from China into the memory market, the industry's sustainable margins (current 80%+ operating margins for DRAM) may be at risk, in our view. We currently forecast DRAM operating margin over a cycle to normalize to 50% in the outer years in our 10-year residual income forecast. We model a more rapid rate of decline for NAND margins to 15-20% given more severe competitive dynamic with recent progress made by YMTC in China.

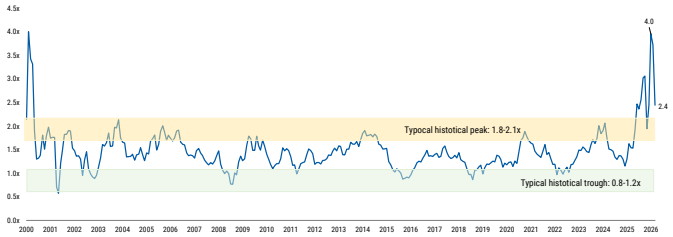
Valuation multiples – peak re-rating can overshoot, but China may lower the normalized range. Strong memory price hikes, AI demand and improved earnings visibility through LTAs have driven the average 12M forward P/B for DRAM companies to a recent peak of ~4.0x, well above the typical historical peak range of ~1.8-2.1x, before moderating to ~2.4x currently which compares to a ~50% ROE (2026) vs sustainable ROE of 10-15%, in-line with 2010-2025 average. This suggests China's entry does not necessarily cap peak-cycle valuation when pricing and earnings revisions are sufficiently strong. The more relevant longer-term question is where multiples normalize once pricing moderates as CXMT and YMTC gain scale, greater competition in mainstream DRAM and NAND pressuring through-cycle margins, sustainable ROE and capital returns. In our view, China is therefore more relevant to the long-term normalized valuation range and downside floor than to peak-cycle re-rating. The key upside/downside debate for memory stocks is shifting from whether China can produce mainstream memory to how quickly CXMT and YMTC become meaningful swing suppliers, impact pricing and how quickly their more aggressive expansion lowers sustainable returns for the incumbents.

Exhibit 14: Memory returns could normalize from current highs…



Source: Morgan Stanley Research. Note: Samsung return on net operating asset (RONA) as proxy of memory industry

Exhibit 15: … as well as valuation can mean-revert to a lower normalized range

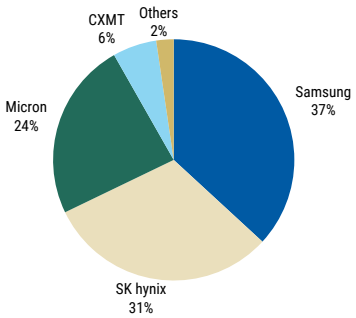


Source: Factset, Morgan Stanley Research.

China memory ambition and industry dynamics

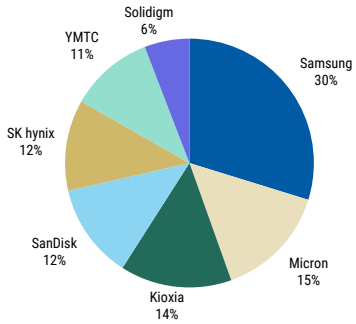
Global DRAM memory market is currently dominated by Samsung, Hynix and Micron, which hold 80%+ global market share in terms of revenue, followed by CXMT's 6%. NAND market is more fragmented, with Samsung being the distant leader, followed by Micron, with the rest of major players taking 11-14% market share respectively.

Exhibit 16: Global DRAM market share (2026E)



Source: Trendforce

Exhibit 17: Global NAND market share (2026E)



Source: Trendforce

Key entry barriers – why China is not yet a leading-edge player

Cost structure remains the key differentiator for commodity memory. Therefore, apart from fundamental IP, manufacturing know-how relating to process integration, defect control, yield improvement and technology migration is also important. Process recipes, reliability data and defect-control expertise accumulate over repeated production cycles and cannot be replicated quickly through capital spending alone. Customer qualification also creates additional barriers because switching suppliers involves engineering, reliability and inventory costs.

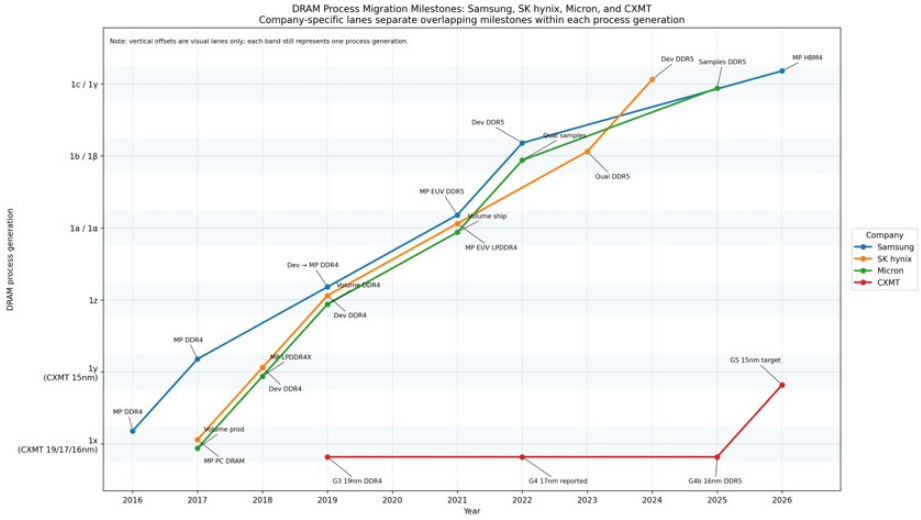
Equipment access has become another important entry barrier. CXMT continues to scale conventional DRAM using immersion DUV and multi-patterning rather than EUV. This approach extends its process roadmap but requires more masks and process steps, lengthens cycle times, raises overlay and defect risk, and can increase cost per good die. Even when visible DDR5 specifications are competitive, the leading suppliers retain advantages in bits per wafer, yield, cycle time and premium-grade output.

HBM introduces an additional set of barriers. Competitive HBM production requires high-yield advanced DRAM dies, TSV formation, wafer thinning, thermal and warpage control, logic-base-die integration, advanced packaging, testing and qualification with accelerator developers. As a result, producing a functional HBM stack is not sufficient to establish a commercially competitive HBM business.

Where has China made breakthroughs?

On DRAM, the breakthrough is mainly process and density. CXMT completed its transition to 100% Gen4B (~16nm-class) production in 2Q26, with yields we estimate above 90%. Reverse-engineering suggests that the die density of CXMT's 16Gb DDR5 is comparable with some older-generation products from leading suppliers. Node labels are not directly comparable and similar die specifications do not imply equivalent cost, yield or reliability. CXMT may be about one visible product generation behind in DDR5 while remaining several years behind in HBM packaging, qualification and ecosystem capabilities. CXMT's DDR5 reportedly supports 5600MHz with a roadmap to 6400MHz, broadening its relevance to mainstream server DRAM, and it has developed 3D stacking capability for a 12H HBM3E package, for which we forecast mass production in 1H27, and ultimately HBM4 in 2028.

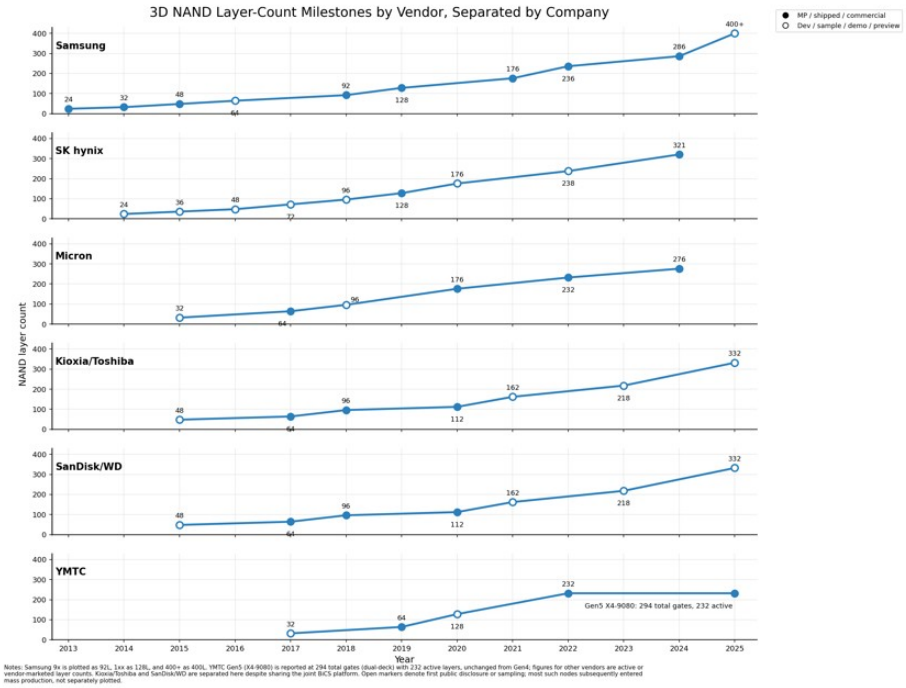
Exhibit 18: DRAM process migration milestones



Source: Company data, TechInsights, Morgan Stanley Research

On NAND, YMTC's breakthrough remains architectural. Hybrid bonding is the basis of Xtacking, now in mass production at 267 layers with 300+ layer devices in development. Its PCIe 5.0 enterprise portfolio now includes TLC SSDs delivering up to 14.0–14.2GB/s of sequential-read bandwidth and 3.1–3.3mn random-read IOPS. Its QLC-based PE501 scales to 122.88TB per drive and is rated at 5,000 program/erase cycles and 0.6 drive writes per day, indicating meaningful progress in storage density, endurance and system-level integration. However, product availability should not yet be equated with broad hyperscaler qualification or an enterprise product mix comparable with the global leaders.

Exhibit 19: NAND process migration milestones



Source: Company data, Morgan Stanley Research

Equipment and materials localization is the third breakthrough and the least visible.

Gen 4B was built primarily with domestic equipment, with the exception of DUV tools, and Chinese suppliers are qualifying across etch, deposition, cleaning, CMP, coating and developing, and inspection. What has not broken through: immersion DUV, advanced metrology, manufacturing software, spare parts and servicing, and HBM packaging and qualification. Those remain among the least localized parts of China's semiconductor equipment chain.

2028 NAND scenario stress test

We think the biggest swing factor for 2028 and beyond could be related to YMTC. This Chinese NAND supplier currently has Fab4 and Fab5 under construction at the same time, with each having 100kwpm planned capacity. If they decide to dedicate all of that to NAND, with the announced five fabs, they could eventually take 24% market share in the global NAND industry (see Chipflation – Navigating A Memory Crisis).

In our base case scenario, we assume 5% YoY growth for non-AI NAND demand, and the rest of the supplier's capacity expansion to be the same as in our US Semiconductor team's forecast. We assume AI SSD demand will grow by another 30-60% YoY in 2028 vs. YMTC capacity ranging from 310kwpm (current base case) to 470kwpm (potential max capacity of all announced five fabs).

In conclusion, we still see NAND supply remaining tight into 2028 if AI capex continues to grow and YMTC keeps a more disciplined approach on expansion. However, if YMTC or other players accelerate greenfield expansion, there will likely be potential risks of oversupply.

Exhibit 20: YMTC capacity expansion vs. AI SSD growth scenario test

YMTC Capacity kwpm in 2028e	AI SSD demand growth % YoY in 2028			
	30%	40%	50%	60%
310	-1%	-2%	-4%	-6%
330	0%	-1%	-3%	-4%
350	2%	0%	-2%	-3%
370	3%	1%	-1%	-2%
390	4%	2%	1%	-1%
410	5%	4%	2%	0%
430	7%	5%	3%	1%
450	8%	6%	4%	3%
470	9%	7%	5%	4%

Source: Company data, Morgan Stanley Research

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(as of July 31, 2026)

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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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